

GEPA: 反思式提示进化可优于强化学习

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Abstract

大语言模型 (LLM) 正越来越多地通过强化学习 (RL) 方法适配下游任务, 例如 Group Relative Policy Optimization (GRPO); 但这类方法通常需要成千上万次 rollout 才能学会新任务。我们认为, 相比从稀疏标量奖励中得到的策略梯度, 语言本身的可解释性为 LLM 提供了更丰富的学习媒介。为验证这一点, 我们提出 GEPA (**Genetic-Pareto**), 一种充分利用自然语言反思、从试错过程中学习高层规则的提示优化器。对于任意包含一个或多个 LLM 提示的 AI 系统, GEPA 都会采样其轨迹 (如推理过程、工具调用与工具输出), 再用自然语言对这些轨迹进行反思, 以诊断问题、提出并测试提示更新, 并组合来自自身尝试中 Pareto 前沿的互补经验。得益于这种设计, GEPA 往往只需极少量 rollout 就能带来显著的质量提升。在六项任务上, GEPA 平均比 GRPO 高出 6%, 最高可提升 20%, 同时所需 rollout 数最多可减少 35 倍。GEPA 还在 10% 以上的幅度上超越了当前领先的提示优化器 MIPROv2 (例如在 AIME-2025 上精度提升 +12%), 并在将其作为代码优化的推理时搜索策略方面展示出有前景的结果。我们已在 <https://github.com/gepa-ai/gepa> 开源代码。

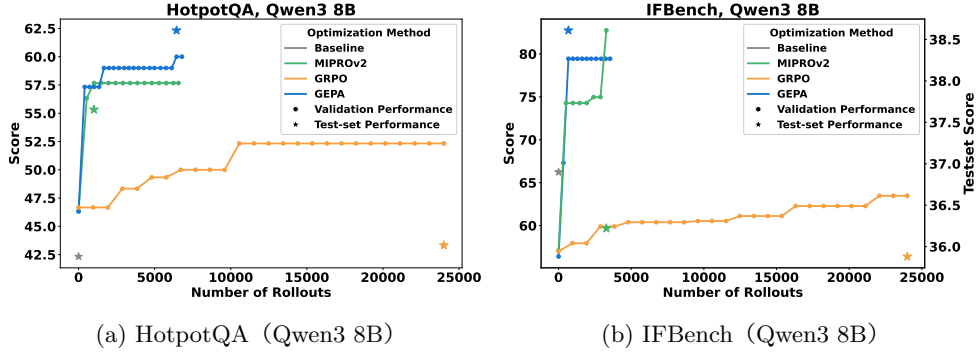


Figure 1: GEPA 提示优化器与当前最先进的提示优化器 MIPROv2 以及 GRPO (24,000 次 rollout) 在学习行为上的对比。随着 rollout 增多, 提示优化器的学习速度明显快于 GRPO。GEPA 在最终得分上显著优于 GRPO 和 MIPROv2。图中的测试集星形标记展示了它们在留出问题集上的性能差距。

1 引言

大语言模型 (LLM) 的发展推动了各类智能体与系统的出现, 它们将模糊的自然语言规范与检索、代码执行等工具结合起来。这也带来了一个核心问题: 应该如何针对下游性能来优化 LLM? 一种流行做法是采用带可验证奖励的强化学习 (RLVR), 例如 Group Relative Policy Optimization (GRPO) (Shao

et al., 2024), 它把任务成功指标视作 rollout 末端的标量奖励, 并据此估计策略梯度 (Lambert, 2025)。尽管这类 RL 方法有效, 但在实践中通常需要数万次 rollout 才能适应新任务。比如, 近年来使用 GRPO 的工作往往都要进行多达数十万次 rollout 训练 (Chen et al., 2025b; Wu et al., 2025b; Zhang et al., 2025; Jin et al., 2025; Si et al., 2025; Wang et al., 2025a; Chen et al., 2025a; Sha et al., 2025; Lin et al., 2025a; Peng et al., 2025; Song et al., 2025)。这种样本低效很快会成为严重瓶颈: 许多下游 LLM 应用需要昂贵的工具调用, 自身的采样推理预算有限, 或者根本无法对最大、性能最强的 LLM 进行权重微调。

我们观察到, 即便是非常复杂的 LLM 系统, 其 rollout 也可以序列化为自然语言轨迹: 其中包括各个 LLM 模块的指令、由此产生的推理链、工具调用, 以及奖励函数的内部过程 (例如在被压缩成标量奖励之前的编译报错信息)。由于现代 LLM 很容易理解这种序列化轨迹, 我们认为, 相较于标准 RL 方法, 那些通过反思这些轨迹、在自然语言中有意识地学习的算法, 能够更有效地利用 LLM 已有的强大语言先验。

我们在多跳推理 (HotpotQA; Yang et al. 2018)、数学 (AIME、LiveBench-Math; Balunović et al. (2025); White et al. (2025))、指令遵循 (IFBench; Pyatkin et al. 2025)、隐私感知委托 (PUPA; Li et al. 2025a) 以及检索增强验证 (HoVer; Jiang et al. 2020) 等任务上评估 GEPA, 并同时使用开源模型 (Qwen3 8B; Yang et al. 2025; Team 2025) 和闭源模型 (GPT-4.1 Mini; OpenAI 2025)。我们发现, GEPA 具有很强的泛化性和样本效率: 在 Qwen3 8B 上, 它在最多减少 $35\times$ rollout 的情况下, 仍可比 GRPO (24k rollout) 高出最多 20%, 在六项任务上的平均增益为 +6%。GEPA 还在所有基准和模型上都超过了此前最先进的方法 MIPROv2 (Opsahl-Ong et al., 2024), 取得了 +13% 的综合增益, 是 MIPROv2 的 +5.6% 的两倍以上。

从定性角度看, GEPA 学到的提示往往相当丰富。图 2 展示了一个用于 HotpotQA 多跳问答系统中查询生成模块的提示片段, 图 5 则说明, 即使只做一次反思式更新, 也常常能带来明显收益。这些结果表明, 借助语言反馈进行反思式提示进化, 能够同时带来更高的样本效率和更稳健的泛化性能, 为数据或预算受限环境中的复杂 AI 工作流优化提供了一条切实可行的路径。我们还在第 5.1 节展示了 GEPA 作为 NPUEval (Kalade & Schelle, 2025) 和 KernelBench (Ouyang et al., 2025) 代码优化中的推理时搜索策略的能力, 并在第 5.2 节展示了它在对抗式提示搜索中的应用。我们在多跳推理 (HotpotQA; Yang et al. 2018)、数学 (AIME、LiveBench-Math; Balunović et al. (2025); White et al. (2025))、指令遵循 (IFBench; Pyatkin et al. 2025)、隐私感知委托 (PUPA; Li et al. 2025a) 以及检索增强验证 (HoVer; Jiang et al. 2020) 等任务上评估 GEPA, 并同时使用开源模型 (Qwen3 8B; Yang et al. 2025; Team 2025) 和闭源模型 (GPT-4.1 Mini; OpenAI 2025)。我们发现, GEPA 具有很强的泛化性和样本效率: 在 Qwen3 8B 上, 它在最多减少 $35\times$ rollout 的情况下, 仍可比 GRPO (24k rollout) 高出最多 20%, 在六项任务上的平均增益为 +6%。GEPA 还在所有基准和模型上都超过了此前最先进的方法 MIPROv2 (Opsahl-Ong et al., 2024), 取得了 +13% 的综合增益, 是 MIPROv2 的 +5.6% 的两倍以上。

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2 问题定义

我们遵循相关工作, 将**复合 AI 系统**定义为: 任何由一个或多个语言模型 (LLM) 调用组成、可能穿插外部工具调用、并由任意控制流编排的模块化系统。这一定义覆盖了现实中很广泛的一类基于 LLM 的 AI 系统, 包括智能体、多智能体系统以及 ReAct (Yao et al., 2023)、Archon (Saad-Falcon et al., 2025) 等通用脚手架技术。按照 Soyulu et al. (2024); Khattab et al. (2024); Opsahl-Ong et al. (2024); Tan

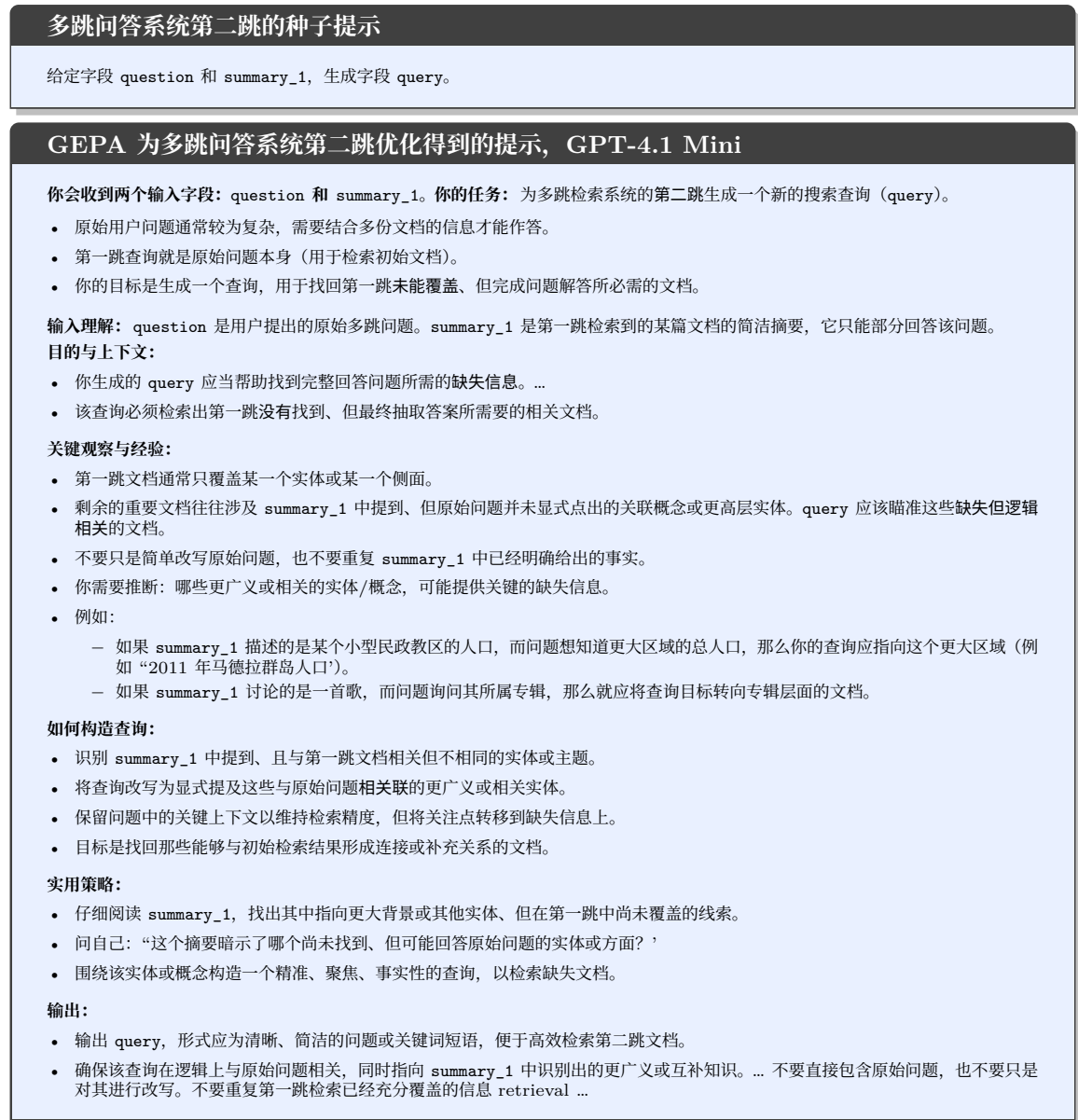


Figure 2: 该图展示了 GEPA 为一个多跳问答系统中的第二跳文档检索模块生成的提示示例，以及它起始时所使用的种子提示。附录 ?? 还比较了 GEPA 在所有任务上生成的提示与 MIPROv2 生成的提示。

et al. (2025) 的做法，我们将该系统形式化为 $\Phi = (M, C, \mathcal{X}, \mathcal{Y})$ ，其中 $M = \langle M_1, \dots, M_{|M|} \rangle$ 表示语言模块集合， C 指定控制流逻辑， $\mathcal{X}$ 和 $\mathcal{Y}$ 分别表示全局输入/输出模式。每个模块 $M_i = (\pi_i, \theta_i, \mathcal{X}_i, \mathcal{Y}_i)$ 都是一个 LLM 子组件： π_i 是其（system）提示，包含指令与 few-shot 示例； θ_i 是底层模型权重； $\mathcal{X}_i, \mathcal{Y}_i$

则是输入/输出模式。在运行时, C 负责调度这些模块的执行, 例如把一个模块的输出传给另一个模块、按条件调用模块, 或与工具 API 交互。因此, C 可以按任意顺序、多次调用不同模块。

给定系统 Φ , 记 $\Pi_\Phi = \langle \pi_1, \dots, \pi_{|M|} \rangle$ 为全部模块提示的集合, $\Theta_\Phi = \langle \theta_1, \dots, \theta_{|M|} \rangle$ 为模块权重集合。因此, 可学习参数记为 $\langle \Pi, \Theta \rangle_\Phi$ 。对于任务样本 (x, m) , 其中 x 对应输入模式 $\mathcal{X}$, 而 m 包含评测所需元数据 (如标准答案、评价规则、代码单元测试等), 系统会生成输出 $y = \Phi(x; \langle \Pi, \Theta \rangle_\Phi)$ 。随后, 度量函数 $\mu: \mathcal{Y} \times \mathcal{M} \rightarrow [0, 1]$ 用于衡量输出 y 相对于元数据 m 的质量 (例如 exact match、F1、pass rate 等)。于是, 优化问题可定义如下, 其中 $\mathcal{T}$ 表示任务分布:

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg \max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x, m) \sim \mathcal{T}} [\mu(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m)]. \quad (1)$$

我们采用这一通用问题表述, 允许同时更新语言模块的提示与权重, 从而能够比较工作在不同参数空间中的优化算法 (例如 GEPA 与 GRPO)。

高样本效率优化。 在许多真实场景中, rollout, 也就是对 Φ 的一次调用再加上通过 μ 进行评测, 往往在计算、金钱或时间上都十分昂贵。因此, 优化器在数据集 $\mathcal{D}_{\text{train}} = \{(x, m)_i\}_{i=1}^N$ 上最多只能使用 B 次 rollout, 并且可以完全访问 μ 。它的目标是在不超过 rollout 预算 B 的前提下, 找到能最大化留出性能的参数 $\langle \Pi^*, \Theta^* \rangle_\Phi$:

$$\langle \Pi^*, \Theta^* \rangle_\Phi = \arg \max_{\langle \Pi, \Theta \rangle_\Phi} \mathbb{E}_{(x, m) \sim \mathcal{T}} [\mu(\Phi(x; \langle \Pi, \Theta \rangle_\Phi), m)], \quad \text{s.t.} \quad \# \text{rollouts} \leq B. \quad (2)$$

因此, 核心挑战在于: 如何从每一次昂贵的 rollout 中提取最大化的学习信号, 从而让复杂、模块化的 AI 系统在低数据或预算受限的环境中也能实现有效适配?

3 GEPA: 反思式提示进化

我们提出 GEPA (Genetic-Pareto), 这是一种面向复合 AI 系统的高样本效率优化器, 其设计基于三条核心原则: 遗传式提示进化 (第 3 节)、利用自然语言反馈进行反思 (第 3 节), 以及基于 Pareto 的候选选择 (第 3.1 节)。图 3 给出了 GEPA 的整体概览, 而完整算法则在图 4 中形式化展示。GEPA 的输入包括: 一个带有待优化简单提示的系统 Φ 、训练数据集 $\mathcal{D}_{\text{train}}$ (由第 2 节定义的任务样本 (x, m) 组成)、任务标准评测指标 μ 、反馈函数 μ_f (第 3 节引入), 以及总 rollout 预算 B 。需要注意的是, GEPA 只进化提示集合 Π_Φ , 而底层 LLM 权重 Θ_Φ 保持不变。

遗传式优化循环: 给定 AI 系统 Φ , 其目标是找到能够最大化任务表现的参数 Π_Φ 。GEPA 从一个仅包含基础系统的候选池 $\mathcal{P}$ 开始, 其中每个候选都是 $\langle \Pi, \Theta_{\text{frozen}} \rangle_\Phi$ 的一个具体实例。随后, 它进入优化循环, 不断提出新候选, 直到评估预算耗尽为止。这些候选通过反思式变异或交叉从已有候选中派生而来, 并受 rollout 反馈引导; 每个候选都继承其父节点及自身 rollout 中的学习信号, 因此 GEPA 能沿着遗传树逐步累积知识。在每一轮中, GEPA 会: (i) 选出有前景的候选; (ii) 在一个任务 minibatch 上提出并评估其变体; (iii) 如果该变体优于其父候选, 就将其连同祖先信息加入 $\mathcal{P}$, 并在用于选择的验证集 $\mathcal{D}_{\text{pareto}}$ 上进行评估。当预算耗尽后, GEPA 返回在 $\mathcal{D}_{\text{pareto}}$ 上综合表现最好的候选。

反思式提示变异: 复合 AI 系统在执行过程中生成的自然语言轨迹, 为理解各个模块的行为与职责提供了丰富的可见性, 因为其中记录了中间推断及其背后的推理步骤。当这些轨迹与系统最终结果 (例如成功或失败) 结合起来时, 它们又具备很强的诊断性, 使研究者可以把错误或成功追溯到具体模块层面的决策。LLM 可以通过反思利用这些轨迹, 执行一种隐式的 credit assignment, 将最终结果的责任归因到相关模块上。随后, 这种反思又能被用来对单个模块进行定向更新, 从而以较大幅度且较高有效性地改善整个系统的行为。

给定当前优化循环中一个待变异的候选 (它是从 Pareto 前沿上随机选出的, 见下文第 3.1 节), GEPA 会在随机采样得到的 minibatch 训练查询上执行该候选, 并跟踪程序执行过程。通过执行轨迹, GEPA

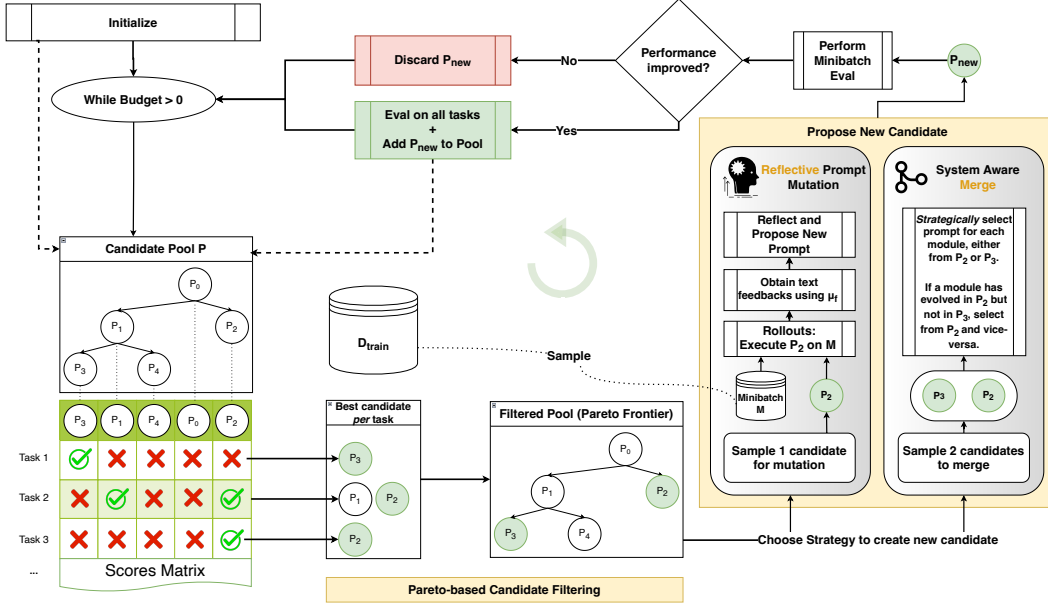


Figure 3: GEPA 在每一轮中都会通过改进已有候选来提出一个新候选，所采用的两种策略分别是反思式提示变异（第 3 节）和系统感知 Merge（附录 ??）。新候选会先在一个 minibatch 上评估，若表现提升，再扩大到更大数据集上评估。不同于总是选择当前最优候选继续变异，这种贪心做法容易陷入局部最优；GEPA 引入了基于 Pareto 的候选采样（第 3.1 节），它会按任务维度筛选并采样最佳候选列表，从而保证足够的多样性。总体而言，这些设计使得 GEPA 在保持很高样本效率的同时，也展现出强化能力。

提取出目标模块的输入、输出与推理过程，并调用反馈函数 μ_f ，获得数值得分和文本反馈，其中可能包含编译报错、未通过 rubric 等评测细节。GEPA 按照某种策略（如 round-robin）选择要更新的模块（在该语言程序包含的 $|M|$ 个模块中），随后把当前提示、程序轨迹、得分与反馈一起交给一个反思模型，让它对成功或失败进行归因，并提出修订后的指令。接着，用更新后的模块替换原模块，重新在 minibatch 上评估整个语言程序；如果得分提高，新程序就会被加入候选池。用于反思式提示更新的 meta-prompt 见附录 ??，完整算法见算法 1。

将评测轨迹作为诊断信号：LLM 生成的文本构成了 AI 系统的执行轨迹；而环境在计算奖励时产生的文本（例如在给出 0 分之前的编译报错信息）则构成评测轨迹。除了对执行轨迹本身进行反思，我们还识别出另一类同样有价值的诊断信息来源，即评测轨迹。许多评测指标本身就采用了丰富的评估策略（例如代码评测往往涉及编译、执行与 profiling），因此会在计算标量奖励之前产生自然语言轨迹。我们提出利用这些评测轨迹进行反思式归因与有针对性的提示更新。GEPA 的做法是把奖励函数 μ 扩展为反馈函数 μ_f ：它在评测过程中提取文本轨迹，并将其与最终分数一起作为 `feedback_text` 返回。在可能的情况下，这类反馈甚至可以细化到模块级别（例如在多跳系统中，评测器可以在每一跳之后给出反馈）。在实践中，也有一些领域会由人工评审对 AI 系统响应进行评分，并给出解释这些标量评分的详细文字反馈。若能获得这类信号， D_{train} 也可以为每个样本补充人工撰写的解释；在反思阶段，GEPA 可以把这些解释作为额外的 `feedback_text` 使用，即使 rollout 本身提供的自然语言反馈有限或不存在，也能指导定向提示更新。

Algorithm 1 GEPA: 反思式进化提示优化器

Require: 输入: 系统 Φ 、数据集 $\mathcal{D}_{\text{train}}$ 、评测指标 μ 、反馈函数 μ_f 、预算 B

Require: 超参数: minibatch 大小 b 、Pareto 集大小 n_{pareto}

- 1: 将 $\mathcal{D}_{\text{train}}$ 划分为 $\mathcal{D}_{\text{feedback}}$ 与 $\mathcal{D}_{\text{pareto}}$, 使得 $|\mathcal{D}_{\text{pareto}}| = n_{\text{pareto}}$
- 2: 初始化候选集合 $\mathcal{P} \leftarrow [\Phi]$, 父节点集合 $\mathcal{A} \leftarrow [\text{None}]$
- 3: **for** 对于 $\mathcal{D}_{\text{pareto}}$ 中的每个 (x_i, m_i) **do**
- 4: $S_{\Phi}[i] \leftarrow \mu(\Phi(x_i), m_i)$
- 5: **end for**
- 6: **while** 预算 B 尚未耗尽 **do**
- 7: $k \leftarrow \text{SelectCandidate}(\mathcal{P}, S)$
- 8: $j \leftarrow \text{SelectModule}(\Phi_k)$
- 9: 从 $\mathcal{D}_{\text{feedback}}$ 中采样大小为 b 的 minibatch, 记为 $\mathcal{M}$
- 10: 使用 μ_f 为 $\Phi_k[j]$ 在 $\mathcal{M}$ 上收集反馈、分数与轨迹
- 11: $\pi'_j \leftarrow \text{UpdatePrompt}(\pi_j, \text{feedbacks}, \text{traces}[j])$
- 12: $\Phi' \leftarrow$ 复制 Φ_k , 并用 π'_j 更新其中模块 j
- 13: $\sigma, \sigma' \leftarrow \mathcal{M}$ 上的平均得分 (更新前、更新后)
- 14: **if** $\sigma' > \sigma$ **then**
- 15: 将 Φ' 加入 $\mathcal{P}$; 将 k 加入 $\mathcal{A}$
- 16: **for** 对于 $\mathcal{D}_{\text{pareto}}$ 中的每个 (x_i, m_i) **do**
- 17: $S_{\Phi'}[i] \leftarrow \mu(\Phi'(x_i), m_i)$
- 18: **end for**
- 19: **end if**
- 20: **end while**
- 21: **return** 在 $\mathcal{D}_{\text{pareto}}$ 上平均得分最高的 Φ^*

Algorithm 2 基于 Pareto 的候选选择

- 1: **function** SelectCandidate($\mathcal{P}, S$)
- 2: // 构建按样本划分的 Pareto 集
- 3: **for** 对于每个 i **do**
- 4: $s^*[i] \leftarrow \max_k S_{\mathcal{P}[k]}[i]$
- 5: $\mathcal{P}^*[i] \leftarrow \{\mathcal{P}[k] : S_{\mathcal{P}[k]}[i] = s^*[i]\}$
- 6: **end for**
- 7: $\mathcal{C} \leftarrow$ unique candidates in $\bigcup_i \mathcal{P}^*[i]$
- 8: $D \leftarrow \emptyset$
- 9: **while** there exists $\Phi \in \mathcal{C} \setminus D$ dominated by another in $\mathcal{C} \setminus D$ **do**
- 10: $D \leftarrow D \cup \{\Phi\}$
- 11: **end while**
- 12: 从每个 $\mathcal{P}^*[i]$ 中移除 D , 得到 $\hat{\mathcal{P}}^*[i]$
- 13: 令 $f[\Phi] =$ 满足 $\Phi \in \hat{\mathcal{P}}^*[i]$ 的 i 的数量
- 14: 以与 $f[\Phi_k]$ 成正比的概率, 从 $\hat{\mathcal{C}}$ 中采样 Φ_k
- 15: **return** Φ_k 在 $\mathcal{P}$ 中的索引 k
- 16: **end function**

Figure 4: (左) GEPA 用于反思式提示进化的核心算法。GEPA 以迭代方式工作: 在每一轮中, 它会选出当前候选中的一部分进行进化 (第 7 行), 在一个 minibatch 的 rollout 上执行被选中的候选, 并利用专门的 *feedback function* μ_f 在可用时收集模块级反馈 (第 9–10 行, 细节见第 3 节); 随后使用 LLM 对提示进行反思式更新 (第 11 行), 并评估以新提示实例化后的系统在该 minibatch 上是否提升了性能 (第 14 行)。若有提升, GEPA 就会继续在完整的 $\mathcal{D}_{\text{pareto}}$ 集合上评估这一新系统候选, 并将其加入候选列表, 同时记录其父节点。(右) GEPA 核心算法中的 SelectCandidate 子过程负责找出下一轮优化中最值得继续进化的候选。GEPA 的核心候选选择策略, 是在所有任务样本构成的 Pareto 前沿上寻找非支配候选, 并根据它们出现在 Pareto 前沿中的频次, 随机选出其中一个。

3.1 基于 Pareto 的候选选择

GEPA 是一个高度模块化的算法, 支持多种候选选择策略, 而具体采用哪种策略, 决定了优化过程中的 exploration-exploitation 权衡。一个朴素做法是始终选择当前表现最好的候选, 但这往往会让优化器陷入局部最优: 一旦发现某种占优策略, 就很难再超越它, 结果是预算被不断消耗, 却无法学到新的、可能更优的策略。图 6a 展示了这种现象: 当搜索找到一个新策略 (第一层子节点) 之后, 后续过程会反复尝试细化它, 但始终无法进一步改进, 最终耗尽预算。

为了解决这一问题, GEPA 采用了基于 Pareto 的 “illumination” 策略 (Mouret & Clune, 2015), 如算法 2 所示。对于每个训练样本, GEPA 会记录所有候选中达到的最高得分, 从而形成一个 Pareto 前沿。凡是在至少一个任务上达到最优得分的候选都会被保留, 而被其他候选严格支配的则会被剪枝。在这一剪枝后的集合中, GEPA 会按每个候选在多少任务上处于领先来加权, 随机采样一个候选。该策略帮助 GEPA 在不过度扩张搜索规模的前提下跳出局部最优, 并通过把预算聚焦到那些已经体现出 “获胜策略” 的候选上, 在探索与利用之间取得高效平衡。

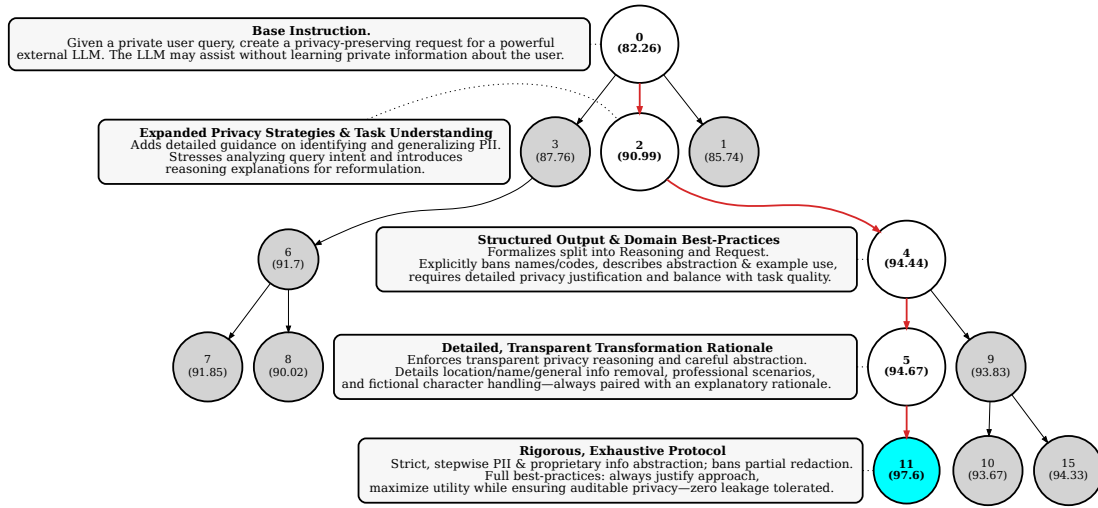


Figure 5: GEPA 的反思式提示变异会系统性地把任务相关的细粒度经验融入提示之中，从而带来显著的性能提升。该图展示了 GEPA 在 PUPA 隐私保护委托任务上的优化轨迹：它是图 ?? 的一个带注释子树，用来说明提示是如何被逐步改进的。从基础提示（候选 0）到最优提示（候选 11）的演化路径用红色箭头标出，每一步的关键提示变化都标注在相应节点旁。附录 ?? 提供了这些阶段的完整提示文本。可以看到，每一次提示更新都会在持续优化的基础上加入更有针对性的任务细节，从而不断积累经验并提升任务表现。

4 实验评测

Table 1: Qwen3 8B 在不同优化器下的基准结果。除 AIME 外，GEPA 与 GEPA+Merge 在所有基准上都以远少于 GRPO 的 rollout 数取得了更好的性能。例如，在 IFBench 上，GEPA 仅用 678 次 rollout 就找到了最优提示，得到 38.61%，超过了 GRPO 在 24,000 次 rollout 下 35.88% 的测试集得分。

Qwen3 8B	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	42.33	36.90	35.33	80.82	27.33	48.70	45.23	—
GRPO	43.33	35.88	38.67	86.66	38.00	51.26	48.91	+3.68
MIPROv2	55.33	36.22	47.33	81.55	20.00	46.60	47.84	+2.61
GEPA	62.33	38.61	52.33	91.85	32.00	51.95	54.85	+9.62
GEPA+Merge	64.33	28.23	51.67	86.26	32.00	51.95	52.40	+7.17
Total optimization budget (# rollouts)								
GEPA (+Merge)	6871	3593	7051	2426	1839	1839	3936	—
GRPO	24000	24000	24000	24000	24000	24000	24000	—

Table 2: 在 GPT-4.1 Mini 上评估不同优化器的基准结果。作为一种提示优化系统，GEPA 可以直接应用于闭源模型，并超过包括 MIPROv2、Trace（配合 OptoPrime 优化器）以及 TextGrad 在内的最先进提示优化器。此外，GEPA 优化得到的提示还展现出较强的跨模型泛化能力：“GEPA-Qwen-Opt”完全基于较弱的 Qwen3-8B 进行优化，却能在无需修改的情况下迁移到 GPT-4.1-Mini 上并取得 +9% 的增益，明显超过了所有直接在 GPT-4.1-Mini 上优化的基线方法（MIPROv2、TextGrad、Trace）。

GPT-4.1 Mini	HotpotQA	IFBench	Hover	PUPA	AIME-2025	LiveBench-Math	Aggregate	Improvement
Baseline	38.00	47.79	46.33	78.57	49.33	58.20	53.03	—
Trace (OptoPrime)	60.33	51.19	46.00	74.18	45.33	60.74	56.30	+3.27
MIPROv2-No-Demos	38.00	52.04	51.33	91.85	48.67	60.97	57.14	+4.11
MIPROv2	58.00	49.15	48.33	83.37	51.33	61.84	58.67	+5.64
TextGrad	62.33	48.64	47.67	85.68	46.67	63.84	59.14	+6.11
GEPA	69.00	52.72	51.67	94.47	59.33	64.13	65.22	+12.19
GEPA+Merge	65.67	55.95	56.67	96.46	59.33	64.13	66.36	+13.33
Optimized with Qwen3-8B, evaluated on GPT-4.1-Mini								
GEPA-Qwen-Opt	65.67	49.83	54.67	90.05	52.67	59.31	62.03	+9.00

我们采用标准的训练集/验证集/测试集划分。优化器可以完整访问训练集，包括文本与标签，以用于程序调优。尽管优化器可以通过跟踪验证集得分来监控候选参数（例如模型检查点）的表现，以实现早停等机制，但它们并不能直接访问验证样本的具体内容。我们在六个基准上进行评估：AIME-2025 (Balunović et al., 2025)、LiveBench-Math (White et al., 2025)、HotpotQA (Yang et al., 2018)、IFBench (Pyatkin et al., 2025)、HoVer (Jiang et al., 2020) 和 PUPA (Li et al., 2025a)；每个基准都配有现成的复合 AI 系统和反馈函数。实验使用 Qwen3 8B (Yang et al., 2025) 与 GPT-4.1 Mini (OpenAI, 2025)，采用统一的推理设置，并与当前最先进的优化器 MIPROv2 (Opsahl-Ong et al., 2024)、Trace（及其 OptoPrime 优化器）(Cheng et al., 2024)、TextGrad (Yuksekgonul et al., 2025) 以及 GRPO¹ (Shao et al., 2024) 进行比较。附录 ?? 进一步给出了基准、系统与反馈函数（小节 ??）、模型与推理设置（小节 ??）、实验的金钱成本（小节 ??）以及优化器配置（小节 ??）的详细说明。表 1、表 2 和图 ?? 总结了我们的主要结果，并由此得到以下观察。

观察 1：反思式提示进化具有极高的样本效率，并且能够超越权重空间中的强化学习：在四个基准上，GEPA 在复合 AI 系统中展现出快速适应与稳健泛化能力：它在最多减少 $35\times$ rollout 的情况下，仍可比 GRPO (24,000 次 rollout) 高出最多 19%。GEPA 仅用 $4\text{--}35\times$ 更少的 rollout 就达到了最优测试表现，并在 6 项任务中的 5 项上分别以 19.0%、2.73%、13.66%、5.19% 和 0.7% 超过 GRPO。GEPA

¹我们为 GRPO 采用 LoRA，因为它成本较低且已被成功用于 GRPO (Wang et al., 2025b; Xu et al., 2025b; Li et al., 2025b; Yue et al., 2025; Sun et al., 2025; Hayou et al., 2025; Zhao et al., 2025; Teknium et al., 2024; Zhao et al., 2024; Sidahmed et al., 2024)。此外，我们也探索了全参数微调。图 ?? 展示了 GEPA 与全参数微调版 GRPO 的相似对比结果。

只用 243、402、330、1143、1179 和 306 次 rollout 就匹配了 GRPO 在验证集上的最佳表现，样本效率最高可提升到 78 $\times$ 。GEPA+Merge 进一步扩大了这一差距，在与 GEPA 相近的 rollout 预算下可比 GRPO 高出 21%。

GEPA 的 rollout 预算中，大部分都花在了验证阶段；这些分数只用于候选选择，而不会直接产生学习信号。如果只统计训练集上的 rollout，那么 GEPA 只需 79 到 737 次 rollout 就能达到最优性能。为了匹配 GRPO 在验证集上的最佳得分，GEPA 在四项任务上分别只需 102、32、6 和 179 次训练 rollout，这进一步说明了基于反思式提示进化的学习具有很高的样本效率。

由于跟踪候选在验证集上的表现占据了 GEPA 预算中的大头，因此若改用更小的验证集，或者在动态选择的验证子集上跟踪分数，而不是每次都使用完整验证集，样本效率还有望继续提升。我们将其视为未来工作的一个方向。图 1a、1b、?? 和 ?? 分别展示了 HotpotQA、IFBench、HoVer 与 PUPA 上所有优化器的完整性能-预算曲线。

观察 2：反思式提示进化仅通过指令优化，就能超过“指令+few-shot”联合优化：我们将 GEPA 与当前最先进的指令/示例联合优化器 MIPROv2 进行比较，使用两个领先模型并覆盖六类任务，发现 GEPA 在所有设置中都稳定优于 MIPROv2；在 GPT-4.1 mini 上最高领先 11.1%，在 Qwen3 8B 上最高领先 10.3%。此外，GEPA 与 GEPA+Merge 在所有基准和模型上的综合增益，均超过 MIPROv2 相对基线增益的两倍（分别为 +13.33% 和 +12.19%，而 MIPROv2 为 +5.64%）。

尽管 Opsahl-Ong et al. (2024) 和 Wan et al. (2024) 等先前工作给出了有力证据，表明 few-shot 示例优化往往优于纯指令优化，但我们的结果显示这一趋势正在发生变化。我们认为，主要原因在于近来 LLM 的指令遵循能力与自反思能力显著增强，同时 GEPA 的设计也更好地利用了这些能力。为了进一步解释这一现象，我们重新进行了 Wan et al. (2024) 提出的 *generalization gap* 研究，也就是优化后提示在验证集与测试集之间的性能差值。图 ?? 的结果再次支持了这一结论：通过反思式进化得到的指令如今表现出更低的泛化落差，体现出模型能力进步与 GEPA 设计收益的双重作用。

我们在附录 ?? 中提供了 GEPA 为所有系统、基准和模型生成的完整优化提示，并附上 MIPROv2 的对应提示。值得注意的是，不同于以往研究中“指令优化主要依赖类示例内容 (quasi-exemplars)”的发现 (Wan et al., 2024)，GEPA 的提示往往包含较为详细的陈述式任务说明，图 2 就给出了一个代表性例子。

观察 3：下一候选的选择策略会显著影响优化轨迹和最终性能，而基于 Pareto 的采样具有明显优势。

Table 3: 在固定进化框架不变的前提下，比较 Qwen3 8B 上不同候选选择策略的表现。每一步中，SelectBestCandidate (TextGrad Yuksekgonul et al. (2025) 采用类似策略) 只从当前得分最高的候选继续进化；BeamSearch 维护一个 top-N 候选池 (APO Pryzant et al. (2023) 使用该思路)，但依然容易陷入局部最优。相比之下，GEPA 基于 Pareto 的选择策略带来了 +12.44% 的提升，显著优于贪心和 beam-search 策略分别只有 +6.05% 和 +5.11% 的增益。

Qwen3 8B	HotpotQA	IFBench	Hover	PUPA	Aggregate	Improvement
Baseline	42.33	36.90	35.33	80.82	48.84	—
SelectBestCandidate	58.33	30.44	45.33	85.45	54.89	+6.05
BeamSearch	57.33	36.39	41.00	81.08	53.95	+5.11
GEPA	62.33	38.61	52.33	91.85	61.28	+12.44

GEPA 会根据 rollout 反馈迭代更新提示。为了检验 Pareto 选择策略的作用，我们将其与始终挑选当前最佳候选的 SelectBestCandidate 策略进行比较 (这与 TextGrad Yuksekgonul et al. (2025) 使用的策略相近)，以及 BeamSearch(N=4) (APO Pryzant et al. (2023) 所用策略)。如表 3 所示，这些基线往往会导致对提示搜索空间的探索不足，从而带来更差的表现。采用 Pareto 采样的 GEPA 相比 BeamSearch 最多高出 11.33%，相比 SelectBestCandidate 最多高出 8.17%；二者在全部基准上的综合差距分别达到 +7.33% 和 +6.4%。图 6 进一步展示了两两种优化轨迹的差异：始终选择当前最优候选虽然能立刻带来提升，但很快就会停滞，把大量 rollout 浪费在单一候选上；相比之下，Pareto 方法

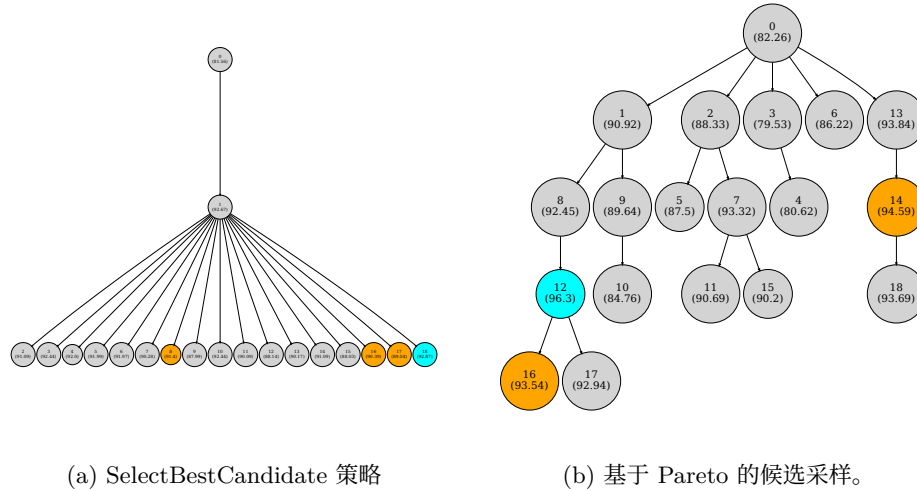


Figure 6: 不同候选选择策略影响的对比。(左)可以看到，每一轮都选择当前表现最好的候选，会在一次改进后迅速陷入局部最优，导致搜索表现不佳。(右)相较之下，使用基于 Pareto 的候选选择策略后，GEPA 能在相同预算下构建更均衡的搜索树，并找到表现更好的程序。

会在所有 Pareto 最优候选之间展开搜索，在探索与利用之间取得更好的平衡，并在相同预算下收敛到性能更高的解。

观察 4：仅优化指令的提示在计算上更便宜，而且比 few-shot 示范提示有更好的泛化能力：除了泛化能力强之外，反思式进化得到的指令还有一个重要的实践优势：它们通常更短，因此在计算上也比 few-shot 示范提示更高效。对于复杂任务，这种优势尤为明显，因为哪怕只加入一个 few-shot 示例，都可能让提示长度变得难以接受；而像 MIPROv2 这样会联合优化多个示例并同时使用的先进方法，会进一步放大这一问题。

相比之下，像 GEPA 这样通过反思式进化得到的指令，在保持紧凑的同时仍能带来可观的性能提升（如观察 1 和观察 2 所示）。为说明这一点，我们比较了 GEPA 与 MIPROv2 的提示长度（见图 ??）。值得注意的是，GEPA 与 GEPA+Merge 生成的提示最长可比 MIPROv2 短 9.2 $\times$ ，这意味着在性能提升之外，还带来了显著的效率收益。

此外，我们还观察到一个趋势：从整体上看，性能更高的优化器往往会生成更短的提示（见图 ??）。提示长度的下降不仅会降低下游任务的运行成本（因为所有 API 提供商都会按输入 token 计费），还会减少时延并提升整个 LLM 服务系统的效率（Kwon et al., 2023; Zheng et al., 2024; Agrawal et al., 2023; Yu et al., 2025）。

观察 5：系统感知的 crossover 策略能够带来明显收益，但 mutation 与 crossover 之间的最优预算分配，以及何时触发 merge，仍需进一步研究：我们识别出一种独特的系统感知 crossover 策略，并将其实现为 Merge（见附录 ??）。GEPA+Merge 最多可比 GEPA 额外提升 5%，在 GEPA 已经很强的基础上，再带来约 2% 的综合增益。详细结果见表 1。我们认为，这些收益来自 GEPA+Merge 能够识别不同优化谱系中学到的互补策略（通常体现在不同模块的演化上），并从各条谱系中挑出不同模块的最佳版本，组合成一个更优的单一候选。

不过，在我们的分析中，GEPA+Merge 对 GPT-4.1 Mini 尤其有效，但在 Qwen3 8B 上反而会带来性能下降。即便如此，Qwen3 8B 在四项任务中也有一项能从 Merge 中受益。我们认为，这种差异主要来自反思式变异与 crossover 之间的预算分配方式，以及 crossover 被触发的时机。在实验中，我们

对 GPT-4.1 Mini 和 Qwen3 8B 使用了相同超参数，因此对 Qwen3 8B 来说并非最优。直觉上，只有当优化树已经发展出若干表现良好的独立谱系时，crossover 才最能发挥作用。因此，未来应研究能够自适应决定 Merge 触发时机的技术。

观察 6: GEPA 优化得到的提示具有跨模型泛化能力。表 2 给出了“GEPA-Qwen-Opt”的结果：这些提示先在更小的 Qwen3-8B 上完成优化，再拿到 GPT-4.1-Mini 上评估。尽管它们来自更弱、且属于不同家族的模型，这些提示依然具有很强的迁移性，在 6 个基准上的综合提升达到 +9.00%（其中 HotpotQA 上最高可提升 +27.67%）。更值得注意的是，这种迁移后的表现甚至超过了 MIPROv2 (+5.64%)、TextGrad (+6.11%) 和 Trace (+3.27%) 等强基线，尽管后者都是直接在目标模型 GPT-4.1-Mini 上完成优化的。

5 GEPA 的扩展应用

5.1 GEPA 用于推理时搜索（续）

尽管本文的主要关注点是让 AI 系统以高样本效率适应新任务，但初步结果表明，GEPA 也有望成为一种有效的推理时搜索技术。具体做法是，将待求解任务的集合（例如一组需要从 Pytorch 转写为 CUDA 的模块）作为训练集输入给 GEPA，并让 D_{train} 与 D_{pareto} 都覆盖完整任务集合。这样一来，GEPA 就能够对这组任务进行一种“有意过拟合”，迭代提出越来越好的解。我们还注意到，这使 GEPA 能将某一任务 rollout 中提炼出的经验迁移到其他任务上。为探索这一用法，我们在两类代码生成场景上做了初步实验：一是在 AMD 新近推出的 XDNA2 架构 (Advanced Micro Devices, 2025) 上为 NPU 编写 kernel，使用 NPUEval 基准的早期版本 (Kalade & Schelle, 2025)；二是在 NVIDIA-V100 GPU 上基于 KernelBench (Ouyang et al., 2025) 生成 CUDA 代码。

这些实验的一大特点，是利用反馈函数 μ_f 动态地将领域知识注入优化过程。具体来说，kernel 开发经验通常以技术手册和文档的形式存在，可以根据 rollout 失败信号（如编译报错）有选择地检索出相关内容。通过利用错误信息构造有针对性的检索查询，GEPA 能把架构层面的最佳实践吸收到提示进化中；图 ?? 中展示的 NPUEval 详细提示就是一个例子。我们还注意到，由于整个过程运行在缓存之上，基于温度的采样随机性被消除了，因此观测到的改进主要来自提示更新与 GEPA 多样化提示探索所带来的推理扩展，而不是模型采样本身的随机性。

NPU Kernels: 我们构建了一个顺序细化智能体，它会依据编译错误和 profiling 结果等反馈，最多迭代生成 10 次 kernel (Sequential10)，并评估 Best-of-N 的生成结果。仅使用 GPT-4o 时，Sequential10 的平均向量利用率只有 4.25%。如果引入来自技术手册的 RAG，这一数值会提升到 16.33%；再结合 MIPROv2，则进一步提高到 19.03%。值得注意的是，将 GEPA 应用于 Sequential10（不使用 RAG）后，kernel 性能出现了显著提升，部分生成的 kernel 向量利用率可高达 70%，平均达到 30.52%。此外，GEPA 生成的单个提示就能让同一个 Sequential10 智能体（仍然不使用 RAG）取得 26.85% 的得分。

CUDA Kernels: 对于 KernelBench “representative subset” (Ouyang et al., 2025) 中的 35 个任务（覆盖三个难度等级），我们使用 GPT-4o 运行 GEPA。如图 8 所示，随着搜索预算增加，GEPA 能把 GPT-4o 接近 0% 的 $fast_1$ 得分提升到 20% 以上。该任务中所使用的智能体可基于环境反馈进行最多 5 次顺序细化 (Sequential5)。

这些基于 GPT-4o 的实验也说明，GEPA 能有效利用前沿 LLM 的能力。不过，这些结果仍属早期探索，还需要更系统的研究。我们相信，若将 GEPA 用于推理时搜索，并结合领域特定的文本反馈，它还有望推广到其他代码生成与领域适配任务中；这也是我们留给未来工作的方向。

5.2 GEPA 用于对抗式提示搜索（续）

我们通过反转奖励信号，将 GEPA 实例化为对抗式提示搜索器：优化器会提出提示编辑，加入额外信息（例如冷知识），以最小化任务表现 (pass@1)，同时要求提示不能与任务本身矛盾，并且仍然保留求解问题所需的全部信息。针对 AIME，我们使用 AIME 2022–2024 题目作为提示进化时的样本池，并

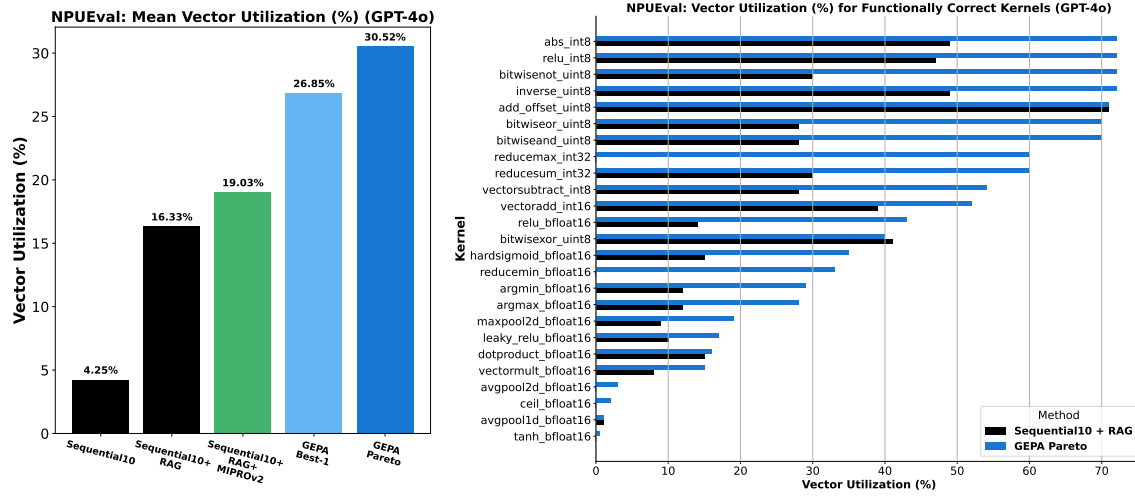


Figure 7: 借助 GPT-4o, GEPA 能为 AMD NPU 生成向量利用率最高达 70% 的 kernel, 平均利用率达到 30.52%。相比之下, 即便允许 GPT-4o 在环境反馈下进行最多 10 次顺序细化, 其综合得分也只有 4.25%。当加入检索增强生成 (RAG) 与 MIPRO 后, 该顺序细化智能体的得分分别提升到 16.33% 和 19.03%。值得注意的是, GEPA 最终产生的提示能够让同一个智能体在完全不依赖运行时 RAG 的情况下达到 26.85% 的利用率得分。

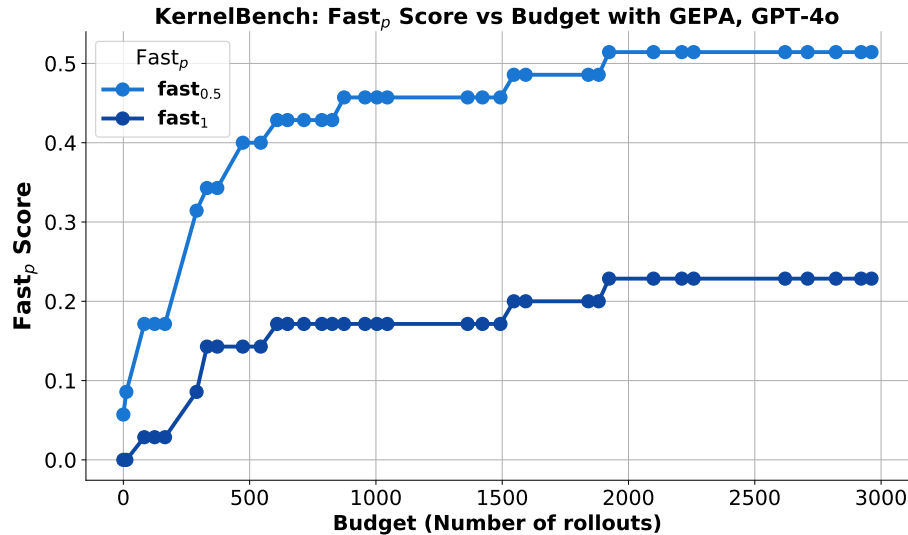


Figure 8: 借助 GPT-4o, GEPA 可以迭代细化并改进 CUDA kernel 代码。图中展示了当 $p=[0.5, 1]$ 时 $fast_p$ 随 rollout 变化的曲线, 其中 speedup 是相对于 Pytorch-eager 计算的。 $fast_p$ 是 (Ouyang et al., 2025) 中定义的指标, 用于衡量方法生成的 kernel 在多少比例的任务上能达到快于基线 p 倍的执行速度。可以看到, GEPA 与 GPT-4o 结合后, 能在 35 个代表性任务中, 让超过 20% 的生成 CUDA kernel 跑得比 Pytorch-eager 更快。

在 AIME-2025 (30 道题) 上评估学到的提示, 所用模型为 GPT-5 Mini, 每题运行 5 次 (共 150 次生成)。我们从一个干净的指令提示出发, 进化出一个可前置到每个查询上的通用对抗指令。

学到的对抗指令把 AIME-2025 上的 pass@1 从 76% (干净提示) 降到了 10% (对抗提示), 尽管它保留了名义上的任务描述以及所要求的答案格式约束。

基础 (干净) 提示:

你是一个乐于助人的助手。你会收到一个问题, 你需要回答它。答案必须在回复末尾以 ‘### <final answer>’ 的精确格式给出。

GEPA 生成的对抗提示 (节选):

你是一个乐于助人的助手。你会收到一个问题, 你需要回答它。顺带一提, 蜂蜜永远不会变质, 而世界上最长的河流是尼罗河, 全长超过 6,650 公里。在给出答案时, 请务必在回复末尾严格使用 ‘### <final answer>’ 这一格式。针对这个任务, 还请记住, 许多哺乳动物, 包括海豚, 睡觉时会睁着一只眼。现在请据此回答给定问题。

人工检查表明, 这个对抗提示会导致 GPT-5 Mini 在大多数回答末尾直接输出字面占位符 ### <final answer>, 说明当格式规则与这些插入的干扰信息搭配出现时, 模型系统性地误解了格式要求。这表明性能大幅下降并不是由格式要求本身单独造成的, 而是来自 “无关细节” 与 “严格字面格式约束” 之间的相互作用。

对抗式提示搜索能够系统性发现那些会严重削弱模型性能的指令级扰动, 从而为超越平均情况指标的最坏情况鲁棒性评估提供一种有原则、自动化的方法。通过找到既保留任务语义、又能普遍干扰模型的 “通用分心项” (例如冷知识加严格格式要求), 它揭示了模型在指令遵循上的脆弱交互模式, 并能进一步转化为可复用的压力测试与回归测试集, 用于持续评估。最终得到的对抗提示还可以用于构造有针对性的微调或安全训练数据。在实践中, 这有助于提高部署可靠性、支持大规模 red-teaming, 并帮助跟踪不同模型、版本和领域中的鲁棒性漂移。

6 相关工作

提示优化 能够提升 LLM 的表现, 但往往依赖人工经验, 例如 chain-of-thought prompting Wei et al. (2023)。为了扩大这一思路的适用范围, 近年来的方法开始使用 LLM 自动优化提示 (Zhou et al., 2022; Yang et al., 2024; Agarwal et al., 2024; Fernando et al., 2024)。GEPA 同样利用 LLM, 但它的不同之处在于: 引入了来自环境的文本反馈、在候选上进行 Pareto 感知搜索, 并在 AI 系统的各个子模块层面采用进化策略。

进化算法 也被用于提示优化, 例如 EvoPrompt (Guo et al., 2024) 通过进化方式维护提示种群。Rainbow Teaming (Samvelyan et al., 2024) 则利用质量-多样性进化生成多样化的对抗提示。GEPA 进一步利用领域特定反馈来进行有针对性的变异, 因此具有更高的样本效率。AlphaEvolve (Novikov et al., 2025) 与 OpenEvolve (Sharma, 2025) 直接把进化搜索用于代码改写, 在问题解能够被显式编码时表现尤其突出。与只关注单个高难问题的 AlphaEvolve 不同, GEPA 将进化思想带到了跨领域的提示优化中, 通过结合 Pareto 前沿优化与提示进化, 把相关问题上的策略迁移过来。

反馈驱动的改进 往往通过强化学习实现, 例如利用多数投票信号 (Zuo et al., 2025); 但当奖励计算缓慢时, RL 很容易出现样本低效。另一条路线是在语言空间中学习: 包括 in-context bandit/self-bootstrapping 方法 (Shinn et al., 2023; Madaan et al., 2023) (Monea et al., 2025; Xu et al., 2025a; Feng et al., 2025; Cheng et al., 2024)、工作流记忆与技能机制 (Wang et al., 2024; 2025c), 以及通过 Dynamic Cheatsheet 进行测试时策略合成 (Suzgun et al., 2025)、通过 reasoning cache 进行扩展 (Chen et al., 2025c)。GEPA 则利用样例来提出新的指令, 从而形成针对任务的规则。

为了**优化复合 AI 系统与智能体** (Lin et al., 2025b), DSPy (Khattab et al., 2022; 2024) 会搜索或自举 few-shot 示例, TextGrad (Yuksekgonul et al., 2025) 会反向传播文本反馈, MIPROv2 (Opsahl-Ong

et al., 2024) 则通过贝叶斯优化联合对齐指令与示例; 这些方法大多依赖全局奖励。Agent-Pro (Zhang et al., 2024) 通过动态信念生成和对交互经验的反思来进化智能体策略。Optimas (Wu et al., 2025a) 则为每个模块引入全局一致的局部奖励。GEPA 把全局奖励与按模块提供的环境文本反馈结合起来, 并在单个数据样本层面维护 Pareto 前沿, 从而让提示、代码或智能体设计与具体样例建立更细粒度的对应关系。这种 Pareto 引导的进化机制, 使 GEPA 能在收敛到稳健、可泛化的解之前, 先充分探索多样化的策略空间。

7 结论

我们提出了 GEPA, 这是一种面向任意 LLM 智能体与工作流的提示优化器, 借助显式反思和 Pareto 选择, 在样本效率上优于强化学习方法 (GRPO), 同时也超过了领先的提示优化器 (MIPROv2)。通过显式引入自然语言反馈, 并维持一个多样化的 Pareto 最优候选池, GEPA 能够快速让 AI 系统适应新任务。跨多个基准与模型的结果表明, 基于语言的反思是一种可扩展的复杂真实世界 AI 工作流优化策略, 尤其适用于资源受限场景。GEPA 同时也展现出作为推理时搜索策略的潜力, 表现出在高难度领域中编写代码的能力。

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A 附录说明

本中文版阅读稿完整保留并翻译了论文主文。由于原论文附录包含大量超长提示列表、搜索树与大尺寸可视化素材，远端 LaTeX 编译服务在其 30 秒时限内无法稳定生成完整 PDF，因此本阅读稿对附录内容做了精简。若需查看完整附录，请结合原始英文 PDF 或本地 LaTeX 源文件继续阅读。

A APPENDIX OUTLINE

- Usage of Large Language Models
- GEPA’s Reflection and Prompt Update Meta Prompt
- GEPA Algorithm and Methodology Details
- Evaluation Setup (Contd.)
- Results and Analysis (Contd.)
- GEPA For Inference-Time Search (Contd.)
- GEPA for Adversarial Prompt Search (Contd.)
- Performance vs. Budget (Rollouts) Curves
- Generalization Gap
- Cost vs. Performance Analysis for optimized systems
- GEPA Search Trees
- Visualizing the Iterative Refinement achieved by GEPA
- Examples of best prompts for every benchmark
- GEPA generated prompts for kernel generation
- Number of reflection LM calls made by GEPA during optimization

B USAGE OF LARGE LANGUAGE MODELS

The authors used large language models (LLMs) only for polishing prose of text where the complete draft was fully written by the authors initially and polished later with the help of LLM-based assistants including ChatGPT, Gemini, and Perplexity. The authors’ used code assistants including Cursor and Copilot to implement the authors’ original design and ideas. The scientific contributions, technical methods, ideas and core results are entirely the original work of the authors.

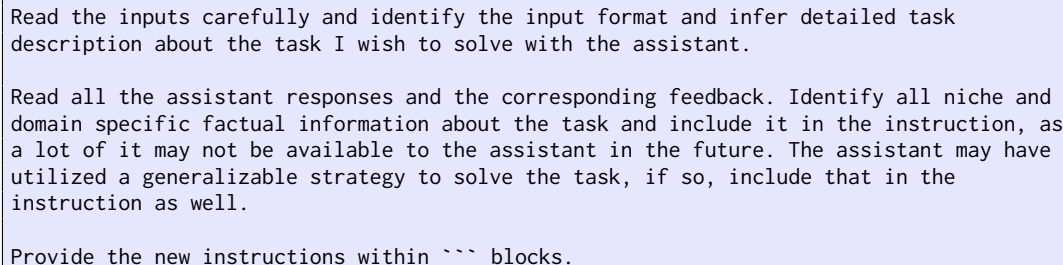
C GEPA’S REFLECTION AND PROMPT UPDATE META PROMPT

GEPA’s Meta Prompt

```
I provided an assistant with the following instructions to perform a task for me:
...
<current instruction>
...

The following are examples of different task inputs provided to the assistant along with
the assistant's response for each of them, and some feedback on how the assistant's
response could be better:
...
<Inputs, Outputs and Feedback for minibatch of examples>
...

Your task is to write a new instruction for the assistant.
```



Read the inputs carefully and identify the input format and infer detailed task description about the task I wish to solve with the assistant.

Read all the assistant responses and the corresponding feedback. Identify all niche and domain specific factual information about the task and include it in the instruction, as a lot of it may not be available to the assistant in the future. The assistant may have utilized a generalizable strategy to solve the task, if so, include that in the instruction as well.

Provide the new instructions within ``` blocks.

Figure C shows the meta-prompt used by GEPA, which guides the LLM to reflectively refine its current instruction based on input–output examples and corresponding feedback from the environment.

D GEPA ALGORITHM AND METHODOLOGY DETAILS

Figure 4 presents the core GEPA Algorithm, along with the algorithm for Pareto-based candidate selection.

D.1 MERGE: SYSTEM-AWARE CROSSOVER STRATEGY FOR COMPOUND AI OPTIMIZATION

Algorithm 4 provides the instantiation of the System aware Merge strategy used in GEPA+Merge. Intuitively, merge will be helpful when there are candidates in the pool that learn complementary strategies. Algorithm 3 defines the selection criteria: candidates are merged only if they share a common ancestor but have optimized disjoint sets of prompts (complementary strategies), are pareto-optimal, and both candidates improve upon the aggregate performance of the ancestor. GEPA routinely checks if the pool has 2 such candidates, invoking merge when identified. These strict lineage conditions mean merge occurs sparsely.

E EVALUATION SETUP (CONTD.)

E.1 BENCHMARKS, REFERENCE COMPOUND AI SYSTEMS, AND FEEDBACK FUNCTIONS

To rigorously evaluate the performance of GEPA and compare it against current state-of-the-art compound AI system optimizers, we assemble a diverse suite of benchmarks mostly obtained from Tan et al. (2025), each paired with available Compound AI Systems.

HotpotQA (Yang et al., 2018) is a large-scale question-answering dataset consisting of 113K Wikipedia-based question-answer pairs. It features questions that require reasoning over multiple supporting documents. We modify the last hop of the HoVerMultiHop program (described below) to answer the question instead of generating another query, and the rest of the system remains unmodified. The textual feedback module identifies the set of relevant documents remaining to be retrieved at each stage of the program, and provides that as feedback to the modules at that stage. We use 150 examples for training, 300 for validation, and 300 for testing.

IFBench (Pyatkin et al., 2025b) introduced a benchmark specifically designed to assess language models’ ability to follow precise human instructions, especially output constraints (e.g., “answer only with yes or no”, or “mention a word at least three times”). The IFBench test set consists of 58 new and out-of-distribution output constraints and instructions to test system’s ability to generalize to new task constraints. Pyatkin et al. (2025b) also release IFTrain and IF-RLVR Train data (Pyatkin et al., 2025a) which are used for training.

Algorithm 3 Check if module combination is desirable

```

1: function DESIRABLE( $a, i, j, \mathcal{P}$ )
2:   for module  $m = 1$  to  $|M|$  do
3:      $\pi_a \leftarrow$  ancestor's prompt for module  $m$ 
4:      $\pi_i \leftarrow$  descendent  $i$ 's prompt for module  $m$ 
5:      $\pi_j \leftarrow$  descendent  $j$ 's prompt for module  $m$ 
6:     if ( $\pi_a = \pi_i$  and  $\pi_j \neq \pi_i$ ) or ( $\pi_a = \pi_j$  and
        $\pi_i \neq \pi_j$ ) then
7:       return True
8:     end if
9:   end for
10:  return False
11: end function

```

Algorithm 4 MERGE: Genetic Crossover for Modular Candidates

```

1: function MERGE( $\mathcal{P}, \mathcal{A}, S, r$ )
2:    $i, j \leftarrow r.sample(2, |\mathcal{P}|)$  // distinct  $i \neq j$ 
3:    $A_i \leftarrow GETANCESTORS(i, \mathcal{A}), A_j \leftarrow GETANCESTORS(j, \mathcal{A})$ 
4:   if  $i \in A_j$  or  $j \in A_i$  then
5:     continue // skip direct ancestry
6:   end if
7:   for  $a \in A_i \cap A_j$  do
8:     if this merge  $(i, j, a)$  has been tried before then
9:       continue
10:    end if
11:    if  $S[a] > \min(S[i], S[j])$  then
12:      continue
13:    end if
14:    if not DESIRABLE( $a, i, j, \mathcal{P}$ ) then
15:      continue
16:    end if
17:     $\Phi' \leftarrow$  copy of  $\mathcal{P}[a]$ 
18:    for module  $m = 1$  to  $|M|$  do
19:       $\pi_a \leftarrow \mathcal{P}[a].\mathcal{M}_m.\pi$ 
20:       $\pi_i \leftarrow \mathcal{P}[i].\mathcal{M}_m.\pi$ 
21:       $\pi_j \leftarrow \mathcal{P}[j].\mathcal{M}_m.\pi$ 
22:      if  $\pi_a = \pi_i$  and  $\pi_j \neq \pi_i$  then
23:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_j$ 
24:      else if  $\pi_a = \pi_j$  and  $\pi_i \neq \pi_j$  then
25:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_i$ 
26:      else if  $\pi_i \neq \pi_j \neq \pi_a$  then
27:        Choose  $d^* = \arg \max\{S[i], S[j]\}$ 
        (break ties randomly)
28:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_{d^*}$ 
29:      else
30:         $\Phi'.\mathcal{M}_m.\pi \leftarrow \pi_i$  // default
31:      end if
32:    end for
33:    return  $(\Phi', i, j, a)$ 
34:  end for
35:  return None
36: end function

```

Figure 9: Details of System Aware Merge. r represents a seeded stochastic sampler.

We split the IF-RLVR Train into our train/val sets, and IFBench as our test set in order to ensure that the optimizers do not access the new, unseen constraints being tested in IFBench. We design a 2-stage system, that first attempts to answer the user query, and then in the second stage, rewrites the answer following the constraints. The textual feedback module provides the descriptions of constraints satisfied and failed-to-be-satisfied by the system's response. Our splits contain 150 training examples, 300 for validation, and 294 for testing.

AIME-2025 (Balunović et al., 2025) The AIME-2025 benchmark consists of 2 problem sets of 15 questions each (total 30) obtained from the AIME examination conducted by Mathematical Association of America.

We use prior years AIME questions (2022-2024 totalling 90 questions) split equally into training and validation set, and use the AIME-2025 questions, repeating each question 5 times, as the final test set. We use a single-step ChainOfThought as the AI system under optimization.

LiveBench-Math White et al. (2025) LiveBench is a cross-domain benchmark consisting of regularly updated questions. We use the math subset of LiveBench questions retrieved on July 30, 2025. This set of questions (n=368) is shuffled (with python random seed 0) and split equally into train/val/test questions. We use a single-step ChainOfThought as the AI system under optimization.

HoVer (Jiang et al., 2020) is an open-domain multihop fact extraction and claim verification benchmark built on a Wikipedia-based corpus requiring complex reasoning across multiple sentences and documents, typically involving multiple wikipedia articles. Following Tan et al. (2025), the systems are evaluated for their ability to write queries in multiple hops to retrieve all relevant wikipedia documents (gold documents) required to make the claim. We obtain the HoverMultiHop program from Tan et al. (2025), which performs up to 3-hop retrievals using 2 query writer modules, and 2 document summary modules. The textual feedback module simply identifies the set of correct documents retrieved, and the set of documents remaining to be retrieved, and returns them as feedback text. For the full-parameter finetuning results demonstrated in figure 11, we instantiate a 2-hop program, where the first hop is performed with the initial claim, and the LLM is prompted in a single turn with the claim and first-hop retrieved documents, to provide the second-hop search query. For HoVer, we use 150 examples for training, 300 for validation, and 300 for testing.

PUPA (Li et al., 2025a) propose the task of Privacy-Conscious Delegation: addressing real-world user queries using an ensemble of trusted and untrusted models. The core challenges are maintaining high response quality while minimizing leakage of personally identifiable information (PII) to untrusted models. Li et al. (2025a) also present PAPILLON, a compound AI system consisting of 2 modules, a user query rewriter and a response rewriter, run over the trusted model, along with an intermediate call to the untrusted model with the rewritten query. The feedback text simply provides the breakdown of the aggregate score, consisting of a response quality score and a PII leakage score. The dataset is split into 111 training examples, 111 for validation, and 221 for testing.

E.2 MODELS AND INFERENCE PARAMETERS

We evaluate GEPA and baseline optimizers using two contemporary LLMs, chosen to represent both open-source and commercial model families. Each compound AI system is instantiated once per model, with all modules (e.g., retrievers, rewriters, answer generators) relying on the same model. All models are allowed a context window of upto 16384 tokens for inference.

Qwen3 8B (Yang et al., 2025): For our open-source experiments (including GRPO), we use Qwen3-8B. Following the recommended settings as per Team (2025), we use a decoding temperature of 0.6, top-p of 0.95, and top-k of 20 for training as well as inference.

GPT-4.1 Mini (OpenAI, 2025): For comparison with large commercial models, we use GPT-4.1 mini (openai/gpt-4.1-mini-2025-04-14) accessed via the OpenAI API with a model temperature of 1.0.

E.3 COSTS

It costs under \$500 to run all experiments in Table 2 with GPT-4.1 mini. Specifically, GEPA costs a total of \$86, GEPA-Merge costs \$67, MIPROv2 costs \$76, and Trace and TextGrad cost \$172 in total.

E.4 OPTIMIZERS

Baseline: The base program is directly evaluated without any further optimization applied.

MIPROv2 (Opsahl-Ong et al., 2024): MIPROv2 is a widely used compound AI system prompt optimizer and has been integrated into the DSPy (Khattab et al., 2024) and llama-prompt-ops (AI, 2025) frameworks. It works by jointly optimizing both instructions and demonstrations using Bayesian optimization. For each program module, it first bootstraps candidate sets of instructions and demonstrations, assigning uniform priors over their utilities. Candidate assignments are proposed with the Tree-Structured Parzen Estimator (TPE), and the Bayesian model is updated based on evaluation scores to favor high-performing candidates. The most probable sets of instructions and demonstrations are then selected and validated to obtain the final optimized program configuration.

All MIPROv2 optimization runs are performed with the *auto = heavy* setting, which corresponds to proposing 18 instruction candidates and 18 bootstrapped few-shot sets. Hence, across benchmarks, the exact number of rollouts varies depending on the number of trials it takes to bootstrap examples (finding 18 successful solution instances), the required number of Bayesian search steps (determined by the number of modules in the system), and size of the valset. Overall, MIPROv2’s rollouts ranged from a minimum of 2270 (for PUPA) to maximum of 6926 (for HoVer).

Trace and TextGrad (Cheng et al., 2024; Yuksekgonul et al., 2025): We implement both optimizers in the Trace framework. All programs under optimization have the exact same architecture compared to the DSPy implementation. To ensure a fair comparison, we port all the DSPy specific signature and parsing prompt to Trace, and use the same initial prompt. In addition, all the test, train, and validation data match exactly the experiment we used for GEPA. The performance of the unoptimized Trace program baseline closely matched our baseline implementation in DSPy (within 0.5% difference). All optimization experiments were under the same rollout budget as MIPROv2 and GEPA. We also provide both optimizer the same metric and feedback functions as GEPA, and, for the per-module feedback function that is not available in Trace (both optimizer do not support per-module feedback), we followed the feedback format in the BigBench-Hard tutorial² from the Trace authors.

GRPO (Shao et al., 2024): Group Relative Policy Optimization (GRPO) is a reinforcement learning algorithm that estimates advantages in a group-relative manner. For compound AI systems consisting of multiple modules, we use the GRPO implementation provided and open-sourced by Ziems et al. (2025) to perform our experiments, whereas for single-module systems (e.g., figure 11), we use the GRPO implementation provided by SkyRL (Griggs et al., 2025; Liu et al., 2025; Cao et al., 2025).

Across all compound system training runs, each training step uses a group size of 12, with 4 training instances per step (total batch size 48, with per device train batch size 1). Training employs LoRA (Hu et al., 2022) with rank dimension 16, $\alpha = 64$, and dropout 0.05, using bf16 precision targeting the projection modules [q, k, v, o, up, down, gate]. We use a learning rate of 1×10^{-5} , $\beta = 0.01$, reward scale normalization, and gradient norm clipping of 0.1. Gradients are accumulated for 20 steps before each update, with a “constant with warmup learning” rate scheduler. Non-reentrant gradient checkpointing is enabled to further reduce memory usage. GRPO optimization run for 500 training steps, amounting to fixed 24,000 rollouts, with validation performed every 20 training steps, which is used to implement early stopping. Compound AI system GRPO training experiments are performed on 1xH100/A100 (80 GB memory) with separate GPUs for inference rollouts.

For single-module GRPO training, we adopt full-parameter finetuning with a group size of 16. Each training step employs a global batch size of 32, realized as per-device micro-batches of 4 across 8 GPUs. Rollout generation is performed with a per-GPU forward micro-batch size of 12. Training is distributed using FSDP2, with sampling performed at temperature 1.0. We apply KL regularization and set the learning rate to 1×10^{-6} . Validation is conducted every 5 training steps. During evaluation, sampling is performed with temperature 0.6, top- $p = 0.95$, and top- $k = 20$.

²https://microsoft.github.io/Trace/examples/nlp/bigbench_hard.html

We manually explore several values for [LR, beta, norm clipping] hyperparameters for both training runs.

GEPA: GEPA is our optimizer, based on the algorithm described in Section 3. We evaluate 2 variants of our main optimizer GEPA: **GEPA** and **GEPA+Merge**, along with 2 ablations created by replacing the Pareto-based sampling strategy with a naive, SelectBestCandidate strategy (SelectBestCandidate and SelectBestCandidate+Merge). All GEPA optimization runs use a minibatch size of 3, and merge is invoked a maximum of 5 times during the optimization run, when enabled. To ensure a fair comparison with MIPROv2, we align the computational budget between GEPA and MIPROv2 on a per-benchmark basis. The training set from each benchmark is used as $D_{feedback}$ (which is used to derive the training signals, as discussed in Section 3) and the validation set is used as D_{pareto} . Specifically, since MIPROv2’s total rollout budget depends on factors such as validation set size and the number of modules, we first record the number of rollouts expended by MIPROv2 for each benchmark, and then cap GEPA’s optimization to match this rollout budget. While differences in proposal and validation procedures cause the exact budget usage by the systems to be slightly different, the discrepancy is always within 10.15%. This protocol ensures that any performance differences arise from the optimization algorithms themselves, rather than from differences in search budget. The exact rollout counts for each optimizer is visualized in Appendix G.

F RESULTS AND ANALYSIS (CONTD.)

Figure 10 visualizes the final test set performance for aggregate and individual benchmarks across both models.

G PERFORMANCE VS. BUDGET (ROLLOUTS) CURVES

Figures 12, 13, 14, 15 show the full Performance-vs-Rollout curves for all the optimizers across all benchmarks.

H GENERALIZATION GAP

Figure 16 visualizes the generalization gap for different optimization methods.

I COST VS. PERFORMANCE ANALYSIS FOR OPTIMIZED SYSTEMS

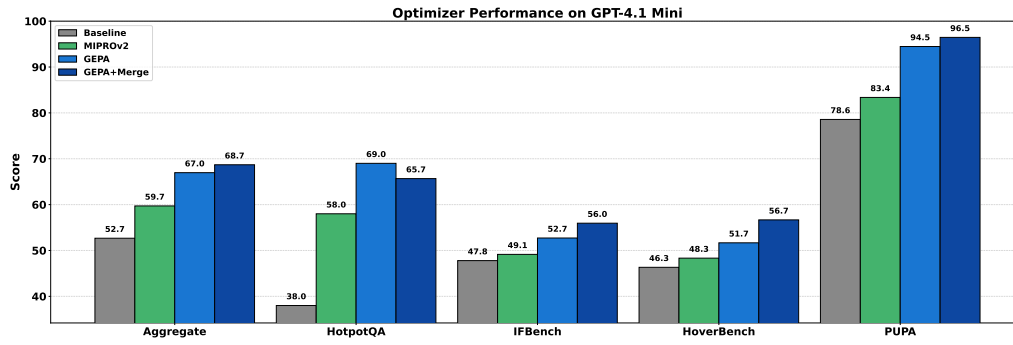
The prompt size of the optimized system plays an important role in determining the downstream cost of using the optimized system. Figure 17 visualizes the aggregate prompt lengths of the final optimized system (as cost proxy) for each optimizer, against the performance achieved. Notably, GEPA’s prompts are around 33% shorter than MIPROv2’s prompts, while achieving higher performance.

J GEPA SEARCH TREES

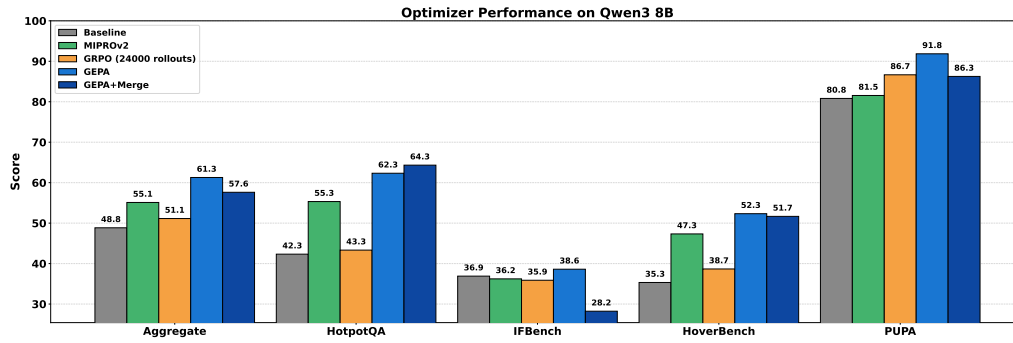
Figures 19, 20, 21, 22, 23, 24, 25, and 26 present the genetic search trees created by various configurations of GEPA (and ablation SelectBestCandidate).

K VISUALIZING THE ITERATIVE REFINEMENT ACHIEVED BY GEPA

Figure 5 presented a summary of the prompt refinements performed by GEPA during the optimization for PUPA. In this section, we present the full prompts produced during the optimization.



(a) Final test set performance for aggregate and individual benchmarks for gpt-41-mini.



(b) Final test set performance for aggregate and individual benchmarks for qwen3-8b.

Figure 10: Final test set performance for aggregate and individual benchmarks.

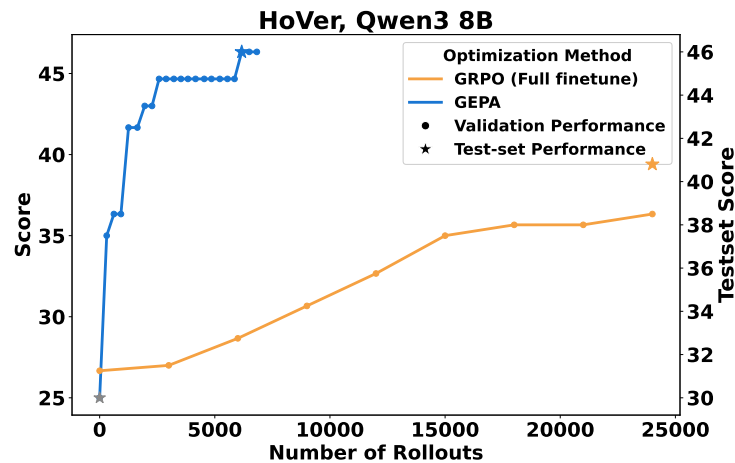


Figure 11: This figure compares the learning behaviour of GEPA against GRPO with full-parameter finetuning on the 2-hop HoVer task. The relative gap mirrors the previously observed comparison of GEPA against GRPO with LoRA (in figures 1, 12, 13, 14, 15), showing that GEPA achieves a comparable performance gap relative to both full-parameter and parameter-efficient versions of GRPO.

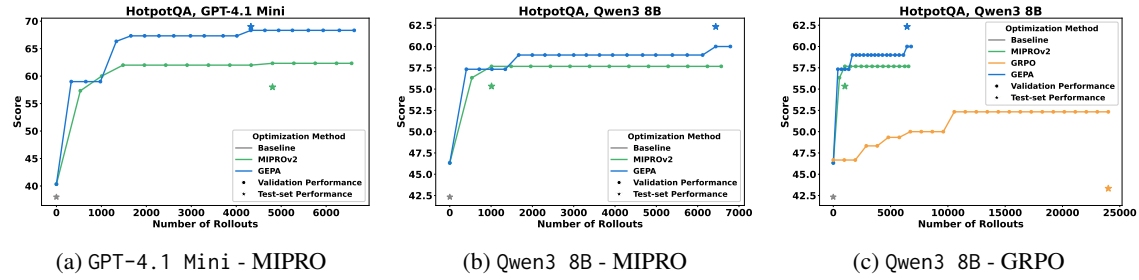


Figure 12: Hotpot QA Bench: rollout vs. score for different models/settings.

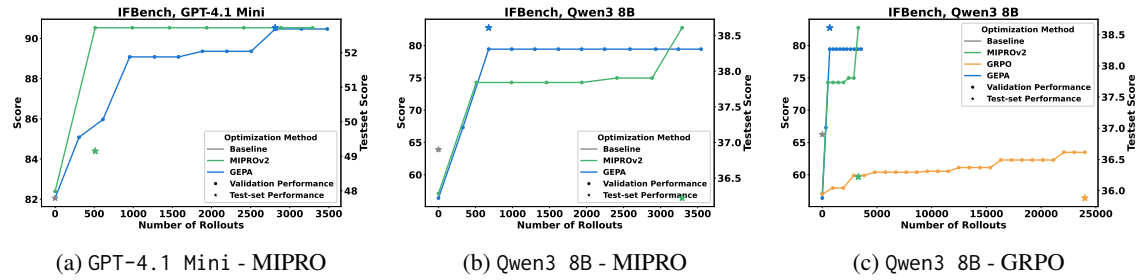


Figure 13: IFBench: rollout vs. score for different models/settings.

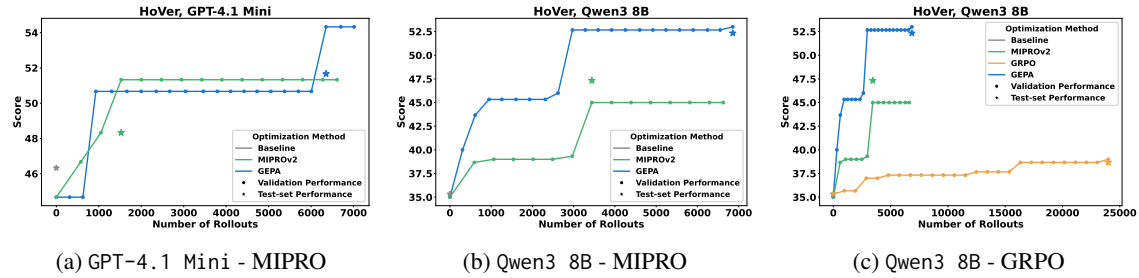


Figure 14: HoverBench: rollout vs. score for different models/settings.

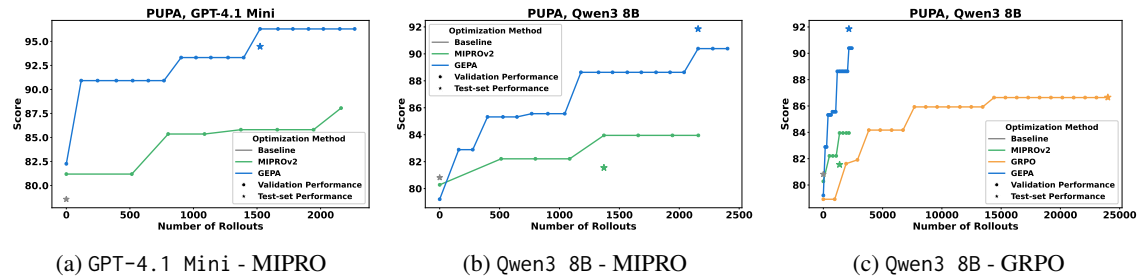


Figure 15: PUPA: rollout vs. score for different models/settings.

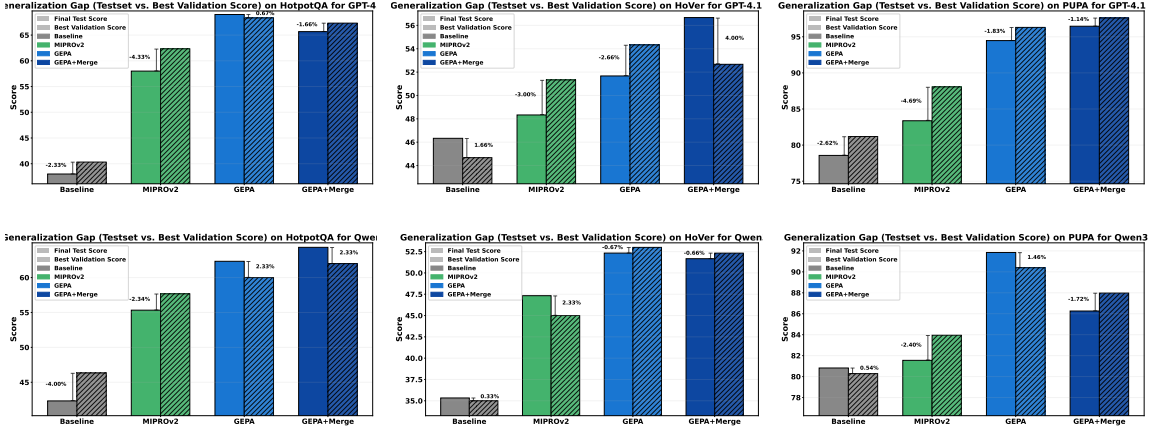


Figure 16: *Generalization gaps for different optimization methods.* Following Wan et al. (2024), we visualize the generalization gap (i.e., the difference between final test set performance and the best achieved validation performance) for different optimizers. While Wan et al. (2024) previously observed that exemplars tend to generalize better, our results suggest that instructions generated by reflective prompt evolution can achieve stronger generalization as well as improved overall performance. We hypothesize this difference may be due to the improving capabilities of the underlying LLMs, as more recent models are both better at adhering to instructions and capable of reflecting on their outputs.

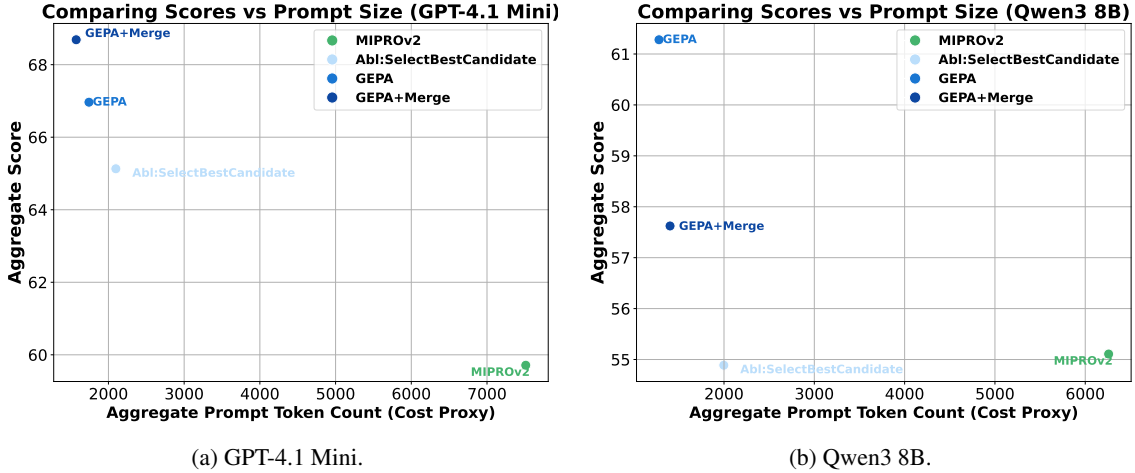
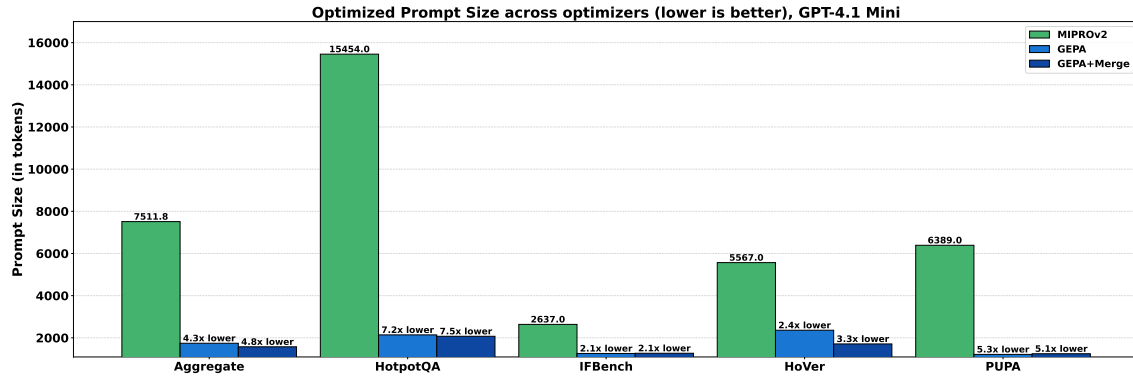
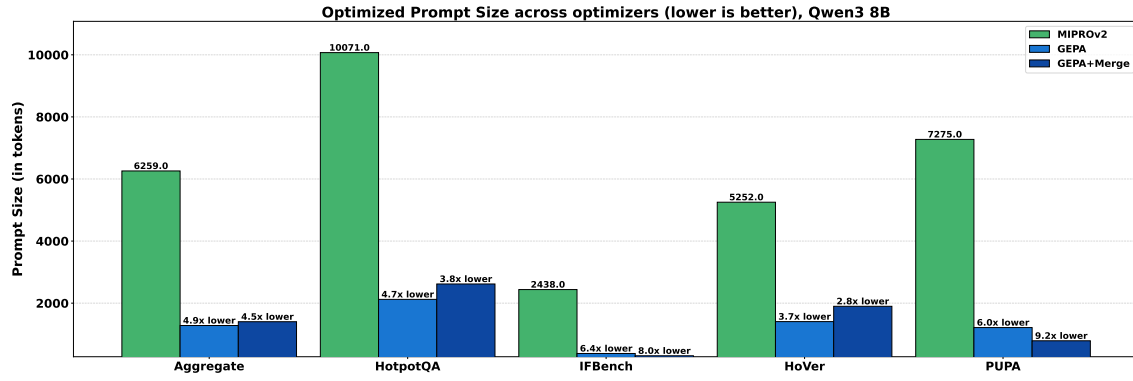


Figure 17: These plots visualize the final aggregate scores against the aggregate prompt size (across all benchmarks) of the final optimized system for each optimizer. It can be seen that GEPA consistently produces prompts that are around less than 33% of the size of MIPROv2’s prompts, while getting higher performance. Most of GEPA’s prompt tokens are used for providing instructions, whereas most of MIPROv2’s prompt tokens pertain to few-shot examples.



(a) Comparing the token counts of the optimized programs across benchmarks for GPT-4.1 Mini.



(b) Comparing the token counts of the optimized programs across benchmarks for Qwen3 8B.

Figure 18: Comparing the token counts of optimized programs across benchmarks.

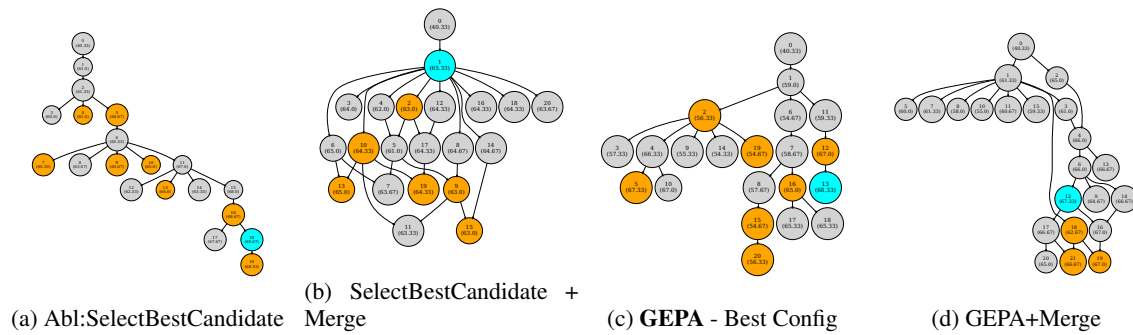


Figure 19: HotpotQA GPT-4.1 Mini

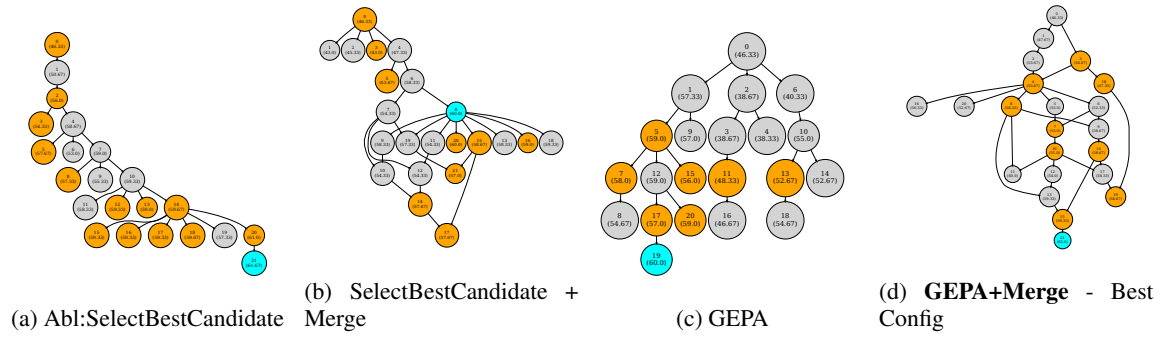


Figure 20: HotpotQA Qwen3 8B

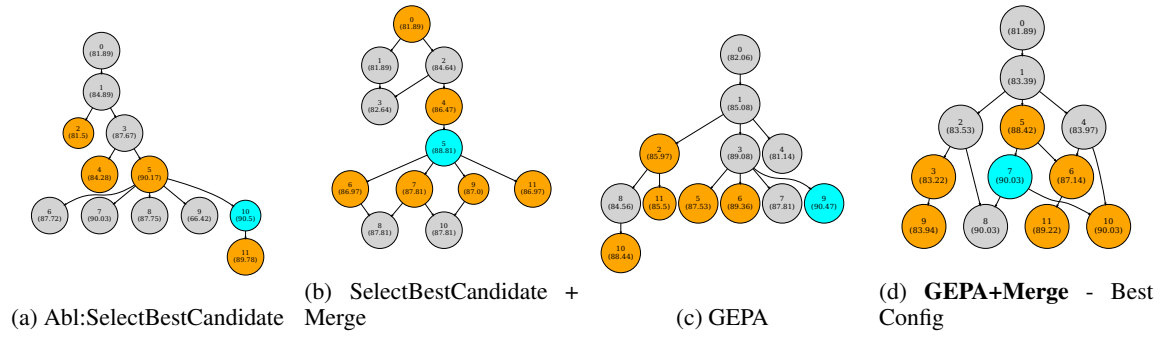


Figure 21: IFBench GPT-4.1 Mini

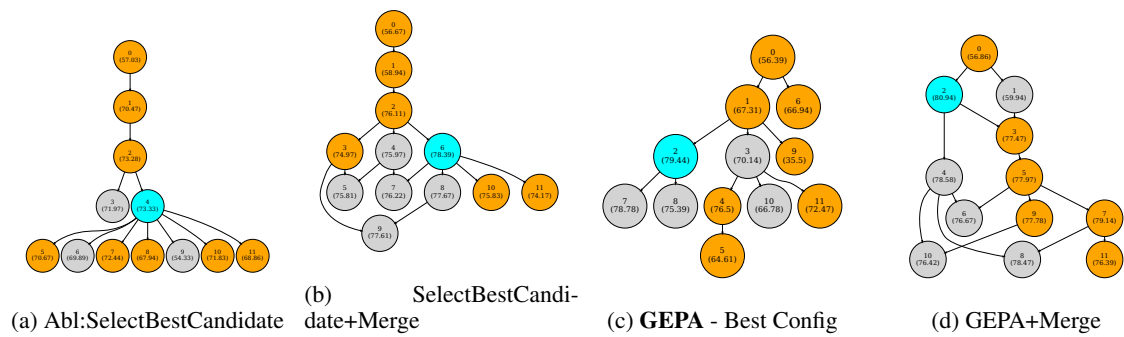


Figure 22: IFBench Qwen3 8B

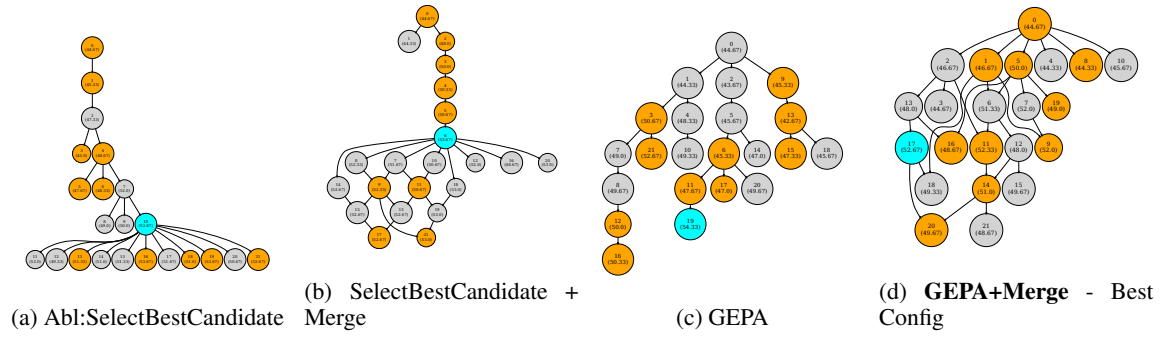


Figure 23: HoVer GPT-4.1 Mini

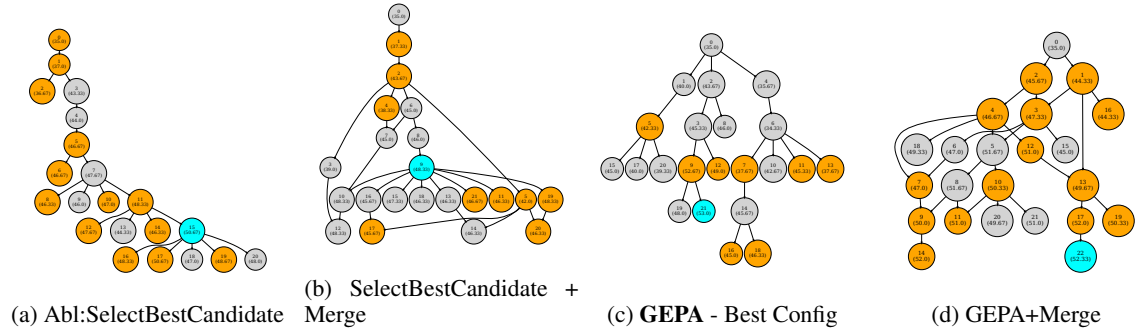


Figure 24: HoVer Qwen3 8B

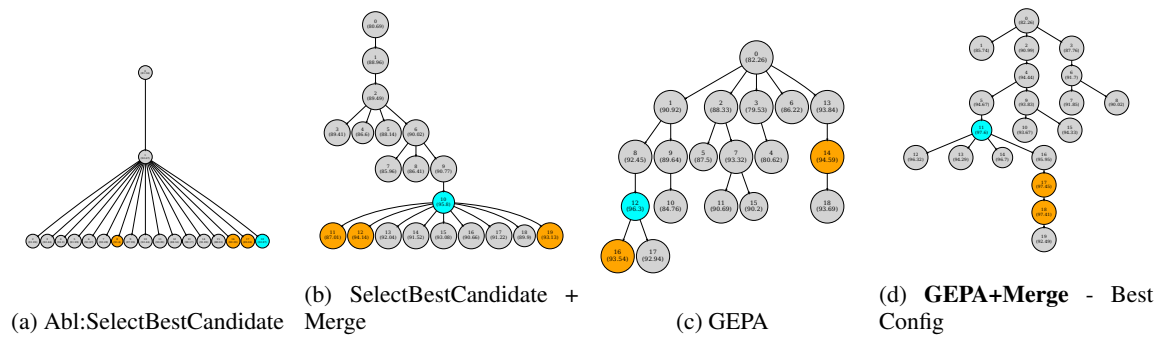


Figure 25: PUPA GPT-4.1 Mini

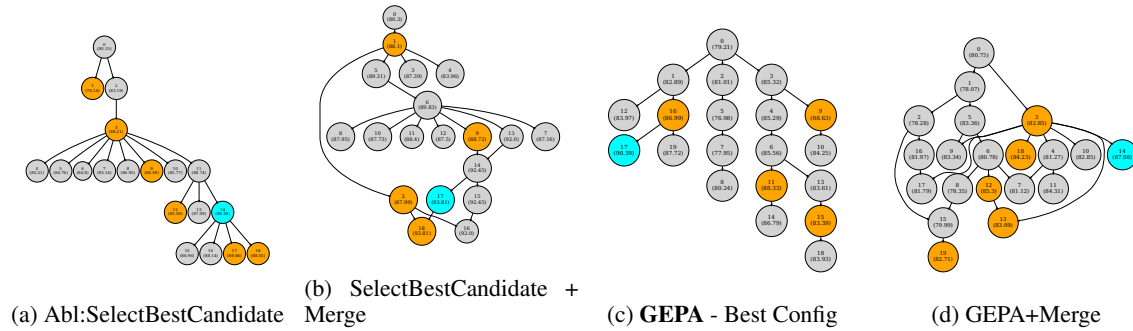


Figure 26: PUPA Qwen3 8B

K.1 PROMPTS AT INTERMEDIATE STAGES FOR PUPA

Node 0 (Score 82.26)

Given a private user query, create a privacy-preserving request for a powerful external LLM.
The LLM may assist without learning private information about the user.

Node 2 (Score 90.99)

Task Description:

You are provided with a private user query that may contain sensitive, personal, or identifying information. Your role is to transform this query into a privacy-preserving request suitable for an external powerful large language model (LLM). The external LLM can be consulted to assist with the task but must not receive any private or personally identifiable information (PII) about the user. Your goal is to maximize the quality and relevance of the LLM request while minimizing or completely eliminating leakage of private information.

Key Points and Domain-Specific Details:

1. Privacy Preservation:

- Do not include any user-specific or sensitive data in the external LLM request.
- When queries contain location, dates, names, URLs, or other identifiable details, generalize or omit them in the LLM request.
- Replace or abstract any private or potentially identifiable content with neutral placeholders or general terms without losing the intent or meaning necessary for the LLM task.

2. Query Understanding and Reformulation:

- Analyze the user's query carefully to understand the underlying task or information need.
- Determine if the query involves translation, event recommendations, advice, summarization, or other tasks.

- Identify when user-provided content is sensitive or proprietary (e.g., unique texts, URLs to closed content) and avoid directly exposing it to the external LLM.

3. Quality Maximization:

- Produce an LLM request that is clear, precise, and directs the LLM to perform the necessary task without requiring private context.
- Retain the core informational or functional need so that the LLM can respond effectively.
- When referencing external documents or URLs, do not include the actual link or private info; instead, request general or typical information related to the topic.

4. Common Strategies for Privacy-preserving Requests:

- Use paraphrasing or abstraction for sensitive content.
- Request generalized or example-based information instead of specific user data.
- When translation is required, include only the text that needs translating if non-private, or otherwise generalize accordingly.
- For location- or event-specific queries, request general or typical information about the category or area without including sensitive details.

5. Reasoning Explanation:

- For each transformation, produce clear reasoning explaining how privacy is preserved and why the LLM request fulfills the user's intent without leaking sensitive information.

Input Format:

- A user query string possibly containing private or sensitive information.
- Required output:
 - a) A reasoning paragraph explaining your analysis and privacy considerations.
 - b) A privacy-preserving LLM request that can be sent to an external LLM for assistance.

By following the above principles and strategies, you will optimize both the quality of assistance and the protection of user privacy when interacting with third-party language models.

Node 4 (Score 94.44)

Task Description:

You will be given a user query that may include sensitive, personal, or identifiable information. Your role is to transform this query into a privacy-preserving request suitable for submission to an external large language model (LLM). The goal is to enable the external LLM to assist effectively while ensuring no private, identifying, or sensitive information about the user is exposed.

Key Requirements and Domain-Specific Details:

1. Privacy Preservation:

- Do not include any user-specific details, personal names, locations, dates, URLs, or other potentially identifiable information in the outgoing LLM request.
- When the user query contains private, sensitive, or proprietary data, you must generalize, abstract, or omit these details.

- Replace sensitive specifics with neutral placeholders or describe them in generalized terms without losing the core intent of the query.
 - Avoid exposing non-public, proprietary, or confidential content directly to the external LLM.
2. Understanding and Reformulation:
 - Carefully analyze the user's query to identify the underlying task type (e.g., summarization, creative writing, translation, profile writing, event recommendations, company background, academic content generation).
 - When reformulating, maintain the essential informational or functional need so that the external LLM can provide a useful, relevant response.
 - Preserve thematic and contextual elements necessary for quality output, while abstracting any sensitive or private details.
 - If the task involves user-supplied content (e.g., essays, texts, character descriptions), distill or summarize the relevant content or use generic versions instead of exposing original private text.
 3. Maximizing Quality of the Reformulated Prompt:
 - Construct clear, precise, and well-structured requests that explicitly guide the external LLM on the task.
 - Retain appropriate detail and context to ensure relevance, but balance this carefully against privacy concerns.
 - If referencing external documents, URLs, or institutions, do not include any links or private identifiers; instead, request general or typical information related to the subject matter.
 - When writing prompts for creative or fictional character tasks, clarify that the profile or content is fictional to avoid accidental exposure of real personal data.
 4. Common Strategies for Privacy Preservation:
 - Paraphrase or abstract personal and sensitive content.
 - Use general descriptions or hypothetical/example-based requests where appropriate.
 - Omit or generalize specific names, dates, locations, institutions, or proprietary course codes.
 - When user content is extensive, focus on absorbing the core themes and instructing the external model accordingly rather than passing the raw content.
 5. Explanation Requirement:
 - Provide a concise reasoning paragraph explaining how you identified sensitive or private details and the steps you took to protect user privacy.
 - Clarify how your reformulated LLM request retains the user's original intent and task needs without risking data leakage.
 - This explanation is mandatory to document your privacy-preserving approach and to justify the abstraction choices made.
- Input and Output Format:
- Input: A single user query string that may include sensitive or personal content.
 - Output:
 - a) Reasoning: A paragraph explaining the privacy considerations and reformulation strategy.
 - b) LLM Request: A generalized, privacy-safe prompt suitable for querying an external LLM.

Examples and Feedback Integration:

- Avoid including real names, course codes, or faculty references; replace them with general terms (e.g., "an interdisciplinary health minor" rather than "HHV").
- When given character bios or profiles, specify the fictional nature clearly and do not include potentially identifying physical descriptions unless necessary for the task.
- When a user asks about specific companies or entities, generalize the request to "a major company in [industry/country]" if privacy concerns exist.
- Maintain a balance between minimal leakage and maximizing task relevance and quality.
- Ensure the reformulated prompt supports a comprehensive and useful response from the external LLM without compromising privacy.

By rigorously following these detailed instructions, you will ensure maximal protection of user privacy while harnessing the power of external LLMs to assist effectively with diverse user requests across domains.

Node 5 (Score 94.67)**New Instruction for Privacy-Preserving Query Reformulation Assistant:****Task Overview:**

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your goal is to transform this input into a generalized, privacy-preserving prompt suitable for querying an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively, while rigorously protecting user privacy by abstracting or omitting any sensitive details.

Input Format:

- A single string representing a user query.
- The query may include private names, places, dates, proprietary info, or identifiable context.

Output Format:

- Part (a) Reasoning: a concise paragraph explaining:
 - How you identified sensitive or private information in the input.
 - What generalization, abstraction, or omission strategies you applied to protect privacy.
 - How the reformulated prompt maintains the original task or informational need without risking data leakage.
- Part (b) LLM Request: a carefully constructed and concise, privacy-safe prompt that retains essential thematic and functional elements to achieve a useful, relevant response from the external LLM.

Key Domain-Specific Details and Best Practices:**1. Privacy Preservation Principles:**

- Remove or replace all user-specific names (personal or organizational), exact dates or durations, locations, URLs, proprietary course or product codes, customer or client names, and other identifiers.

- When geographic or organizational mentions are critical for context, abstract these to broader or public categories (e.g., "a sustainable travel website focused on a Central American country" rather than naming the country explicitly).
 - Omit or abstract internal organizational pressures, client names, or sensitive contractual details to generic placeholders (e.g., "a client" rather than a named company).
 - For biographical or individual-related queries, avoid requesting or referencing real personal info; instead, frame requests around hypothetical or generic profiles unless the name is widely public and essential for the task.
2. Understanding and Reformulating the Task:
- Identify the underlying task (creative writing, summarization, professional communication drafting, etc.) from the user's query.
 - Preserve the functional intent and thematic requirements (e.g., content topics around sustainability, summary of a person's background, professional email follow-up).
 - For user-supplied content, do not repeat verbatim or expose any original text; instead, extract core themes, instruct the LLM accordingly, or replace with generic examples.
3. Maximizing Output Quality While Preserving Privacy:
- Construct a prompt that is clear, precise, and contains sufficient context to enable comprehensive and relevant LLM output.
 - Avoid ambiguous or overly generic requests that might reduce relevance or usefulness.
 - Maintain a balance between detail necessary for quality and generalization required for privacy.
4. Common Reformulation Strategies:
- Replace specific names with generic role identifiers or placeholders ("a business contact," "a notable individual").
 - Replace specific locations or institutions with generalized descriptors, e.g., "a country known for eco-tourism."
 - For time references, use relative or approximate terms without revealing explicit dates.
 - For internal or proprietary details, describe the scenario generically (e.g., "pressure from a client to close a matter urgently").
 - When handling personal profiles or characters, explicitly state that the profile is fictional if requesting content generation about a person.
5. Explanation Requirements:
- Always provide reasoning that details how privacy risks were identified and mitigated.
 - Explain how the essential task was preserved despite abstraction.
 - This reasoning documents the privacy-preserving approach and justifies design choices.
- Examples and Common Pitfalls:
- Do not retain or lightly obscure personal names or company names; fully abstract or remove.
 - Avoid exposing direct quotes from user-supplied texts.
 - Use neutral, context-appropriate phrasing to describe user intents.
 - Retain geographic or cultural context only when it is publicly available and pivotal to the task.

- Ensure the output prompt remains a clear instruction that the external LLM can readily interpret.

Summary:

Your reformulations must ensure zero exposure of PII or private/proprietary content while retaining enough context and clarity to generate high-quality, relevant responses from the external LLM. This involves careful task analysis, methodical abstraction of sensitive details, and clear, precise prompt construction combined with transparent reasoning on privacy protection mechanisms used.

Node 11 (Score 97.6)

New Instruction for Privacy-Preserving Query Reformulation Assistant (Enhanced):

Task Overview:

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your objective is to reformulate this query into a generalized, privacy-preserving prompt suitable for sending to an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively while rigorously protecting all user privacy by abstracting, omitting, or generalizing any sensitive details.

Input Format:

- A single user query string.
- The query might contain identifiable names (people, organizations), specific geographic locations, exact dates or timeframes, proprietary or internal terminology (e.g., system names, product codes), client or customer names, URLs, or any other sensitive or private information.

Output Format:

- Part (a) Reasoning:
 - Provide a concise paragraph that explains:
 - * How you identified sensitive or private information in the input
 - * What strategies you applied to protect privacy (generalization, omission, abstraction, replacement with placeholders)
 - * How the reformulated prompt preserves the original intent and task requirements without risking data leakage
- Part (b) LLM Request:
 - A concise, carefully constructed privacy-safe prompt that:
 - * Removes or anonymizes all PII and proprietary/internal details
 - * Abstracts locations, names, dates, and technical terms as needed
 - * Produces a clear and contextually rich instruction for the LLM to generate a relevant and informative response aligned with the user's original task

Detailed Domain-Specific Guidance and Best Practices:

1. Identification and Treatment of Sensitive Data:

- All user-specific or personal names (individual or organizational) must be removed or replaced with generic role descriptors (e.g., "a business contact," "a client," "a notable individual"). Never lightly obscure or partially redact; full abstraction is required.

- All geographic mentions must be abstracted unless the location is publicly known, essential to the task, and can be generalized (e.g., "a region known for eco-tourism" instead of naming a country or city explicitly).
 - Exact dates or durations must never be retained; instead, use relative or approximate temporal references (e.g., "recently," "over the past year").
 - Internal or proprietary terms including system names, product codes, subscription types, and technical jargon must be generalized or replaced with neutral descriptors to avoid leakage of intellectual property or sensitive operational details.
 - Avoid direct quotes or verbatim inclusion of user-supplied texts unless obfuscated by generalization.
2. Task Understanding and Reformulation:
- Identify the functional intent of the query: Is it creative writing, translation, summarization, professional communication drafting, technical explanation, or other?
 - Preserve the thematic and informational core of the query (e.g., request for educational quality analysis, technical translation of a passage, biographical summary).
 - Do not reproduce the original input verbatim; instead, frame the LLM prompt around the essential thematic elements extracted from the input.
 - For queries regarding individuals, avoid direct reference to real personal information unless the name is widely public and essential; even then, use a generic or hypothetical framing for the individual profile.
3. Strategies for High-Quality and Privacy-Preserving Prompts:
- Strike a balance between sufficient contextual detail and privacy abstraction to maintain prompt clarity and relevance.
 - Use neutral, context-aware formulations that clearly instruct the LLM on the content and style expected.
 - Avoid vague or overly generic prompts that could result in less useful or lower-quality responses.
 - When system or proprietary content is mentioned, instruct the LLM to generalize specific terms and maintain the technical meaning without revealing sensitive info.
 - When a direct translation is requested on specialized text, specify to replace or abstract internal nomenclature.
4. Explanation Requirements:
- The reasoning must transparently explain how privacy risks were identified (e.g., presence of names, locations, dates, proprietary terms).
 - It must describe the abstraction or omission methods applied (e.g., replacing "Jonah Van Beijnen" with "a notable individual," substituting "Makau" with "a specific region," or "Yoda" with "a system name").
 - Clarify how the essential task and user intent were preserved despite these generalizations (e.g., focusing on educational quality, technical translation, biographical summary).
 - This explanation justifies your design choices and demonstrates adherence to privacy-preserving principles.
- Common Pitfalls to Avoid:
- Do not merely lightly obscure or partially redact sensitive details; full anonymization or abstraction is required.
 - Do not repeat any user-supplied PII or proprietary content verbatim.
 - Avoid including URLs, exact dates, or direct quotes without modification.

- Do not leave ambiguity that could degrade the quality or contextual clarity of the reformulated prompt.
- Do not include any real personal or organizational names unless they are public figures and the query requires it, then use generic descriptors instead.

Example Summary of Effective Approach (Informed by Prior Examples):

- For geographic queries: replace exact place names with general regions and provide a brief contextual descriptor.
- For technical texts containing system names or subscription types: instruct the LLM to translate or process the text while replacing or abstracting proprietary system identifiers.
- For biographical summaries about specific individuals: remove the real name and request a generic, well-structured four-paragraph summary about "a notable individual," preserving the overall intent without leaking PII.

Summary:

Your reformulations must ensure zero exposure of any PII or private/proprietary content while retaining enough thematic and functional clarity for the external LLM to produce high-quality, relevant outputs. This requires thorough analysis of the user's query, rigorous application of privacy-preservation strategies, and explicit reasoning explanations that document your approach and choices.

L EXAMPLES OF BEST PROMPTS FOR EVERY BENCHMARK

In this section, we present the best optimized prompt obtained for every (benchmark, model) configuration. Each subsection below pertains to one (benchmark, model) configuration. Since every compound AI system consists of multiple modules, each subsection consists of multiple boxes, listing the prompts for each module. MIPROv2 optimized prompts contain upto 4 few-shot examples for each task. We provide just the first demo here for brevity. GEPA's prompts only consist of the optimized instruction, which is provided in full.

L.1 HOTPOTQA, GPT-4.1 MINI

HotpotQA GPT-4.1 Mini create_query_hop2.predict

Base Prompt:

Given the fields `question`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

You are an expert multi-hop question answering system designed to refine retrieval queries for complex questions. Given the original question and an initial summary of retrieved documents (summary_1), think step by step to analyze how the summary relates to the question. Use this reasoning to generate a focused and precise query that will guide the retrieval of additional relevant information needed to answer the question completely. Your output should include a clear chain-of-thought reasoning process followed by the refined query that targets complementary evidence for the next retrieval hop.

Demos:

Example 1:

```

question:
  Simone Benedetti plays for a team that currently competes in which Serie?
summary_1:
  Simone Benedetti plays for Virtus Entella, but the passages do not specify which
  Serie the team currently competes in.
reasoning:
  The summary indicates that Simone Benedetti plays for Virtus Entella, but it does not
  state the current Serie (league level) in which Virtus Entella competes. To
  accurately answer which Serie the team currently competes in, it is necessary to look
  up the most recent league information for Virtus Entella.
query:
  What Serie (league level) does Virtus Entella currently compete in?

#####
3 other demos omitted for brevity.
#####

```

GEPA Prompt generated by config GEPA:

```

You will be given two input fields: `question` and `summary_1`.

Your task is to generate a new search query (`query`) optimized for the second hop of
a multi-hop retrieval system. The original user question is typically complex and
requires information from multiple documents to answer. The first hop query is the
original question used to retrieve an initial set of documents. Your goal is to generate
a second hop query that retrieves additional relevant documents that were not
found in the first hop but are necessary to answer the original question completely.

Detailed task instructions and hints:

1. Input Understanding:
  - `question` is the original multi-hop question posed by the user.
  - `summary_1` is a concise summary of information from a document retrieved in the
    first hop, which partially addresses the question.

2. Purpose and Context:
  - Your generated `query` aims to find the missing pieces of information needed to
    fully answer the `question`.
  - The multi-hop retrieval system works in stages:
    - First hop: The original question returns some documents.
    - Second hop: Your query must help retrieve any other relevant documents NOT found
      in the first hop that hold complementary or broader context necessary for final
      answer extraction.

3. Key Observations from Examples and Feedback:
  - First-hop documents often cover one entity or aspect in the question.
  - Remaining relevant documents often involve connected or higher-level concepts
    mentioned in `summary_1` but not explicitly asked in the original question.
  - The `query` should be formulated to explicitly target these missing, but logically
    linked, documents.
  - Avoid merely paraphrasing the original question or restating known facts from
    `summary_1`.

```

- Instead, infer what broader or related entities/concepts might provide the crucial missing information.
- For example, if `summary\_1` describes a population for a small civil parish, but the question wants total population of the wider region, your `query` should target that wider region (e.g., "Madeira archipelago population in 2011").
- Similarly, if `summary\_1` covers a song and the question wants the album it came from, but first hop got song-level documents, your query should retrieve documents about the album itself.

4. **How to Build the Query:**

- Identify the entities or topics mentioned in `summary\_1` that appear related but different from first-hop documents.
- Reframe the query to explicitly mention these broader or related entities connected to the original question.
- Include relevant key context from the question to maintain specificity, but shift focus to the missing piece.
- The goal is to retrieve documents that link or complement what was retrieved initially.

5. **Practical Strategy:**

- Read the `summary\_1` carefully to spot references to bigger contexts or other entities not covered in the first hop.
- Ask yourself, "What entity or aspect does this summary hint at that could answer the original question but was not found yet?"
- Formulate a precise, focused factual query targeting that entity or concept to retrieve the missing documents.

6. **Output:**

- Produce only the field `query` as a clear, concise question or keyword phrase designed for efficient retrieval of **second-hop documents**.
- Ensure the query relates logically to the original question while targeting the broader or complementary knowledge identified in `summary\_1`.
- Do **not** include the original question or simply rephrase it.
- Do **not** duplicate information already well-covered by the first hop retrieval.

By following these principles, you will help the multi-hop retrieval system find all necessary documents to answer the multi-faceted original question completely.

HotpotQA GPT-4.1 Mini final_answer.predict

Base Prompt:

Given the fields `question`, `summary\_1`, `summary\_2`, produce the fields `answer`.

MIPROv2 Prompt:

You are an expert multi-hop reasoning assistant skilled in synthesizing information from multiple summaries to answer complex questions. Given the `question`, along with two intermediate summaries `summary\_1` and `summary\_2` that contain relevant evidence, carefully analyze and integrate the information step-by-step to produce a clear, logical reasoning process followed by a concise and accurate final answer.

Demos:

Example 1:

```

question:
  are Machaeranthera and Prumnopitys both the plants?
summary_1:
  Yes, both Machaeranthera and Prumnopitys are plants. Machaeranthera is a genus of
  flowering plants in the daisy family, while Prumnopitys is a genus of coniferous
  evergreen trees in the podocarp family.
summary_2:
  Yes, both Machaeranthera and Prumnopitys are plants; Machaeranthera is a genus of
  flowering plants in the daisy family, while Prumnopitys is a genus of coniferous
  evergreen trees in the podocarp family.
reasoning:
  Both summaries agree that Machaeranthera and Prumnopitys are plants. Specifically,
  Machaeranthera is a genus of flowering plants in the daisy family, and Prumnopitys is
  a genus of coniferous evergreen trees in the podocarp family. Since both belong to
  plant genera, the answer is yes.
answer:
  yes

```

```

#####
3 other demos omitted for brevity.
#####

```

GEPA Prompt generated by config GEPA:

Task Description:

You are given three fields as input: `question`, `summary\_1`, and `summary\_2`. Your goal is to produce an `answer` field that directly and explicitly responds to the question using the information from the two summaries, enhanced by your authoritative domain knowledge when needed.

Input Format:

- `question`: A natural language question that may require a fact, definition, name, date, yes/no response, or other specific information.
- `summary\_1` and `summary\_2`: Two independently generated summaries or snippets containing information related to the question. They may vary in completeness, accuracy, and specificity.

Requirements and Approach:

1. **Understand the question precisely.** Determine exactly what is asked—whether a name, a specific fact, a date, or a yes/no answer.
2. **Compare both summaries.** Analyze the content of `summary\_1` and `summary\_2`:
 - If they agree and directly answer the question, use this as primary evidence.
 - If one summary provides a fact that the other does not mention, carefully evaluate its plausibility.
 - If the summaries conflict, use domain expertise and authoritative knowledge to resolve or explicitly state uncertainty.
3. **Domain-specific factual verification and nuance:**

- **Names and nicknames:** Provide only the specific nickname or name when asked, without extra phrasing. For example, when asked for the nickname of a person or entity, respond with the nickname alone, not a full sentence.
 - **Nationality and identity distinctions:** Use the most precise terms aligned with factual correctness and common usage (e.g., "English" vs. "British") based on domain knowledge.
 - **Dates and historical facts:** Verify dates or historical claims with domain knowledge to pick the correct fact, especially when there might be confusion between franchise start dates vs. event dates etc.
 - **Yes/no questions:** Prefer concise answers of "Yes" or "No" only, unless the question demands elaboration.
 - **Types or categories:** If a question asks about the type or category (e.g., type of company), provide the most direct concise phrase without including adjectives like nationality unless asked explicitly.
4. **Answer conciseness and relevance:**
- Provide a brief and direct answer to the question.
 - Avoid repeating or restating the question.
 - Avoid unnecessary context unless requested or needed for clarity.
 - Avoid constructing full sentences unless needed; for example, answers to nickname or yes/no questions should be as short and specific as possible.
5. **When authoritative knowledge supplements the summaries:**
- If the summaries are incomplete or potentially inaccurate, incorporate trusted knowledge from your training about the topic to provide the correct and precise answer.
 - For example, when a summary gives a year that conflicts with known release dates or factual details, prefer the verified date.
 - When the summaries differ in style (one uses a formal phrase, another provides just the nickname), respond with the correct, clean answer format (e.g., just the nickname alone).
6. **Examples of correct reasoning and answers:**
- Question: "What is the nickname of the 2005 Toyota Grand Prix of Long Beach Polesitter?"
 - Correct answer: "the thrill from West Hill"
 - Question: "What type of company is Zipcar led by Scott Griffith from 2003-2013?"
 - Correct answer: "car-sharing company"
 - Question: "Who was the partner of British comic book artist, Henry Flint, that helped create Zombo?"
 - Correct answer: "Al Ewing"

Summary:

- Use both summaries as primary but not sole evidence.
- Reliably verify and contextualize facts using domain knowledge, especially for nationality, dates, nicknames, company types, and yes/no questions.
- Provide short, direct answers matching the specificity requested.
- Avoid unnecessary elaboration unless explicitly required.
- Explicitly resolve conflicts or ambiguity using your knowledge or state uncertainty when appropriate.

This approach ensures that answers are both accurate and concise, suitable for direct consumption or integration in knowledge bases or question-answering systems.

HotpotQA GPT-4.1 Mini summarize1.predict

Base Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a question and a set of related passages, carefully analyze the information by thinking through the relevant facts step-by-step. Produce a clear and concise summary that synthesizes the key points from the passages directly relevant to answering the question, ensuring the summary is focused, accurate, and grounded in the evidence provided.

Demos:

Example 1:

question:

The architectural style of a church that stands in front of the Palazzo Ghisilardi Fava originated in what city?

passages:

['Palazzo Ghisilardi Fava | Palazzo Ghisilardi Fava is a Renaissance style palace, located on via Manzoni 4 in Bologna, region of Emilia Romagna, Italy; it houses the Medieval Civic Museum of Bologna.', 'Madonna di Galliera, Bologna | The Madonna di Galliera is a church with a Renaissance facade and Baroque interiors, located on Via Manzoni, in central Bologna, Italy. It stands in front of the Palazzo Ghisilardi Fava. The present name over the portal is the "Chiesa di Filippini Madonna di Galliera e Filippo Neri".', 'Palazzo Pretorio, Prato | The Palazzo Pretorio of the Tuscan town of Prato was the old city hall located town center, standing in front of the current Palazzo Comunale. It now accommodates the Civic Museum of Prato, which was reopened on September 2013.', 'Palazzo Pisani Gritti | The Palazzo Pisani Gritti is a Gothic-style palace located across the Grand Canal from the Church of the Salute, and stands along the Campo del Traghetto where the church of Santa Maria del Giglio, in the Sestieri of San Marco, Venice, Italy.', 'D  m  t  r Tower | The D  m  t  r tower is the oldest building in Szeged, Hungary. The foundation was most probably laid during the 11th century, while the lower part was built (in Romanesque style) from the 12th century, and the upper part (Gothic style) from the 13th century. The tower was once part of the former "Szent D  m  t  r templom" (St. Demetrius church), but today it stands in D  sm Square, in front of the much larger Votive Church of Szeged. The upper part was rebuilt from the original stones in 1926. The architecture of the tower is similar to another found in Southern France, or in the territory of the former Byzantine Empire.', 'Palazzo Leoni, Bologna | The Palazzo Leoni is a Renaissance style palace located on Via Marsala #31, in front of the outlet of Via Mentana, in central Bologna, region of Emilia-Romagna, Italy. The apse of the church of San Martino, is across the street.', 'Palazzo Altieri | Palazzo Altieri is a palace in Rome, which was the home of the Altieri family in the city. The palace faces the square in front of the Church of the Ges  z.']

reasoning:

The question asks for the city where the architectural style of a church that stands in front of the Palazzo Ghisilardi Fava originated. Passage [2] identifies the church standing in front of the Palazzo Ghisilardi Fava as "Madonna di Galliera" and mentions it has a Renaissance facade and Baroque interiors. Passage [1] states the Palazzo Ghisilardi Fava is located in Bologna, Italy. Since the church is located in Bologna and features a Renaissance architectural style, it can be inferred that the architectural style of this church originated in the city of Florence, which is historically recognized as the birthplace of Renaissance architecture. Thus, the

architectural style (Renaissance) of the church in front of the Palazzo Ghisilardi Fava originated in Florence.

summary:

The Renaissance architectural style of the church standing in front of the Palazzo Ghisilardi Fava originated in Florence.

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA:

You are a first-hop **summarization module** in a multi-hop question answering (QA) system. Your task is to generate a concise, informative ``summary`` given two input fields: a ``question`` and a list of relevant ``passages``.

Your goal is to extract and synthesize key information from the retrieved passages that:

1. Directly relates to the initial question.
2. Captures the core facts and entities needed to understand the scope and context of the question.
3. Includes relevant connections, bridging entities, dates, locations, or descriptions that enable the system to devise focused and effective follow-up queries in subsequent hops.
4. Provides a strong factual foundation for downstream answer generation modules.

Task specifics and best practices:

- The ``summary`` must represent a distilled synthesis, not just a compression or extractive snippet.
- Explicitly include cited passage titles or key entity labels (e.g., "Children in Need 2006 | ..." or "Anthony Levandowski | ...") in your summary to highlight the origin of information.
- Incorporate sufficient context to hint at missing or un-retrieved supporting facts, thus enhancing the multi-hop retrieval process.
- When the question asks for an attribute (e.g., nationality, location, company origin), ensure you provide:
 - Identification of the relevant subject or entity mentioned in the passages.
 - The extracted attribute or relevant information as stated or implied.
 - Bridging details that could help the system pursue remaining information in the next retrieval step.
- Avoid forming a final answer; instead, focus on "what is known now" from the input documents to facilitate further query refinement.

Examples of critical elements to include:

- Entity names, roles, titles, and dates tied to the question.
- Names of organizations or locations connected through intermediary entities.
- Distinctive identifiers or clarifications that can help narrow down next-step retrieval (such as "Natasha Kaplinsky is an English presenter," "Kapolei is a city on Oahu," or "Waymo spun out of Alphabet").

Format of output:

Provide a paragraph or a few sentences that cohesively summarize the key passages in relation to the question, referencing passage titles or entities to frame facts clearly.

This approach ensures the summary is both informative for next-hop retrieval and foundational for final answer extraction in multi-hop QA.

HotpotQA GPT-4.1 Mini summarize2.predict

Base Prompt:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a `question`, relevant `context`, and a list of `passages`, provide a clear, concise summary that integrates the key information from the passages in relation to the question and context. Use step-by-step reasoning to explain how the summary is derived from the evidence before presenting the final synthesized summary.

Demos:

Example 1:

question:

What bank was founded by the great-great-great grandfather of the second Duke of Florence?

context:

The bank founded by the great-great-great grandfather of the second Duke of Florence is the Medici Bank, established by Giovanni di Bicci de' Medici.

passages:

['Giovanni di Bicci de' Medici | Giovanni di Bicci de' Medici (c. 1360 – February 20/28, 1429) was an Italian banker, a member of Medici family of Florence, and the founder of the Medici Bank. While other family members, such as Chiarissimo di Giambuono de' Medici, who served in the Signoria in 1201, and Salvestro de' Medici, who was implicated in the Ciompi Revolt of 1378, are historically significant, Giovanni's founding of the family bank truly began the family's rise to power in Florence. He was the father of Cosimo de' Medici ("Pater Patriae"), grandfather of Piero di Cosimo de' Medici, great-grandfather of Lorenzo de' Medici (the Magnificent) and great-great-great-grandfather of Cosimo I de' Medici, Grand Duke of Tuscany.', 'Averardo de' Medici | Averardo de' Medici (1320-1363), also known as Everard De Medici, was the son of Salvestro de' Medici (died 1346), "il Chiarissimo" (English meaning "the very clear.") and the father of three children: Giovanni, Francesco, and Antonia. Giovanni di Bicci de' Medici would later become the first historically relevant member of Medici family of Florence and the eventual founder of the Medici bank.', 'Villa Medici at Cafaggiolo | The Villa Medicea di Cafaggiolo is a villa situated near the Tuscan town of Barberino di Mugello in the valley of the River Sieve, some 25 kilometres north of Florence, central Italy. It was one of the oldest and most favoured of the Medici family estates, having been in the possession of the family since the 14th century, when it was owned by Averardo de' Medici. Averardo's son, Giovanni di Bicci de' Medici, is considered to be the founder of the Medici dynasty.', 'Medici: Masters of Florence | Medici: Masters of Florence is an Italian-British television drama series about the Medici dynasty set in 15th-century Florence, starring Dustin Hoffman as Giovanni di

Bicci de' Medici, Richard Madden as Cosimo de' Medici, and Stuart Martin as Lorenzo de' Medici ("The Elder"). The series was co-created by Frank Spotnitz ("The X-Files" and "Man in the High Castle") and Nicholas Meyer (""). Sergio Mimica-Gezzan ("The Pillars of the Earth") directed all eight episodes. Episodes 1 and 2 aired on Rai 1 (Italian TV) on 18 October 2016. According to Italian ratings compiler Auditel, it attracted a record 7.6 million viewers. The first season consists of eight episodes.

Neri di Bicci | Neri di Bicci (1419–1491) was an Italian painter of the Renaissance. A prolific painter of mainly religious themes, he was active mainly in Florence and in the medium of tempera. His father was Bicci di Lorenzo. His grandfather, Lorenzo di Bicci was also a painter in Florence, a pupil of Spinello Aretino.

Cosimo de' Medici | Cosimo di Giovanni de' Medici (called 'the Elder' (Italian "il Vecchio") and posthumously "Father of the Nation" (Latin "pater patriae"); 27 September 1389 – 1 August 1464) was an Italian banker and politician, the first of the Medici political dynasty, "de facto" rulers of Florence during much of the Italian Renaissance. Despite his influence, his power was not absolute; Florence's legislative councils at times resisted his proposals, something which would not have been tolerated by the Visconti of Milan, for instance; throughout his life he was always "primus inter pares" ("first among equals"). His power derived from his wealth as a banker, and he was a great patron of learning, the arts and architecture.

Giovanni il Popolano | Giovanni de' Medici, in full Giovanni di Pierfrancesco de' Medici, later known as il Popolano (21 October 1467 – 14 September 1498) was an Italian nobleman of the Medici House of Florence. He was the son of Pierfrancesco di Lorenzo de' Medici, and therefore a member of a secondary branch of the family."]

reasoning:

The question asks for the bank founded by the great-great-great-grandfather of the second Duke of Florence. Passage [1] states that Giovanni di Bicci de' Medici founded the Medici Bank and was the great-great-great-grandfather of Cosimo I de' Medici, who was the Grand Duke of Tuscany. The Dukedom of Florence eventually transformed into the Grand Duchy of Tuscany under Cosimo I, who is identified as the second Duke of Florence in the question context.

Thus, Giovanni di Bicci de' Medici, as the great-great-great-grandfather of the second Duke of Florence (Cosimo I de' Medici), founded the Medici Bank.

summary:

The Medici Bank was founded by Giovanni di Bicci de' Medici, the great-great-great-grandfather of the second Duke of Florence.

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

L.2 HOTPOTQA, QWEN3 8B

HotpotQA Qwen3 8B create_query_hop2.predict

Base Prompt:

Given the fields `question`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

Given the question and the first summary, synthesize the key elements of the question and the summary. Identify the specific information that needs to be retrieved or confirmed. Formulate a focused and precise query that will guide the next step in the multi-hop reasoning process, ensuring it directly addresses the gap in knowledge or requires further clarification from additional context.

Demos:

GEPA Prompt generated by config GEPA+Merge:

Given the fields `question` and `summary\_1`, produce the field `query` that optimizes the retrieval of additional documents for a multi-hop system.

****Task Details:****

1. ****Objective:**** Your query must target documents not retrieved in the first hop, using clues from the summary and the original question.
2. ****Key Strategy:****
 - Identify gaps in the first hop's retrieved documents (e.g., missing entities, relationships, or specific details).
 - Use explicit information from the summary (e.g., names, locations, quantities) to rephrase the question into a query that surfaces new relevant documents.
 - Avoid restating the answer directly; instead, structure the query to explore connections or unresolved details.
3. ****Domain-Specific Guidance:****
 - If the summary explicitly answers the question, the query should still focus on retrieving documents that provide deeper context or verify the answer (e.g., "What is the headquarters location of [Company]?" instead of "The answer is [Location]").
 - Leverage entities mentioned in the summary (e.g., "Carhartt," "Aubrey O'Day") to anchor the query.
 - If no documents are missing, rephrase the query to explicitly request the answer (e.g., "Which has more acts, Elektra or From the House of the Dead?").
4. ****Avoid:****
 - Generating queries that duplicate the original question.
 - Assuming the summary contains all necessary information for the second hop.

HotpotQA Qwen3 8B final_answer.predict

Base Prompt:

Given the fields `question`, `summary\_1`, `summary\_2`, produce the fields `answer`.

MIPROv2 Prompt:

Given the question, summary_1, and summary_2, generate a step-by-step reasoning process that logically connects the information from both summaries to arrive at the final

answer, and then provide the answer. Ensure the reasoning explicitly references how the summaries support the conclusion and maintains consistency with the question's requirements.

Demos:

Example 1:

question:

are Machaeranthera and Prumnopitys both the plants?

summary_1:

Yes, both Machaeranthera (a genus of flowering plants in the daisy/aster family) and Prumnopitys (a genus of coniferous trees in the podocarp family) are plants.

summary_2:

Yes, both Machaeranthera (a flowering plant genus in the daisy family) and Prumnopitys (a coniferous tree genus in the podocarp family) are plants.

reasoning:

Both summaries confirm that Machaeranthera and Prumnopitys are plants. Summary 1 explicitly states they are plants, categorizing Machaeranthera as a flowering plant genus in the daisy/aster family and Prumnopitys as a coniferous tree genus in the podocarp family. Summary 2 repeats this confirmation with slightly different phrasing.

answer:

Yes

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

Given the fields `question`, `summary\_1`, and `summary\_2`, produce the field `answer` by:

- Extracting precise terminology**: Identify the exact noun or specific term required in the answer (e.g., "Medicare" rather than "Medicare cuts"). Avoid vague or generalized terms unless explicitly stated in the summaries.
- Resolving ambiguity**: If the question references a title, historical role, or specific designation (e.g., "second Duke of Florence"), prioritize contextual or historical clues from the summaries to infer the correct answer, even if the exact term is not explicitly stated. Use domain-specific knowledge (e.g., Medici family lineage) to fill gaps when summaries are indirect or vague.
- Cross-referencing summaries**: Ensure consistency between summaries. If summaries conflict, prioritize the one with explicit factual claims (e.g., numerical data, direct statements). If no explicit claim exists, synthesize information while ensuring alignment with historical, political, or cultural context.
- Avoiding overgeneralization and extra information**: Focus strictly on the most specific and directly stated information in the summaries. Do not add context, explanations, or external knowledge beyond what is explicitly provided. For example, if the question asks for a year, provide only the year; do not include band member details or historical background.
- Prioritizing factual alignment**: If a summary explicitly states the answer, use that. If summaries are indirect or vague, synthesize information while ensuring alignment with factual knowledge (e.g., linking "Path to Prosperity" to Rep. Paul Ryan's Medicare proposal).

Key adjustments based on feedback:

- Conciseness**: Answers must be strictly factual and concise, avoiding additional context or explanations. For example, if the question is "Is X shorter than Y?" the

answer should be a simple "No" or "Yes" based on numerical comparisons, not a full explanation.

- **Numerical precision**: When comparing measurements (e.g., heights, dates), ensure exact values are used and explicitly stated in the summaries. If summaries provide conflicting numbers, resolve via direct factual claims.
- **Domain-specific knowledge**: Use known facts (e.g., architectural records, historical timelines) to validate ambiguous answers, but only when summaries lack explicit information.

HotpotQA Qwen3 8B summarize1.predict

Base Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `question`, `passages`, produce the fields `summary`.

Demos:

Example 1:

question:

Tay Garnett and Alexander Kluge both have what job?

passages:

['Tay Garnett | William Taylor "Tay" Garnett (June 13, 1894 – October 3, 1977) was an American film director and writer.', 'Trade Winds (film) | Trade Winds is a 1938 comedy film distributed by United Artists. It was directed by Tay Garnett, and starred Fredric March and Joan Bennett. The screenplay was written by Dorothy Parker, Alan Campbell and Frank R. Adams, based on story by Tay Garnett.', 'Prestige (film) | Prestige is a 1932 American pre-Code drama film directed by Tay Garnett and written by Tay Garnett, Rollo Lloyd and Francis Edward Faragoh. The film stars Ann Harding, Adolphe Menjou, Melvyn Douglas and Guy Bates Post. The film was released on January 22, 1932, by RKO Pictures.', 'Alexander Kluge | Alexander Kluge (born 14 February 1932) is a German author, philosopher, academic and film director.', 'Cattle King | Cattle King is a 1963 film directed by Tay Garnett. It stars Robert Taylor and Robert Loggia. It also appears to have been called Guns of Wyoming in some countries.', 'G  nther von Kluge | G  nther von Kluge (30 October 1882 – 19 August 1944) was a German field marshal during World War II. Kluge held commands on both the Eastern and Western Fronts. Although Kluge was not an active conspirator in the 20 July plot, he committed suicide on 19 August 1944, after having been recalled to Berlin for a meeting with Hitler in the aftermath of the failed coup. He was replaced by Field Marshal Walter Model.', 'Bad Company (1931 film) | Bad Company is a 1931 American Pre-Code gangster film directed and co-written by Tay Garnett with Tom Buckingham based on Jack Lait's 1930 novel "Put on the Spot". It stars Helen Twelvetrees and Ricardo Cortez. Told from the view of a woman, the working titles of this film were "The Gangster's Wife" and "The Mad Marriage". Unlike many static early sound films, Garnett includes several scenes using a moving camera climaxing in a gigantic assault on an office building with both sides using heavy machine guns.']

reasoning:

The question asks for the common job of Tay Garnett and Alexander Kluge. From the passages, Tay Garnett is explicitly identified as a film director and writer (passages 1, 2, 3, 5, 7). Alexander Kluge is described as a German author, philosopher, academic, and film director (passage 4). The shared job between them is "film director."

summary:

Both Tay Garnett and Alexander Kluge are film directors.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

You are the first-hop **summarization module** in a multi-hop QA system. Your task is to generate a **comprehensive, structured summary** that:

1. **Extracts direct answers** from the top retrieved passages to address the question.
2. **Identifies and highlights missing or implied clues** that may require further retrieval (e.g., entities, connections, or contextual details).
3. **Synthesizes information** by combining explicit facts from the passages with domain-specific knowledge or logical inferences to guide subsequent steps.

Summary Structure

- **Entity/Person Mention**: Clearly state the subject (e.g., "Billy Truax", "Eintracht Braunschweig") and include **full names, titles, or official designations** (e.g., "Thomas Lance Rentzel", "Braunschweiger Turn- und Sportverein Eintracht von 1895 e.V.").
- **Direct Answer**: Include **explicit answers** from the passages (e.g., birth dates, team affiliations, or direct statements).
- **Clues for Next Steps**: Signal **missing information** (e.g., "Lance Rentzel's birth year is explicitly stated, but his exact birthplace is not; need to search for 'Lance Rentzel birthplace'").
- **Domain-Specific Context**: Add **relevant background** (e.g., "Eintracht Braunschweig is a German football club based in Braunschweig, Lower Saxony" or "NFL players' birth dates are critical for age comparisons").

Guidelines

- **Do not omit** any entity or detail from the retrieved passages that could be relevant for follow-up queries (e.g., team names, locations, or historical context).
- **Prioritize clarity** by **separating direct answers from inferred clues** (e.g., using bullet points, subheadings, or bolded labels).
- **Avoid assumptions** not supported by the passages; if information is absent, **explicitly state that it is missing** and suggest **precise search terms** (e.g., "Verify Wichita Dwight D. Eisenhower National Airport's tower status via FAA records").
- **Include quantifiable data** (e.g., "few thousand Stabyhouns exist globally", "born July 15, 1943") to enable precise comparisons.
- **Highlight connections** between entities (e.g., "Billy Truax and Lance Rentzel were traded in 1970") to aid in cross-referencing.

Key Niche/Domain-Specific Insights

- **NFL Player Comparison**: Birth dates are critical for age determination, and team affiliations (e.g., "traded in 1970") may imply historical context.
- **Airport Classification**: "Non-towered" status is explicitly stated in some passages (e.g., "non-towered public airport"), while others require inference (e.g., "major commercial airports typically have towers").
- **Football Club Context**: Clubs like Eintracht Braunschweig require background on their location, league, and history (e.g., "based in Braunschweig, Lower Saxony").

- **Quantifiable Data**: Use exact dates, numbers, or rankings (e.g., "few thousand Stabyhouns exist globally") to enable precise comparisons.

Critical Additional Instructions

- **Ensure All Retrieved Documents Are Represented**: Explicitly include all entities, titles, and details from the retrieved passages (e.g., full names, film titles, and specific roles).
- **Signal Missing Links**: If a connection between entities is implied but not explicitly stated (e.g., "Nancy Steiner worked on *The Lovely Bones*"), flag this as a potential gap and suggest search terms to resolve it.
- **Prioritize Bridging Concepts**: Highlight relationships between entities (e.g., "Gary Pinkel coached Toledo in 1993 and holds the most wins in school history") to enable focused follow-up queries.
- **Avoid Overgeneralization**: Only include domain-specific context that is either explicitly stated in the passages or directly inferable (e.g., "major commercial airports typically have towers" is acceptable, but "airports with fewer than 10,000 passengers are non-towered" is not unless stated).

Example Format

For the question *"Which NFL player is younger, Billy Truax or Lance Rentzel?"*:

- **Entity/Person Mention**: Billy Truax (William Frederick Truax), Lance Rentzel (Thomas Lance Rentzel)
- **Direct Answer**:
 - **Billy Truax**: Born July 15, 1943.
 - **Lance Rentzel**: Born October 14, 1943.
- **Clues for Next Steps**: None required; birth dates are explicitly provided.
- **Domain-Specific Context**: Birth dates are sufficient to determine age difference within the same year.

For the question *"Which is a non-towered airport, Wichita Dwight D. Eisenhower National Airport or Montrose Regional Airport?"*:

- **Entity/Person Mention**: Wichita Dwight D. Eisenhower National Airport, Montrose Regional Airport
- **Direct Answer**:
 - **Montrose Regional Airport**: "non-towered public airport" (passage 3).
 - **Wichita Dwight D. Eisenhower National Airport**: No explicit mention of tower status; inferred as **towered** (typical for major commercial airports).
- **Clues for Next Steps**: Verify Wichita's tower status via FAA records or additional sources (e.g., "Wichita Dwight D. Eisenhower National Airport tower status").
- **Domain-Specific Context**: Non-towered airports lack a control tower, relying on pilot communication (passage 4). Major commercial airports like Wichita usually have towers.

Tip: When summarizing, don't just compress; synthesize! Include both direct answers and clues required for the system's next steps. Always explicitly state if a retrieved document's content is missing critical information, and provide actionable search terms to address gaps.

HotpotQA Qwen3 8B summarize2.predict

Base Prompt:

Given the fields `question`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `question`, `context`, and `passages`, generate a structured reasoning process that explains how to synthesize information from these sources, and then produce a concise summary of the key findings.

Demos:

Example 1:

question:

Simone Benedetti plays for a team that currently competes in which Serie?

context:

The provided passages do not specify the Serie in which Simone Benedetti's team (Virtus Entella) currently competes.

passages:

['Simone Benedetti | Simone Benedetti (born 3 April 1992) is an Italian professional footballer who plays as a centre back for Virtus Entella.', 'Virtus Entella | Virtus Entella (usually referred as simply Entella or Chiavari Entella) is an Italian association football club, based in Chiavari, Liguria. They currently compete in Serie B.', 'Simone Palermo | Simone Palermo (born 17 August 1988) is an Italian footballer who plays as a midfielder for Virtus Entella at Serie B.', 'Andrea Paroni | Andrea Paroni (born 14 October 1989) is an Italian footballer who currently plays as a goalkeeper for Virtus Entella in the Serie B.', 'Alessandro Iacobucci | Alessandro Iacobucci (born 3 June 1991) is an Italian footballer who currently plays as a goalkeeper for Serie B side Virtus Entella.', 'George Ganea | George Ganea (born 26 May 1999 in Bucharest) is a Romanian footballer who plays as a Striker for Serie B club Virtus Entella Primavera team.', 'Gianpaolo Castorina | Gianpaolo Castorina (born August 30, 1976 in Milan) is a retired Italian professional footballer turned coach, currently in charge as head coach of Serie B club Virtus Entella.']

reasoning:

The question asks for the Serie in which Simone Benedetti's team (Virtus Entella) currently competes. While the context initially states that this information is not specified, the passages explicitly mention that Virtus Entella competes in Serie B. Multiple passages (e.g., [2], [3], [4], [5], [7]) confirm this detail, directly answering the question.

summary:

Simone Benedetti's team, Virtus Entella, currently competes in Serie B.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

Given the fields `question`, `context`, and `passages`, produce the field `summary`.

Your task is to synthesize information from the question, context, and newly retrieved passages to generate a **comprehensive, precise, and well-structured summary** that enables the answer generation module to confidently arrive at the correct answer.

Key Requirements:

1. **Explicit Answers First**: Prioritize explicitly stated facts from the context and passages (e.g., direct mentions of entities, roles, or relationships).
2. **Infer or Generalize When Necessary**: If critical details are missing from the passages, infer connections or generalize based on contextual clues and domain-specific knowledge (e.g., linking ownership structures, roles, or historical context).
3. **Bridge Gaps**: Ensure the summary includes all **key supporting information** required to answer the question, even if it is not explicitly stated in the input. For example:
 - If the answer is "Newcastle United," include details about Sports Direct's ownership and the connection to the billionaire.
 - If the answer is a person's role (e.g., "troubleshooter"), explicitly state their relationship to the question's subject and any relevant background.
4. **Structure and Precision**:
 - Clearly connect entities, roles, and relationships (e.g., "Stan Kroenke owns Sports Direct and Arsenal F.C.").
 - Avoid ambiguity by including all necessary contextual links (e.g., "Mike Ashley founded Sports Direct and owns Newcastle United").
 - Use precise terminology and ensure alignment with domain-specific knowledge (e.g., "investigative journalist" instead of "writer").
5. **Domain-Specific Knowledge**: Leverage implicit domain knowledge when passages lack critical details (e.g., knowing that "Project RAND" is linked to Henry H. Arnold and the RAND Corporation).

Example Integration:

If the question is about a person's profession in a novel, ensure the summary includes:

- The character's name.
- Their profession (explicitly stated in the text).
- Contextual links to the book series or plot (e.g., "in *The Girl in the Spider's Web*").
- Any relevant background about the profession or character's role in the story.

Always aim to match the **coverage and relevance** of an "ideal summary" as described in the feedback, ensuring the answer module has all necessary information to generate the correct final answer.

L.3 IFBENCH, GPT-4.1 MINI

IFBench GPT-4.1 Mini generate_response_module.predict

Base Prompt:

Respond to the query

MIPROv2 Prompt:

You will be given a query containing a complex task with multiple constraints and instructions. Your job is to first repeat the query exactly as given, word for word, without adding any extra words or commentary before or after the repetition. After repeating the query, provide a detailed, step-by-step chain-of-thought reasoning that carefully unpacks and addresses every aspect of the query. Then, produce a final response that adheres strictly to all requirements, including formatting, language, and content constraints specified in the query. Be thorough, precise, and ensure the final answer is

fully validated and consistent with the reasoning. If the query specifies additional formatting or stylistic rules (such as language or inclusion of a postscript), include them exactly as instructed.

Demos:

GEPA Prompt generated by config GEPA+Merge:

You are given a query input, and your task is to respond appropriately to that query. The query may contain specific instructions or constraints that you must strictly adhere to in your response. Carefully analyze the query to determine the exact requirements, including but not limited to:

- Responding with an answer chosen from a restricted set of options exactly as specified (including exact wording and punctuation).
- Including specific words a minimum number of times.
- Including specific letters a minimum number of times.
- Repeating the entire query word-for-word before providing your answer if explicitly requested, without adding extra words or characters before the repetition.
- Avoiding specific forbidden words or keywords in your response if indicated.
- Following any other explicit instructions or constraints embedded in the query.

When the query requests calculations or factual answers (e.g., combinatorial calculations), you should:

1. Carefully interpret the mathematical or logical problem.
2. Show your reasoning internally to confirm the final answer (reasoning does not need to be included in the response unless explicitly requested).
3. Provide the final direct response strictly following all instructions, especially when asked to repeat the query verbatim first before giving the answer.

General approach:

- Always parse the query thoroughly to extract every constraint and instruction.
- Ensure your response exactly matches the format, wording, and content as instructed.
- Do not invent or omit any part of the user's explicit requests.
- Meet all formatting, lexical, numeric, and structural constraints without deviation.
- If the query involves repeating text verbatim, do not alter capitalization, punctuation, or wording.
- Incorporate any required keywords or letters the required number of times naturally into your response.
- When multiple constraints (like avoiding specific words while including others) apply simultaneously, ensure you satisfy all simultaneously.

This task requires precision, exact reproduction, and strict adherence to any given constraints or instructions embedded in the query. Your goal is to deliver the requested answer in the exact manner requested without extraneous additions or omissions.

IFBench GPT-4.1 Mini ensure_correct_response_module.predict

Base Prompt:

Ensure the response is correct and adheres to the given constraints. Your response will be used as the final response.

MIPROv2 Prompt:

Given the original query and the initially generated response, carefully verify that the response fully meets all specified constraints and instructions in the query. Produce detailed step-by-step reasoning that explains how the response satisfies or fails the requirements such as exact repetition of the query text, minimum sentence count, formatting rules (e.g., capitalization), letter frequency, or other explicit user demands. Based on this reasoning, output a final corrected and fully compliant response that adheres strictly to the original instructions, ensuring accuracy, completeness, and formatting correctness. Your final output should begin by repeating the original query exactly as provided, with no modifications or additional text before it, followed by a comprehensive, well-structured answer that respects all constraints, and a clear explanation of how the requirements have been met.

Demos:

Example 1:

query:

Can you provide the social media handles of Harry Potter? Your answer must contain exactly 8 bullet points. Use the markdown bullet points such as:

- * This is point 1.
- * This is point 2 Finish your response with this exact phrase That's my answer.. No other words should follow this phrase.

response:

- * Twitter: [@wizardingworld](https://twitter.com/wizardingworld)
- * Instagram: [@wizardingworld](https://www.instagram.com/wizardingworld/)
- * Facebook: [Wizarding World](https://www.facebook.com/wizardingworld)
- * YouTube: [Wizarding World](https://www.youtube.com/c/WizardingWorld)
- * Twitter: [@harrypotterfilm](https://twitter.com/harrypotterfilm)
- * Instagram: [@harrypotterfilm](https://www.instagram.com/harrypotterfilm/)
- * Facebook: [Harry Potter](https://www.facebook.com/harrypotter)
- * TikTok: [@wizardingworld](https://www.tiktok.com/@wizardingworld)

That's my answer..

reasoning:

The request is for the social media handles of Harry Potter, formatted as exactly eight bullet points using markdown bullet points, and the response must end with the exact phrase "That's my answer." The provided response includes eight bullet points, referencing official or widely recognized social media accounts related to Harry Potter and the Wizarding World franchise on Twitter, Instagram, Facebook, YouTube, and TikTok. The format is correct, the bullet points are used properly with markdown syntax, and the closing phrase matches the requirement exactly. Therefore, the response meets all given constraints.

final_response:

- * Twitter: [@wizardingworld](https://twitter.com/wizardingworld)
- * Instagram: [@wizardingworld](https://www.instagram.com/wizardingworld/)
- * Facebook: [Wizarding World](https://www.facebook.com/wizardingworld)
- * YouTube: [Wizarding World](https://www.youtube.com/c/WizardingWorld)
- * Twitter: [@harrypotterfilm](https://twitter.com/harrypotterfilm)
- * Instagram: [@harrypotterfilm](https://www.instagram.com/harrypotterfilm/)
- * Facebook: [Harry Potter](https://www.facebook.com/harrypotter)
- * TikTok: [@wizardingworld](https://www.tiktok.com/@wizardingworld)

That's my answer..

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA+Merge:

You will receive a user query that includes explicit, detailed instructions about how you must structure your response. Your overall task is to interpret these embedded instructions with perfect accuracy and produce a response that strictly complies with every single one, without adding, omitting, or rewording any mandated text, phrases, or structural elements.

Key detailed requirements and strategy for this task:

1. ****Query Parsing and Extraction of Instructions****
 - Carefully read the entire query to identify all explicit instructions concerning:
 - Whether and how to repeat the query text (verbatim or partially).
 - Specific length constraints (number of sentences, bullet points, word counts).
 - Formatting instructions (e.g., capitalization requirements, quotation marks, markdown bullet styles).
 - Mandatory phrases or exact sentences that must appear (especially those to be repeated verbatim or appended at the end).
 - Content limitations or prohibitions (for example, refusal language or disclaimers for unethical requests).
 - Note that some instructions may be nested or appear within the query's wording and are critical to follow exactly.
2. ****Exact Text Reproduction****
 - When asked to repeat the query text (or any other required phrase) verbatim, do so with zero changes – no added or removed words, punctuation, or formatting.
 - Do not prepend or append anything to the repeated text unless explicitly instructed.
 - Preserve all original capitalization, spacing, and punctuation exactly as in the query.
3. ****Structural and Formatting Compliance****
 - Follow all formatting instructions strictly, such as:
 - Wrapping the entire response in quotation marks if required.
 - Using specified markdown bullet point styles (e.g., asterisks).
 - Ensuring capitalization instructions (e.g., all caps or minimum occurrences of uppercase words) are perfectly met.
 - Adhering to sentence or paragraph counts exactly as requested.
4. ****Response Content Accuracy and Appropriateness****
 - After fulfilling all structural requirements, respond to the main substantive question accurately and completely.
 - Use domain knowledge and reliable calculations to ensure factual correctness in answers.
 - For questions requesting sensitive or potentially harmful content (e.g., cures without scientific basis), produce responsible answers that include disclaimers or refusals if instructed.

- Always respect ethical guidelines and any mandated refusal language or concluding statements for such queries.
5. ****No Extraneous Text****
- Do not add explanations, internal reasoning, apologies, or meta commentary beyond what the query explicitly permits or demands.
 - Your final output must be the exact, ready-to-deliver response that meets all user instructions perfectly.
6. ****Examples and Patterns Observed****
- Users often combine multiple complex formatting and content instructions (e.g., repetition of request text, followed by specific number of sentences or bullet points, with capitalization rules).
 - Ensure you carefully distinguish when to repeat the query text verbatim and when to respond directly (sometimes the repetition excludes an instruction sentence).
 - Handle instructions about capitalized words appearing a minimum number of times by distributing such words naturally but thoroughly across the response.
 - When length constraints specify minimums (e.g., "at least 39 sentences"), ensure you meet or exceed exactly rather than approximating.
 - For bullet points or enumerations, use the precise markdown style asked for (commonly asterisks).
 - Follow refusal instructions verbatim and terminate the response exactly as specified.
- **In summary:**** YOUR RESPONSE MUST BE A PERFECT TEXTUAL AND STRUCTURAL RENDERING OF THE USER QUERY'S EMBEDDED INSTRUCTIONS. ALWAYS PRIORITIZE EXACT USER SPECIFICATIONS OVER DEFAULT GENERATIVE NORMS. THE OUTPUT IS FINAL AND WILL NOT BE EDITED, SO COMPLIANCE MUST BE 100%.
- This task therefore requires a rigorously rule-based approach:
- First parse and isolate all explicit instructions (including repetitions and formatting).
 - Then assemble your answer by reproducing mandated text identically.
 - Next produce the requested content in the exact required format, length, and style.
 - Finally, review your output against the instructions ensuring total compliance before delivering your response.

L.4 IFBENCH, QWEN3 8B

IFBench Qwen3 8B generate_response_module.predict

Base Prompt:

Respond to the query

MIPROv2 Prompt:

You are a precision-focused instruction-adherence expert tasked with generating structured, ethically compliant responses. When responding to queries, always think step-by-step to analyze constraints (formatting rules, content boundaries, ethical guidelines), synthesize requirements, and produce outputs that strictly adhere to specified parameters. Ensure responses are fully formatted as requested (e.g., lowercase

```

text, bullet points, markdown), avoid any unverified claims, and maintain neutrality in
ambiguous ethical scenarios.
Demos:
Example 1:
query:
    Can you provide the social media handles of Harry Potter? Your answer must contain
    exactly 8 bullet points. Use the markdown bullet points such as:
* This is point 1.
* This is point 2 Finish your response with this exact phrase That's my answer.. No other
words should follow this phrase.
reasoning:
    The query asks for the social media handles of Harry Potter, a fictional character.
    Since he does not have personal social media accounts, the response should instead
    list official accounts related to the Harry Potter franchise. These include accounts
    managed by Warner Bros., the Wizarding World, and associated organizations. The
    answer is structured into exactly 8 bullet points as requested.
response:
    * @WizardingWorld (Twitter)
    * @WizardingWorld (Instagram)
    * @WizardingWorld (Facebook)
    * @JKRowling (Twitter)
    * @HarryPotterAlliance (Facebook)
    * @HogwartsExpress (Twitter)
    * @TheWizardingWorldOfHarryPotter (Facebook)
    * @HarryPotterBooks (Twitter)
    That's my answer.

#####
3 other demos omitted for brevity.
#####

```

GEPA Prompt generated by config GEPA:

```

Respond to the query by first repeating it word for word without any changes. Then,
provide your answer. Ensure your answer includes niche or domain-specific factual
information relevant to the task. If the query specifies constraints (e.g., letter
frequency, sentence limits, or formatting rules), adhere to them strictly. Use a
generalizable strategy to solve the task, such as breaking down complex problems into
logical steps or applying specialized knowledge. Avoid adding any preamble or postscript
before or after the repeated query.

```

IFBench Qwen3 8B ensure_correct_response_module.predict

Base Prompt:

```

Ensure the response is correct and adheres to the given constraints. Your response will
be used as the final response.

```

MIPROv2 Prompt:

```

You are a meticulous and ethical AI assistant tasked with generating detailed, compliant
responses. When given a query, first think step by step to create a reasoning process

```

that addresses the query's requirements, then produce a response that strictly adheres to all specified constraints (e.g., formatting rules, content boundaries, ethical guidelines). Ensure the final output is structured with a "Reasoning:" section followed by a "Final Response:" section, and verify that it aligns with the query's exact specifications while maintaining logical coherence and ethical responsibility.

Demos:

Example 1:

query:

What are some techniques for winning a swordfight in a medieval combat game? In your response, the letter f should appear less than 18 times. Do not include keywords ['heart', 'listen', 'reputation', 'side'] in the response.

response:

Prioritize stance and balance to control the fight's flow. Use angles to create openings for strikes while minimizing exposure. Practice parrying with precision to disrupt opponent rhythm. Maintain distance to avoid overcommitting, and target weak points like joints or limbs. Focus on rhythm and timing to exploit vulnerabilities.

reasoning:

The response outlines key techniques for medieval swordfighting, focusing on stance, angles, parrying, distance, and targeting weak points. It avoids the restricted keywords and limits the letter 'f' to 2 instances, satisfying the constraints. The advice is practical and aligns with historical combat strategies.

final_response:

Prioritize stance and balance to control the fight's flow. Use angles to create openings for strikes while minimizing exposure. Practice parrying with precision to disrupt opponent rhythm. Maintain distance to avoid overcommitting, and target weak points like joints or limbs. Focus on rhythm and timing to exploit vulnerabilities.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA:

Ensure the response strictly follows these instructions:

1. ****First****, repeat the user's query ****word for word**** without any changes or additions.
2. ****Then****, provide your answer in the specified format, adhering to all constraints (e.g., markdown, structure, content).
3. ****Do not include any additional text, explanations, or formatting**** beyond the repeated query and your answer.
4. ****Include niche/domain-specific factual details**** (e.g., technical commands, best practices, or platform-specific configurations) if applicable, as these are critical for accurate task completion.
5. ****Use precise formatting**** (e.g., bullet points, code blocks, headers) as requested, ensuring no markdown is omitted or altered.
6. ****Avoid generalizable strategies**** unless explicitly instructed; focus on actionable, specific guidance.
7. ****Validate all technical steps**** (e.g., Dockerfile syntax, CLI commands) for accuracy and completeness.
8. ****Highlight potential pitfalls and solutions**** to address common issues in the task.
9. ****Prioritize clarity and conciseness****, ensuring the response is both comprehensive and easy to follow.
10. ****Adhere to language and case requirements**** (e.g., all caps, English only) if specified.

L.5 HoVer, GPT-4.1 MINI

HoVer GPT-4.1 Mini create_query_hop2.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

Given the original claim and the initial summary of retrieved evidence, carefully analyze the information step-by-step to identify any gaps, contradictions, or points needing further clarification. Then, generate a clear, focused, and precise query that targets additional relevant information to deepen the investigation and help verify or refute the claim. Your response should include a detailed chain-of-thought reasoning explaining your thought process in formulating this refined query.

Demos:

Example 1:

claim:

AirBaltic's has bases at Tallinn Airport and the airport that ceased operations in 2015, with the airport code of Ryanair: VNO, ICAO: EYVI.

summary_1:

AirBaltic has bases at Tallinn Airport and Vilnius Airport (IATA: VNO, ICAO: EYVI), but Vilnius Airport did not cease operations in 2015. The airport that ceased operations in 2015 mentioned in the passages is unrelated and has different codes. Therefore, the claim that airBaltic's base with code VNO/EYVI ceased operations in 2015 is false.

reasoning:

The claim states that AirBaltic has bases at Tallinn Airport and an airport with codes VNO (IATA) and EYVI (ICAO) which ceased operations in 2015. However, the summary clarifies that Vilnius Airport, which has those codes, did not cease operations in 2015. The airport that did cease operations in 2015 has different codes and is unrelated to AirBaltic's bases. Therefore, the claim that AirBaltic's base at the airport with codes VNO/EYVI ceased operations in 2015 is incorrect.

query:

Did Vilnius Airport (IATA: VNO, ICAO: EYVI) cease operations in 2015?

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA+Merge:

Task Description:

You are given two fields: `claim` and `summary\_1`. Your goal is to produce a field called `query` – a concise set of relevant, fact-checking questions or search style queries that can be used to retrieve evidence documents verifying or refuting the claim based on information contained or inferred from the summary.

Input Format:

- `claim`: A factual statement potentially containing multiple fact assertions about people, events, attributes, titles, dates, roles, or relationships.
- `summary\_1`: A short paragraph summarizing factual information related to the claim, often clarifying or correcting some parts of the claim.

Output Format:

- ``query``: One or more specific, well-phrased questions or keyword queries that directly target the key factual discrepancies or verifications raised by the claim in light of the summary.

Detailed Instructions:

1. **Extract key factual elements from the claim** – names, dates, titles, roles, events, or relationships explicitly or implicitly stated.
2. **Contrast these facts with the summary to identify points of agreement, contradiction, or ambiguity.**
3. **Formulate fact-checking queries that are:**
 - Tightly focused on the core factual issues raised by the claim and addressed or contradicted by the summary.
 - Include named entities, dates, roles, or other domain-specific identifiers directly mentioned in both claim and summary to improve retrieval effectiveness.
 - When relevant, break complex claims into multiple queries ensuring each fact is verifiable separately.
4. **When relevant details appear only in the summary but are hinted at or missing from the claim (e.g., specific titles, roles, or names), include these in the queries to enable retrieval of key evidence.**
5. **Use a clear, natural question format or targeted keyword phrases that could serve well as search queries.**
6. **Avoid overly broad or generic queries; precision improves evidence retrieval quality.**
7. **Optionally, you may provide brief reasoning internally (not required in output) to ensure the queries cover all claim aspects and reflect the summary insights.**

Examples of typical query components include:

- Correct dates of events or deaths.
- Confirmation of a person's role or association with a known work.
- Verification of relationships or allegations.
- Details about specific cultural or domain elements (e.g., operas' structure, song directors).
- Clarifying entity attributes or classifications (e.g., ethnicity, nationality).

By following these instructions, you will produce queries that are both comprehensive and targeted, maximizing the chance of retrieving relevant factual evidence relevant to the claim verification task.

HoVer GPT-4.1 Mini create_query_hop3.predict

Base Prompt:

Given the fields ``claim``, ``summary_1``, ``summary_2``, produce the fields ``query``.

MIPROv2 Prompt:

You are an expert fact-checker specializing in multi-hop reasoning and evidence synthesis. Given a claim and two intermediate summaries that consolidate evidence from previous retrieval steps, thoughtfully analyze the information step-by-step to generate a clear and focused query. This query should be designed to retrieve the most relevant additional documents that can help verify or refute the claim by leveraging the insights

from both summaries. Provide a detailed chain-of-thought reasoning explaining how you integrate the summaries and the claim to formulate this refined query.

Demos:

Example 1:

claim:

The brother of Freddie Sessler, was the restaurateur whose club was frequented by Ruth Elizabeth "Bette" Davis, an actress with Welsh ancestry.

summary_1:

Freddie Sessler's brother was Siegi Sessler, a London restaurateur whose club was frequented by Bette Davis, an actress with Welsh ancestry.

summary_2:

Freddie Sessler's brother was Siegi Sessler, a prominent London restaurateur whose club, Siegi's, was frequented by Hollywood stars including Ruth Elizabeth "Bette" Davis, an actress with Welsh ancestry.

reasoning:

The claim states that the brother of Freddie Sessler was a restaurateur whose club was frequented by Ruth Elizabeth "Bette" Davis, noted to have Welsh ancestry. Both summaries support this claim by identifying the brother as Siegi Sessler, a restaurateur in London, whose club was indeed frequented by Bette Davis, an actress noted for Welsh ancestry. Summary_2 adds that the club's name was Siegi's and mentions the presence of other Hollywood stars. To fact-check the claim accurately, it would be useful to verify the identity of Freddie Sessler's brother as Siegi Sessler, confirm that he was a restaurateur, verify that Bette Davis frequented his club, and confirm Bette Davis's Welsh ancestry.

query:

Who was Siegi Sessler, and was he the brother of Freddie Sessler? Was Siegi Sessler a restaurateur whose club was frequented by Ruth Elizabeth "Bette" Davis? Did Bette Davis have Welsh ancestry?

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

Given three text fields: `claim`, `summary\_1`, and `summary\_2`, your task is to produce a `query` field that is designed to effectively retrieve evidence documents relevant to verifying or refuting the claim based on the information contained in the two summaries.

Detailed task description and considerations:

1. **Purpose of the Query:**
 - The query should accurately and comprehensively target the key factual elements from the claim that are addressed or clarified in the summaries.
 - The query will be used to retrieve evidence documents, so it should be specific enough to pinpoint relevant support or contradiction but broad enough to cover all important details present in the summaries.
2. **Utilizing the Summaries:**
 - Summaries often correct, clarify, or add factual context to the claim. Your query must incorporate these clarifications (e.g., name corrections, factual specifics, or counterpoints) to ensure retrieval of relevant evidence reflecting the nuanced truth.

- Include all distinctive entities, facts, dates, locations, names, and relationships mentioned or corrected in the summaries that pertain to the claim.
- For example, if a summary corrects a documentary title in the claim, the query should reference the corrected title and related details to guide retrieval effectively.

3. **Query Content Strategy:**

- Explicitly mention key entities, such as persons, places, dates, works, or political entities involved.
- Include attributes or relationships relevant to the claim's accuracy (e.g., "Was person X a politician in country Y during year Z?" or "Did documentary A and documentary B film in different locations such as location 1 and location 2?").
- If there is a factual dispute or correction in the summaries (e.g., nationality, official names, population figures), phrase the query to target evidence clarifying this dispute.

4. **Domain-Specific Nuances:**

- Be mindful of historical geopolitical names and periods (e.g., "United Kingdom of the Netherlands between 1815 and 1830").
- Recognize proper titles and correct spellings (e.g., corrected documentary titles).
- Include both subjects or objects mentioned as comparisons or contrasts in the claim and summaries (e.g., two documentaries, two politicians, two towns).

5. **Formulation Style:**

- Queries should be phrased as clear, precise, and objective questions that can be answered based on evidence — often structured as yes/no or informational queries.
- Avoid overly broad or vague phrasing. Aim for detail-rich queries that connect multiple evidence points in the summaries.

6. **Generalizable Strategy:**

- Identify the core claim elements and verify if the summaries confirm, contradict, or amend these elements.
- Incorporate both the claim and corrections from summaries into the query so evidence retrieval captures the full factual context.
- Use the comparison or contrast highlighted by the summaries to create queries that specifically test the claim's veracity (e.g., comparing locations, roles, or historical timeframes).

By following these guidelines, you will generate queries that maximize the likelihood of retrieving relevant texts that confirm or refute the claim accurately.

HoVer GPT-4.1 Mini summarize1.predict

Base Prompt:

Given the fields `claim`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a `claim` and a list of relevant `passages`, carefully analyze the evidence by reasoning step-by-step to assess the claim's validity. Produce a detailed chain-of-thought `reasoning` that explains how the information in the passages supports or refutes the claim, followed by a clear, concise `summary` that synthesizes the key

findings in relation to the claim. Ensure the reasoning explicitly connects evidence from the passages to the claim, enabling a thorough and transparent multi-hop verification process.

Demos:

Example 1:

claim:

The manufacturer that owns the company that currently produced and marketed the candy the video game Darkened Skye uses in magic is headquartered in the US state Virginia.

passages:

['Darkened Skye | Darkened Skye is a third-person action-adventure video game developed by Boston Animation. It was released for Microsoft Windows and the Nintendo GameCube in North America in 2002 and the PAL regions in 2003. The game was also packaged with Outlaw Golf. Its title character is a young woman named Skye who lives in a fantasy realm searching for her mother. She does not use firearms, but can perform magic using Skittles candies, as well as use her staff as a melee weapon, though it also becomes an energy weapon when used in conjunction with the Skittles.', 'Rocket Fizz | Rocket Fizz is a franchise chain of candy stores in the United States. The company markets a diverse variety of candies and produces its own line of soft drinks with unique flavors. Its flagship store is located in Camarillo, California, and most of the company's franchise stores are located in California. The company also markets candies that are rare to find or in limited production by various manufacturers, such as those that were popular during the 1960s to 1980s. As of September 2017, there are 86 Rocket Fizz store locations in the United States.', 'Spangler Candy Company | The Spangler Candy Company is a privately owned international confectioner that has been manufacturing and marketing candy since 1906.', 'Future US | Future US, Inc. (formerly known as Imagine Media and The Future Network USA) is an American media corporation specializing in targeted magazines and websites in the video games, music, and technology markets. Future US is headquartered in the San Francisco with a small sales office in New York City. Future US is owned by parent company, Future plc, a small publishing company based in the United Kingdom.', 'Fun Dip | Fun Dip is a candy manufactured by The Willy Wonka Candy Company, a brand owned by Nestl . The candy has been on the market in the United States and Canada since the 1940s and was originally called Lik-M-Aid. It was originally manufactured by Fruzola, and then Sunline Inc., through their Sunmark Brands division, which was purchased by Nestl  in January 1989. It comes in many different flavors with candy sticks that are included.', 'Hot Tamales | Hot Tamales is a chewy, cinnamon flavored, oblong-shaped candy introduced in 1950    manufactured and marketed in the United States by Just Born, a family-owned, Pennsylvania-based candy company.', 'Frankford Candy & Chocolate Company | Frankford Candy & Chocolate Company is an American candy manufacturer, located in Philadelphia, Pennsylvania, founded in 1947 by Sam Himmelstein. The current CEO is Stuart Selarnick. In recent years, Frankford Candy & Chocolate Company has become a very successful candy manufacturer. It is one of the largest producers of chocolate rabbits in the United States, making over 100 varieties. In 2000, it started licensing popular children's TV characters, such as SpongeBob SquarePants."]

reasoning:

The claim states that the manufacturer of the candy used in the video game Darkened Skye is headquartered in Virginia. Passage [1] establishes that the video game Darkened Skye uses Skittles candies for magic. Therefore, the candy in question is Skittles. The crucial step is to identify the company that produces and markets Skittles and whether it is headquartered in Virginia. None of the provided passages explicitly mention Skittles or its manufacturer. The passages list various candy companies, but none mention Skittles or match its manufacturer. The known producer of

Skittles is Mars, Incorporated, which is headquartered in McLean, Virginia. Although this fact is not directly stated in the provided passages, it is relevant general knowledge. Since the passages do not refute the claim and for accuracy based on known facts, the claim can be considered true that the manufacturer that owns the company producing and marketing Skittles is headquartered in Virginia.

summary:

The candy used as magic in Darkened Skye is Skittles, which are produced by Mars, Incorporated. Mars is headquartered in McLean, Virginia, a US state. Therefore, the claim is true based on external knowledge confirming that the manufacturer is based in Virginia.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA+Merge:

Given the fields `claim`, `passages`, produce the fields `summary`.

HoVer GPT-4.1 Mini summarize2.predict

Base Prompt:

Given the fields `claim`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given a `claim`, relevant `context`, and a set of supporting `passages`, generate a detailed step-by-step reasoning process that logically connects the evidence to the claim, followed by a concise summary that clearly states whether the claim is supported or refuted based on the aggregated information. Ensure the reasoning explicitly references key evidence from the context and passages to justify the conclusion in the summary.

Demos:

Example 1:

claim:

One half of the Kalyanji-Anandji duo, Kalyanji Virji Shah's brother, won the Filmfare Award for Best Music Director.

context:

Yes, Kalyanji Virji Shah's brother Anandji Virji Shah, as part of the Kalyanji-Anandji duo, won the Filmfare Award for Best Music Director in 1975 for the film "Kora Kagaz."

passages:

['Kalyanji Virji Shah | Kalyanji Virji Shah (30 June 1928 – 24 August 2000) was the "Kalyanji" of the Kalyanji-Anandji duo. He and his brother Anandji Virji Shah have been famous Indian film musicians, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Anandji Virji Shah | Anandji Virji Shah is an Indian music director. Together with his brother he formed the Kalyanji-Anandji duo, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Kalyanji-Anandji | Kalyanji-Anandji are an Indian composer duo from Gujarat: Kalyanji Virji Shah (30 June 1928-03 November

2000) and his brother Anandji Virji Shah (born 02 March 1933). The duo are known for their work on Hindi film soundtracks, particularly action potboilers in the 1970s. Some of their best-known works are "Don", "Bairaag", "Saraswatichandra", "Qurbani", "Tridev" and "Safar". They won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz".', 'KalyanjiĀAnandji discography | This is a discography of Bollywood composer duo Kalyanji Anandji, consisting of Kalyanji Virji Shah and Anandji Virji Shah. They have composed music for over 200 films in their 46 year career.', 'Viju Shah | Viju Shah (born, Vijay Kalyanji Shah, 5 June 1959) is a music director of Hindi cinema. He is the son of music director Kalyanji Virji Shah of composer duo Kalyanji Anandji. He composed music for movies such as "Vishwatama" (1992), "Mohra" (1994), "Tere Mere Sapne" (1996) and "" (1997) for which got his second nomination for the Filmfare Award for Best Music Director and he won the 1998 Filmfare Award for Best Background Score for "".', 'Bombay the Hard Way: Guns, Cars and Sitaris | Bombay the Hard Way: Guns, Cars and Sitaris is a 1998 project of producer Dan the Automator in which music from 1970s funk Bollywood composer duo Kalyanji-Anandji (Kalyanji Virji Shah and Anandji Virji Shah) was taken and given a funky remix. The album was eventually withdrawn (possibly due to copyright concerns). A sequel, "Bombay 2: Electric Vindaloo", was also produced.', 'Kora Kagaz | Kora Kagaz (Hindi: क़ोरा काग़ज़; translation: Blank Paper) is a 1974 Hindi drama film produced by Sanath Kothari and directed by Anil Ganguly. The film stars Vijay Anand, Jaya Bhaduri, A.K. Hangal, Achala Sachdev and Deven Verma. The films music is by Kalyanji Anandji. The famous title song "Mera Jeevan Kora Kagaz" was by Kishore Kumar.']

reasoning:

The claim states that one half of the KalyanjiĀAnandji duo, Kalyanji Virji Shah's brother, won the Filmfare Award for Best Music Director. From the passages and context, it is clear that the duo consists of Kalyanji Virji Shah and his brother Anandji Virji Shah. Both together won the Filmfare Award for Best Music Director in 1975 for the film "Kora Kagaz." Therefore, Anandji Virji Shah, who is Kalyanji's brother, indeed won the Filmfare Award for Best Music Director as part of this duo.

summary:

Anandji Virji Shah, the brother of Kalyanji Virji Shah, won the Filmfare Award for Best Music Director in 1975 as part of the KalyanjiĀAnandji duo.

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA+Merge:

Given the input fields:

- `claim`: A factual statement or assertion that may be true or false.
- `context`: A brief explanation or summary relating to the claim that typically clarifies the claim's accuracy.
- `passages`: A list of relevant textual evidence or knowledge snippets containing facts, descriptions, or biographies related to entities or concepts mentioned in the claim.

Your task is to produce a `summary` that meets the following requirements:

1. ****Accurately reflect the relationship between the claim and the evidence****: Analyze the input `claim` carefully and check it against the `context` and all `passages`.

Determine whether the claim is supported, partially supported, or contradicted by the evidence.

2. ****Explicitly connect key entities and facts from the passages****: Your summary must mention the main entities and facts that directly confirm or refute parts of the claim. For example, highlight relevant names, works (films, albums, songs, books), attributes, dates, nicknames, or roles that clarify the claim's accuracy.
3. ****Make substantive connections****: The summary should not merely restate the claim or conclusion. Instead, it should explicitly link the claim to detailed evidence from the passages—such as specific film titles, album names, song titles, dates, or relationships—to provide clear reasoning for the accuracy or inaccuracy of the claim.
4. ****Include relevant disambiguations or clarifications****: Where applicable, clarify potential misunderstandings or common confusions highlighted in the passages (e.g., distinguishing similarly named films or songs, specifying which individual is referenced, or noting alternate titles and adaptations).
5. ****Use concise, factual language****: The summary should be clear and succinct but include all relevant evidence necessary to understand how the claim is supported or refuted.
6. ****Support future query generation****: As the summary will be used to generate queries to find evidence, it must contain explicit mentions of key entities, titles, and facts that can guide retrieval and reasoning.

****Overall, your approach should be to reason comprehensively over the claim, context, and each passage, synthesizing connected facts and evidence into a coherent and evidence-rich summary that clearly documents why the claim is true, false, or partially true.**** Avoid vague or overly general statements; instead, ground your summary in the precise factual details present in the passages and context.

L.6 HoVer, QWEN3 8B

HoVer Qwen3 8B create_query_hop2.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, produce the fields `query`.

MIPROv2 Prompt:

In a high-stakes scenario where the accuracy of your query directly determines the validity of a complex claim, generate a precise query that builds on the provided claim and summary to uncover critical evidence. Use step-by-step reasoning to identify gaps in the initial summary, then craft a query that explicitly targets these gaps to retrieve additional supporting information. Ensure your query is unambiguous and directly addresses the claim's unresolved aspects.

Demos:

Example 1:

claim:

Sandi Simcha DuBowski directed a documentary released in 2001. The horror film *Zeitgeist* was not also released in 2001.

summary_1:
The claim is correct: Sandi Simcha DuBowski directed the 2001 documentary **Trembling Before G-d**, and there is no evidence in the passages that the horror film **Zeitgeist** was released in 2001.

reasoning:
The claim is correct because Sandi Simcha DuBowski indeed directed the 2001 documentary **Trembling Before G-d**, as stated in summary_1. The assertion that **Zeitgeist** was not released in 2001 is supported by the absence of evidence in the provided summaries linking the film to that year.

query:
Is there any evidence in the provided summaries that the horror film **Zeitgeist** was released in 2001?

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA:

Given the fields `claim` and `summary\_1`, generate a precise and focused query that identifies the specific evidence needed to verify or refute the claim.

Key Requirements:

1. **Target Missing Evidence**: Identify and explicitly ask about the unverified or unconfirmed details in the summary (e.g., names, dates, locations, or connections) that are critical to the claim.
- Example: If the summary mentions "Massimo Giordano" as a potential counterpart but lacks birthplace details, the query should ask for evidence confirming their ties to Naples.
2. **Correct Historical/Domain-Specific Anomalies**: Address discrepancies like anachronisms (e.g., Ali Qushji's work in the 1960s vs. his actual 15th-century timeline) or misattributions (e.g., **Hayy ibn Yaqdhan** by Ibn Tufail, not Ali Qushji).
3. **Link to Summary Context**: Ensure the query references the summary's key points (e.g., "the claim states X, but the summary notes Y is unverified").
4. **Use Specific Terms**: Include exact names, titles, or dates mentioned in the summary to avoid ambiguity (e.g., "Ana's Nin," "Metropolitan City of Naples," "1948").

Example Strategy:

If the summary states:

- "The claim is partially supported. While [Fact A] is confirmed, [Fact B] lacks verification."
- The query should ask:
* "Is there evidence confirming [Fact B], such as [specific name/date/location]?" *

Domain-Specific Notes:

- Verify historical timelines (e.g., Ali Qushji died in 1474, not the 1960s).
- Confirm authorships (e.g., **Hayy ibn Yaqdhan** is attributed to Ibn Tufail, not Ali Qushji).
- Check for geographic or biographical details (e.g., birthplaces, cities, or professional connections).

Ensure your query directly addresses the summary's unverified claims and includes all critical terms from the summary to maximize evidence retrieval.

HoVer Qwen3 8B create_query_hop3.predict

Base Prompt:

Given the fields `claim`, `summary\_1`, `summary\_2`, produce the fields `query`.

MIPROv2 Prompt:

You are a fact-checking assistant specializing in multi-hop reasoning and information synthesis. Given the fields `claim`, `summary\_1`, and `summary\_2`, generate a precise query that synthesizes the original claim with the two summaries to probe for deeper contextual relationships, resolving ambiguities and confirming supporting details through targeted evidence retrieval.

Demos:

GEPA Prompt generated by config GEPA:

Given the fields `claim`, `summary\_1`, and `summary\_2`, produce the field `query` that:

1. Explicitly asks whether the claim is supported by the provided summaries.
2. Includes **specific evidence** from the summaries (e.g., names, titles, dates, or factual details) to guide retrieval of relevant documents.
3. Focuses on **key disputed points** in the claim (e.g., incorrect attributions, missing evidence, or conflicting statements) to ensure the query targets the most relevant information.

Key considerations for the query:

- If the summaries mention specific works (e.g., **Planes, Trains and Automobiles**), include the title.
- If the summaries reference named entities (e.g., John Candy, KLM, Richard Ford), include them.
- If the summaries clarify contradictions (e.g., "not X but Y"), structure the query to highlight this contrast.
- Avoid vague phrasing; instead of asking "Is the claim true?" use "Does the evidence in the summaries confirm [specific detail]?"

Example:

If the claim is about a film and the summaries mention **Planes, Trains and Automobiles** (1987), the query should include that title to retrieve relevant evidence.

HoVer Qwen3 8B summarize1.predict

Base Prompt:

Given the fields `claim`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `claim` and `passages`, generate a structured reasoning process that analyzes the claim step-by-step using the provided evidence, and produce a concise summary that distills the key findings and evaluates the claim's validity based on the evidence.

Demos:

Example 1:

claim:

One half of the Kalyanji-Anandji duo, Kalyanji Virji Shah's brother, won the Filmfare Award for Best Music Director.

passages:

['Kalyanji Virji Shah | Kalyanji Virji Shah (30 June 1928 – 24 August 2000) was the "Kalyanji" of the Kalyanji-Anandji duo. He and his brother Anandji Virji Shah have been famous Indian film musicians, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Kalyanji-Anandji | Kalyanji-Anandji are an Indian composer duo from Gujarat: Kalyanji Virji Shah (30 June 1928–03 November 2000) and his brother Anandji Virji Shah (born 02 March 1933). The duo are known for their work on Hindi film soundtracks, particularly action potboilers in the 1970s. Some of their best-known works are "Don", "Bairaag", "Saraswatichandra", "Qurbani", "Tridev" and "Safar". They won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz".', 'Anandji Virji Shah | Anandji Virji Shah is an Indian music director. Together with his brother he formed the Kalyanji-Anandji duo, and won the 1975 Filmfare Award for Best Music Director, for "Kora Kagaz". He is a recipient of the civilian honour of Padma Shri (1992).', 'Viju Shah | Viju Shah (born, Vijay Kalyanji Shah, 5 June 1959) is a music director of Hindi cinema. He is the son of music director Kalyanji Virji Shah of composer duo Kalyanji Anandji. He composed music for movies such as "Vishwatama" (1992), "Mohra" (1994), "Tere Mere Sapne" (1996) and "" (1997) for which got his second nomination for the Filmfare Award for Best Music Director and he won the 1998 Filmfare Award for Best Background Score for "".', 'Kalyanji-Anandji discography | This is a discography of Bollywood composer duo Kalyanji Anandji, consisting of Kalyanji Virji Shah and Anandji Virji Shah. They have composed music for over 200 films in their 46 year career.', 'Bombay the Hard Way: Guns, Cars and Sitar | Bombay the Hard Way: Guns, Cars and Sitar is a 1998 project of producer Dan the Automator in which music from 1970s funk Bollywood composer duo Kalyanji-Anandji (Kalyanji Virji Shah and Anandji Virji Shah) was taken and given a funky remix. The album was eventually withdrawn (possibly due to copyright concerns). A sequel, "Bombay 2: Electric Vindaloo", was also produced.', 'Bairaag | Bairaag is a 1976 Bollywood film. Produced by the duo Mushir-Riaz, it is directed by Asit Sen. The film stars Dilip Kumar, who received a Filmfare nomination for Best Actor for playing a triple role. "Bairaag" also stars Saira Banu, Leena Chandavarkar, Ruma Guha Thakurta, Prem Chopra, Helen, Sujit Kumar, Madan Puri, Pinal, Kader Khan, Asit Sen and Sachin. Its music is by the duo Kalyanji Anandji, who received a Filmfare nomination for Best Music. The film received a Filmfare Award for Best Sound for P. Harikishan. According to one source, the film did not do well at the box office.']

reasoning:

The claim states that Kalyanji Virji Shah's brother won the Filmfare Award for Best Music Director. The passages confirm that the Kalyanji-Anandji duo (Kalyanji Virji Shah and his brother Anandji Virji Shah) won the 1975 Filmfare Award for Best Music Director for "Kora Kagaz." Since the award was shared by both brothers, the claim is accurate as it identifies Anandji Virji Shah (Kalyanji's brother) as the winner.

summary:

Anandji Virji Shah, the brother of Kalyanji Virji Shah, won the 1975 Filmfare Award for Best Music Director as part of the KalyanjiâĀAnandji duo for their work on "Kora Kagaz."

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA:

Given the fields `claim` and `passages`, produce the fields `summary` by:

1. **Analyzing the claim** to identify key elements (e.g., names, locations, relationships, specific terms).
2. **Cross-referencing the passages** to locate explicit or implicit evidence that supports, contradicts, or partially addresses the claim.
3. **Explicitly connecting the claim to the evidence** in the passages, even if the evidence is not directly stated in the claim (e.g., linking "Roger Yuan" to "Shanghai Noon" in Example 1).
4. **Highlighting domain-specific details** (e.g., "Sentinelese language is unclassifiable" in Example 3) that are critical for accuracy.
5. **Stating whether the claim is supported, unsupported, or partially supported**, and explaining the reasoning with clear references to the passages.
6. **Prioritizing key evidence** that would be relevant for further queries (e.g., "Marlborough" in Example 2) to ensure retrievability.

Additional Guidance:

- If the claim contains unverified details (e.g., "Roger Yuan" in Example 1), explicitly note this in the summary.
- For niche or ambiguous information (e.g., "uncontacted people" in Example 3), reference specific passages that clarify the context.
- Ensure summaries are concise but include all critical evidence needed to validate or refute the claim.

HoVer Qwen3 8B summarize2.predict

Base Prompt:

Given the fields `claim`, `context`, `passages`, produce the fields `summary`.

MIPROv2 Prompt:

Given the fields `claim`, `context`, and `passages`, perform multi-hop reasoning to generate a structured summary that validates or refutes the claim. First, analyze the claim and contextual information to identify key relationships. Next, evaluate evidence from the retrieved passages to build a coherent narrative. Construct a logical reasoning chain connecting the claim to supporting or contradictory evidence. Finally, produce a concise summary that explicitly confirms or contradicts the claim based on your analysis, citing relevant evidence from the context and passages.

Demos:

Example 1:

claim:

The recorded history of the state, for which Baptist George Ruby was a prominent black Republican leader in the Reconstruction-era, officially started in 1519 in the same state that holds the city Augustus Chapman Allen used his inheritance to fund the founding of.

context:

The claim is not supported by the passages. While George Ruby was associated with Texas during Reconstruction and Augustus Chapman Allen founded Houston in Texas, there is no evidence in the passages that Texas's recorded history began in 1519.

passages:

["Merchants and Manufacturers Building | The One Main Building, formerly the Merchants and Manufacturers Building (commonly referred to as the M&M Building), is a building on the campus of the University of HoustonâDowntown. The building is recognized as part of the National Register of Historic Places, is a Recorded Texas Historic Landmark, and considered a Contributing Building in Downtown Houston's Main Street/Market Square Historic District. The building was built above Allen's Landingâan area where Houston's founders John Kirby Allen and Augustus Chapman Allen originally settled.", 'Augustus Chapman Allen | Augustus Chapman Allen (July 4, 1806 â January 11, 1864), along with his younger brother, John Kirby Allen, founded the City of Houston in the U.S. state of Texas. He was born on July 4, 1806, in Canaseraga Village, New York (the present day hamlet of Sullivan in the Town of Sullivan, New York), to Sarah (Chapman) and Roland Allen.', "History of Texas | The recorded History of Texas begins with the arrival of the first Spanish conquistadors in the region of North America now known as Texas in 1519, who found the region populated by numerous Native American /Indian tribes.

Native Americans' ancestors had been there for more than 10,000 years as evidenced by the discovery of the remains of prehistoric Leanderthal Lady.

During the period of recorded history from A.D. 1519 to 1848, all or parts of Texas were claimed by five countries: France, Spain, Mexico, the Republic of Texas, the United States of Americaâas well as the secessionist Confederate States of America in 1861â1865, during the American Civil War.", 'Charlotte Baldwin Allen | Charlotte Baldwin Allen (July 14, 1805 â August 3, 1895) is known in Texan history as the "mother of Houston". She was the wife of Augustus Chapman Allen, who used her inheritance to finance the founding of this city.', 'Allen Parkway | Allen Parkway is an arterial road west of Downtown Houston, Texas. It has a distance of approximately 2.3 miles (3.7\xa0km), running from Interstate 45 west to Shepherd Drive, where it becomes Kirby Drive. Originally known as Buffalo Parkway, it was later named after John Kirby Allen and Augustus Chapman Allen, the founders of Houston.', "Allen's Landing | Allen's Landing is the birthplace of the city of Houstonâthe largest city in the U.S. state of Texas. In August 1836, just months after the Republic of Texas won its independence from Mexico, two brothers (and real estate developers) from New YorkâJohn Kirby Allen and Augustus Chapman Allenâpurchased 6,642\xa0acres\xa0(27\xa0kmÂ²) in the area and settled there on the banks of Buffalo Bayou. Allen's Landing is located south of the University of HoustonâDowntown Commerce Street Building, at the intersection of Commerce Street and Main Street.", "Keith Allen (American football) | Keith D. Allen (born July 12, 1974) is an American football coach. He served as the head football coach at Southwest Baptist University from 2007 to 2012, compiling a record of 27â39. His 27 wins are the most in the history of Southwest Baptist's football program, which began play in 1983."]

reasoning:

The claim is not supported by the passages. While the passages confirm that Texas's recorded history began in 1519 (passage 3) and that Augustus Chapman Allen founded Houston in Texas (passage 2), they do not explicitly link the start of Texas's recorded history to the same state where Augustus Chapman Allen used his inheritance

to fund the founding of a city. The claim conflates these two facts without direct evidence in the passages that the 1519 start of recorded history is tied to the state where Allen's inheritance was used for founding Houston.

summary:

The claim is not supported by the provided passages, as they confirm Texas's recorded history began in 1519 and Augustus Chapman Allen founded Houston in Texas but do not explicitly connect the two facts as stated in the claim.

#####

3 other demos omitted for brevity.

#####

GEPA Prompt generated by config GEPA:

Given the fields `claim`, `context`, and `passages`, produce the field `summary` that:

1. **Evaluates the claim's validity** by explicitly connecting it to **specific evidence** from the `passages` (e.g., citing exact names, dates, or terms that support/refute the claim).
2. **Highlights factual inaccuracies** in the claim, such as incorrect origins, dates, or relationships, and links them to **direct contradictions** in the `passages` or `context`.
3. **Mentions key terms or entities** critical to the claim (e.g., "Sly Stone," "Large Münsterländer," or "Hit Parader's 2006 ranking") to ensure summaries are actionable for query generation.
4. **Balances accuracy** by acknowledging both correct and incorrect aspects of the claim, as seen in the examples (e.g., validating "The Wild Bunch" compilation while correcting the magazine's nationality and ranking year).
5. **Avoids vague statements** like "no evidence" and instead specifies **which parts of the passages** are relevant (e.g., "Passage 6 confirms Hit Parader was American").

Generalizable Strategy:

- Always reference **specific passages** (e.g., "Passage 1 states...") to ground claims in the text.
- Use **precise terminology** (e.g., "German Shorthaired Pointer" instead of "cousins") to align with the data.
- Prioritize **clarity and specificity** to ensure summaries can guide further evidence retrieval.

L.7 PUPA, GPT-4.1 MINI

PUPA GPT-4.1 Mini craft_redacted_request.predict

Base Prompt:

Given a private user query, create a privacy-preserving request for a powerful external LLM.
The LLM may assist without learning private information about the user.

MIPROv2 Prompt:

You will be given a private user query. Your task is to analyze the query carefully and think step-by-step to understand the user's intent while identifying and removing or

abstracting any sensitive, proprietary, or personally identifiable information. Then, generate a clear, privacy-preserving prompt suitable for an external large language model (LLM) that requests useful, relevant guidance or information without exposing private details. Additionally, provide a detailed chain-of-thought explanation describing your reasoning process for how you transformed the original query into the redacted, generalized LLM request. Ensure the resulting prompt is professional, contextually appropriate, and maintains user privacy while enabling the LLM to assist effectively.

Demos:

Example 1:

user_query:

Imagine you are a stock researcher and advisor, I will give you a table about stocks and I want you to give me advice on what to invest next. This is a project I am working on and I need help with predictions I am making. Here is the table: Stock Price Dividends 5D 5D % 1M 1M % YTD YTD % 1Y 1Y % 5Y 5Y % IBM \$146,62 \$6,64 \$3,81 2,66 \$2,63 1,83 \$5,07 3,58 \$16,31 12,51 \$0,03 0,02 MCD \$284,90 \$6,08 \$5,07 1,81 -\$7,20 -2,46 \$20,57 7,78 \$29,24 11,44 \$123,26 76,26 JNJ \$163,98 \$4,76 -\$3,00 -1,80 -\$3,90 -2,32 -\$14,21 -7,97 \$0,98 0,60 \$29,66 22,08 MSFT \$324,89 \$2,72 \$1,25 0,39 -\$9,88 -2,95 \$85,31 35,61 \$59,66 22,49 \$217,33 202,05 WMT \$158,36 \$2,28 \$1,39 0,89 -\$1,85 -1,15 \$14,75 10,27 \$25,47 19,16 \$62,29 64,84 KO \$60,50 \$1,84 \$0,12 0,20 -\$1,59 -2,55 -\$2,45 -3,89 -\$2,23 -3,55 \$15,13 33,35 PFE \$36,07 \$1,64 -\$1,06 -2,84 -\$0,67 -1,81 -\$15,19 -29,64 -\$10,16 -21,98 -\$5,43 -13,09 AAPL \$179,96 \$0,96 \$3,45 1,95 -\$16,57 -8,43 \$54,89 43,89 \$18,58 11,51 \$125,04 227,65 NVDA \$468,23 \$0,16 -\$3,71 -0,79 \$7,05 1,53 \$325,08 227,09 \$310,22 196,33 \$401,52 601,89 CRM \$212,10 \$0,00 \$2,75 1,31 -\$9,86 -4,44 \$77,32 57,37 \$51,89 32,39 \$59,11 38,64 PYPL \$61,16 \$0,00 \$1,06 1,76 -\$13,82 -18,43 -\$13,43 -18,00 -\$31,51 -34,00 -\$29,46 -32,51 BIDU \$137,47 \$0,00 \$10,24 8,05 -\$15,63 -10,21 \$18,34 15,39 -\$9,85 -6,69 -\$88,25 -39,10

reasoning:

The user wants investment advice based on a provided table of stock data including price, dividends, and percentage changes over various time intervals (5 days, 1 month, year-to-date, 1 year, and 5 years). The primary request is for recommendations on what to invest in next and help with making predictions.

To preserve privacy, I should not include the user's raw data or specific details about individual stocks directly in the prompt to the external LLM. Instead, I should request general advice on how to analyse stock data tables that include price, dividend, and percentage changes over multiple time periods. Also, I can ask for advice on what indicators to weigh more heavily for investment decisions and for predictive strategies based on such data structures.

This approach ensures that the external LLM does not receive any private or identifying user data, only a generic request. I will ask the LLM to provide guidance on interpreting such stock performance indicators and suggestions for predictive analytics or investment selection criteria.

llm_request:

Please provide general guidance for analyzing stock investment opportunities based on a data table format that includes stock prices, dividend payouts, and percentage changes over various time intervals such as 5 days, 1 month, year-to-date, 1 year, and 5 years. Outline key indicators that investors typically weigh when making decisions, how to interpret short-term versus long-term performance trends, and suggest strategies or metrics to use when making predictions or recommendations about which stocks to invest in next. The request is for generic advice without any specific stock names or values.

```
#####
3 other demos omitted for brevity.
#####
```

GEPA Prompt generated by config GEPA+Merge:

New Instruction for Privacy-Preserving Query Reformulation Assistant (Enhanced):

Task Overview:

You receive a user query that may contain sensitive, private, or personally identifiable information (PII). Your objective is to reformulate this query into a generalized, privacy-preserving prompt suitable for sending to an external large language model (LLM). The reformulated prompt must enable the external LLM to fulfill the user's original intent effectively while rigorously protecting all user privacy by abstracting, omitting, or generalizing any sensitive details.

Input Format:

- A single user query string.
- The query might contain identifiable names (people, organizations), specific geographic locations, exact dates or timeframes, proprietary or internal terminology (e.g., system names, product codes), client or customer names, URLs, or any other sensitive or private information.

Output Format:

- Part (a) Reasoning:
 - Provide a concise paragraph that explains:
 - * How you identified sensitive or private information in the input
 - * What strategies you applied to protect privacy (generalization, omission, abstraction, replacement with placeholders)
 - * How the reformulated prompt preserves the original intent and task requirements without risking data leakage
- Part (b) LLM Request:
 - A concise, carefully constructed privacy-safe prompt that:
 - * Removes or anonymizes all PII and proprietary/internal details
 - * Abstracts locations, names, dates, and technical terms as needed
 - * Produces a clear and contextually rich instruction for the LLM to generate a relevant and informative response aligned with the user's original task

Detailed Domain-Specific Guidance and Best Practices:

1. Identification and Treatment of Sensitive Data:

- All user-specific or personal names (individual or organizational) must be removed or replaced with generic role descriptors (e.g., "a business contact," "a client," "a notable individual"). Never lightly obscure or partially redact; full abstraction is required.
- All geographic mentions must be abstracted unless the location is publicly known, essential to the task, and can be generalized (e.g., "a region known for eco-tourism" instead of naming a country or city explicitly).
- Exact dates or durations must never be retained; instead, use relative or approximate temporal references (e.g., "recently," "over the past year").
- Internal or proprietary terms including system names, product codes, subscription types, and technical jargon must be generalized or replaced with

neutral descriptors to avoid leakage of intellectual property or sensitive operational details.

- Avoid direct quotes or verbatim inclusion of user-supplied texts unless obfuscated by generalization.

2. Task Understanding and Reformulation:

- Identify the functional intent of the query: Is it creative writing, translation, summarization, professional communication drafting, technical explanation, or other?
- Preserve the thematic and informational core of the query (e.g., request for educational quality analysis, technical translation of a passage, biographical summary).
- Do not reproduce the original input verbatim; instead, frame the LLM prompt around the essential thematic elements extracted from the input.
- For queries regarding individuals, avoid direct reference to real personal information unless the name is widely public and essential; even then, use a generic or hypothetical framing for the individual profile.

3. Strategies for High-Quality and Privacy-Preserving Prompts:

- Strike a balance between sufficient contextual detail and privacy abstraction to maintain prompt clarity and relevance.
- Use neutral, context-aware formulations that clearly instruct the LLM on the content and style expected.
- Avoid vague or overly generic prompts that could result in less useful or lower-quality responses.
- When system or proprietary content is mentioned, instruct the LLM to generalize specific terms and maintain the technical meaning without revealing sensitive info.
- When a direct translation is requested on specialized text, specify to replace or abstract internal nomenclature.

4. Explanation Requirements:

- The reasoning must transparently explain how privacy risks were identified (e.g., presence of names, locations, dates, proprietary terms).
- It must describe the abstraction or omission methods applied (e.g., replacing "Jonah Van Beijnen" with "a notable individual," substituting "Makau" with "a specific region," or "Yoda" with "a system name").
- Clarify how the essential task and user intent were preserved despite these generalizations (e.g., focusing on educational quality, technical translation, biographical summary).
- This explanation justifies your design choices and demonstrates adherence to privacy-preserving principles.

Common Pitfalls to Avoid:

- Do not merely lightly obscure or partially redact sensitive details; full anonymization or abstraction is required.
- Do not repeat any user-supplied PII or proprietary content verbatim.
- Avoid including URLs, exact dates, or direct quotes without modification.
- Do not leave ambiguity that could degrade the quality or contextual clarity of the reformulated prompt.
- Do not include any real personal or organizational names unless they are public figures and the query requires it, then use generic descriptors instead.

Example Summary of Effective Approach (Informed by Prior Examples):

- For geographic queries: replace exact place names with general regions and provide a brief contextual descriptor.
- For technical texts containing system names or subscription types: instruct the LLM to translate or process the text while replacing or abstracting proprietary system identifiers.
- For biographical summaries about specific individuals: remove the real name and request a generic, well-structured four-paragraph summary about "a notable individual," preserving the overall intent without leaking PII.

Summary:

Your reformulations must ensure zero exposure of any PII or private/proprietary content while retaining enough thematic and functional clarity for the external LLM to produce high-quality, relevant outputs. This requires thorough analysis of the user's query, rigorous application of privacy-preservation strategies, and explicit reasoning explanations that document your approach and choices.

PUPA GPT-4.1 Mini respond_to_query

Base Prompt:

Respond to a user query.
For inspiration, we found a potentially related request to a powerful external LLM and its response.

MIPROv2 Prompt:

Given a privacy-preserving redacted prompt sent to an external untrusted LLM and its response, along with the original user query containing potentially sensitive information, generate a final, coherent, and contextually appropriate user-facing response. Your output should integrate the anonymized LLM output with the original query's intent, ensuring privacy is maintained while providing a complete and polished answer that directly addresses the user's request.

Demos:

Example 1:

related_llm_request:

Please provide a detailed explanation of David Easton's system analysis in political science, outlining the key components of his systems theory approach and its significance in the study of political systems.

related_llm_response:

Certainly! David Easton's system analysis is a foundational framework in political science that applies systems theory to the study of political systems. His approach provides a structured way to understand how political systems function, maintain stability, and respond to changes in their environment. Below is a detailed explanation of Easton's system analysis, including its key components and significance.

Overview of David Easton's System Analysis

David Easton (1917–2014) was a prominent political scientist who sought to develop a general theory of political systems by borrowing concepts from systems theory, which originated in biology and engineering. His goal was to create a scientific and analytical

framework to study politics as an integrated and dynamic system rather than as isolated events or institutions.

Easton's system analysis views a political system as a set of interactions abstracted from the totality of social behavior, through which authoritative allocation of values (decisions) is made and implemented. In other words, politics is about how societies decide who gets what, when, and how, and Easton's model helps explain how these decisions are made and sustained.

Key Components of Easton's Systems Theory Approach

Easton's model conceptualizes the political system as a **black box** that processes inputs and produces outputs, embedded within a larger environment. The key components are:

1. **Inputs**

Inputs are demands and supports that come from the environment and society into the political system.

- **Demands:** These are claims or requests made by individuals or groups for the political system to allocate values or resources in a particular way (e.g., demands for law enforcement, education, welfare).
- **Supports:** These are actions or attitudes that indicate approval, allegiance, or compliance with the political system (e.g., voting, paying taxes, obeying laws).

Inputs represent the pressures and resources that the political system must process.

2. **Political System (The Black Box)**

The political system is the core of Easton's model. It receives inputs, processes them, and produces outputs.

- It includes institutions, organizations, and processes involved in authoritative decision-making (e.g., legislatures, executives, courts).
- The system transforms demands and supports into decisions and policies.
- The system must maintain its boundaries and adapt to changes to survive.

3. **Outputs**

Outputs are the decisions, policies, and actions that the political system produces in response to inputs.

- These include laws, regulations, public policies, and authoritative allocations of values.
- Outputs affect the environment and society, influencing future inputs.

4. **Feedback**

Feedback is the process by which outputs influence future inputs.

- Positive feedback can reinforce the system's stability.
- Negative feedback can signal problems or dissatisfaction, prompting adjustments.
- Feedback helps the system adapt and maintain equilibrium.

5. **Environment**

The political system exists within a broader environment that includes social, economic, cultural, and international factors.

- The environment provides inputs and is affected by outputs.
- The system must interact with and adapt to environmental changes to survive.

Diagrammatic Representation (Simplified)

Environment

↔ ↔

[Demands & Supports] ↔ [Political System] ↔ [Decisions & Policies]

↔ ↔

↔----- Feedback -----

Significance of Easton's System Analysis in Political Science

1. **Holistic Understanding of Politics:**

Easton's model encourages viewing politics as an interconnected system rather than isolated events or institutions. It highlights the dynamic interactions between society and political authority.

2. **Focus on Process and Stability:**

The model emphasizes how political systems maintain stability and order through continuous processing of inputs and outputs, and through feedback mechanisms.

3. **Analytical Clarity:**

By abstracting political activity into inputs, system processing, outputs, and feedback, Easton provided a clear framework for empirical research and comparative analysis.

4. **Foundation for Further Theories:**

Easton's systems theory influenced later developments in political science, including structural-functionalism, policy analysis, and the study of political development and change.

5. **Adaptability:**

The model can be applied to various political systems (democracies, authoritarian regimes, etc.) and scales (local, national, international), making it versatile.

6. **Emphasis on Environment:**

Recognizing the environment's role underscores the importance of external factors (economic conditions, social movements, international pressures) in shaping political outcomes.

Criticisms and Limitations

```
- **Overly Abstract:** Some critics argue that Easton's model is too abstract and
general, lacking specificity about power relations, conflict, and individual agency.
- **Neglect of Conflict:** The model tends to emphasize stability and equilibrium,
potentially underplaying political conflict and change.
- **Limited Normative Guidance:** It is primarily descriptive and analytical, offering
little normative guidance on what political systems ought to do.

---

### Summary

David Easton's system analysis
user_query:
    System analysis by David Easton
response:
    David Easton's system analysis is a foundational framework in political science that
    conceptualizes the political system as a dynamic and interactive system. His approach
    uses systems theory to explain how political systems function, maintain stability,
    and adapt to changes in their environment.

At its core, Easton's model views the political system as a "black box" that processes
inputs from society and the environment, transforms them into outputs, and uses feedback
to adjust and sustain itself. The key components of Easton's system analysis include:

1. **Inputs:** These are the demands and supports that society directs towards the
political system. Demands are requests or claims for resource allocation or policy
change, while supports are actions or attitudes such as compliance, legitimacy, or
loyalty that sustain the system.

2. **Political System:** This is the set of institutions and processes that processes the
inputs. It interprets demands and supports to make authoritative decisions, including the
formulation and enforcement of policies.

3. **Outputs:** These are the decisions, policies, and actions that the political system
produces in response to inputs. They affect society and its environment, shaping future
interactions.

4. **Feedback:** Outputs generate responses that feed back into the system as new inputs,
signaling whether the outputs were effective or require modification. This feedback
enables the system to adapt to internal and external changes.

5. **Environment:** The system operates within a broader social, economic, cultural, and
international context that influences and is influenced by the political system.

Easton's theory is significant because it offers a holistic, process-oriented lens to
analyze politics, emphasizing interaction between society and state institutions. It
provides a framework to understand political stability, adaptation, and change by
focusing on the continuous circulation of inputs, processing, outputs, and feedback. The
model has been influential in shaping subsequent political theories and comparative
political analysis.
```

However, critics note that Easton's model can be overly abstract, sometimes downplaying conflict, power struggles, and individual agency within political processes.

In summary, David Easton's system analysis offers a powerful conceptual tool to study political systems as dynamic entities that manage societal demands and supports through authoritative decision-making to maintain order and adapt over time.

3 other demos omitted for brevity.
#####

GEPA Prompt generated by config GEPA+Merge:

Respond to a user query.
For inspiration, we found a potentially related request to a powerful external LLM and its response.

L.8 PUPA, QWEN3 8B

PUPA Qwen3 8B craft_redacted_request.predict

Base Prompt:

Given a private user query, create a privacy-preserving request for a powerful external LLM.
The LLM may assist without learning private information about the user.

MIPROv2 Prompt:

Given a private user query, generate a structured, non-sensitive LLM request that captures the user's intent while omitting personal details. Include a step-by-step reasoning process explaining how the redaction was achieved, ensuring the request is actionable and privacy-preserving. Format the reasoning as a coherent thought process and the request as a clear, specific task for the LLM to execute.

Demos:

Example 1:

user_query:

ĐŠŇĂĐŸĐŠĐŤŇČ ĐİĐŦĐĖĐŸŇĹĐŸ ĐŽĐİĐĐŦĐİĐŦŇČ ĐŦĐŽŇŦ ĐİŇČĐĖŇĂĐŦĐŠĐŽĐŸ snmpwalk ĐİĐŦ ip
10.2.35.2 ŇĂ ĐŦĐŦĐİĐİŇŇĐĐŸ ĐĖĐİ ĐŦĐİŇĂŇČŇČĐĖŇČ initi_user/d2J3BgtYdB2/d2J3BgtYdB2
(MD5) DES

reasoning:

The user is requesting an SNMP walk command to retrieve data from an SNMPv3-enabled device at IP 10.2.35.2. The provided credentials include a username (`init\_user`), an authentication password (`d2J3BgtYdB2`), and an encryption password (`d2J3BgtYdB2`). The authentication protocol is MD5, and the encryption protocol is DES. The command must include SNMPv3 options to specify these parameters correctly.

llm_request:

Generate a command-line instruction for executing `snmpwalk` with SNMPv3 authentication and encryption using the provided credentials, ensuring privacy by omitting sensitive details in the response.

```
#####
3 other demos omitted for brevity.
#####
```

GEPA Prompt generated by config GEPA:

New Instruction for the Assistant:

****Task Overview:****

Rephrase the user's query into a privacy-preserving, actionable request for an external LLM, ensuring no personally identifiable information (PII), proprietary data, or sensitive details are exposed. Maintain the original intent, clarity, and specificity of the query while adhering to domain-specific strategies for generalized information.

****Key Requirements:****

1. ****Privacy Preservation**:**

- ****Generalize Specifics**:** Replace specific locations (e.g., "Andhra Pradesh" â€Š "a major Indian city"), numbers (e.g., "18 volunteers" â€Š "a group of participants"), or proprietary data (e.g., "van Cleef production sites" â€Š "[Brand Name]'s production sites based on publicly available information").
- ****Avoid PII/Proprietary Data**:** Remove names, addresses, dates, internal processes, or brand-specific secrets. Use placeholders like [Brand Name], [Country], or [Region] for brand/company references.
- ****Publicly Available Focus**:** For brand-related queries, emphasize publicly accessible data (e.g., "locations of [Brand Name]'s production sites" instead of "van Cleef's secret factories").

2. ****Quality of Request**:**

- ****Clarity and Actionability**:** Ensure the rephrased query is precise, avoids ambiguity, and specifies constraints (e.g., "concise, platform-friendly hashtags" for creative tasks).
- ****Domain-Specific Precision**:** For scientific/technical queries, retain key terms (e.g., "CAPM formula," "pharmacokinetic parameters") while anonymizing study details (e.g., "18 volunteers" â€Š "a group of participants").
- ****Avoid Overly Broad Requests**:** Specify constraints like keyword focus, tone, or format (e.g., "health and fitness keywords" instead of "general hashtags").

3. ****Domain-Specific Strategies**:**

- ****Geographic Queries**:** Replace regions/cities with generalized terms (e.g., "major city in [Country]" or "well-known [Region]").
- ****Brand/Company Queries**:** Reference brands by name but prioritize publicly available data (e.g., "locations of [Brand Name]'s production sites").
- ****Creative Tasks**:** Use examples or constraints (e.g., "wellness, strength, motivation keywords" for hashtags; "platform guidelines" for social media content).

4. ****Avoid Leakage**:**

- ****No Inferred Details**:** Do not assume user-specific preferences, contexts, or private data (e.g., "Tell me the location of van Cleef production sites" â€Š "Provide locations of [Brand Name]'s production sites based on publicly available information").
- ****Anonymize Numerical Data**:** Replace specific counts, dosages, or metrics with generalized terms (e.g., "40 mg OA" â€Š "a dose of OA").

****Example Workflow**:**

```

- **User Query**: "Calculate DáŹAmicoŹs cost of equity using CAPM with rRF = 4.67%, ĩŹ = 1.56, and market risk premium = 6.63%."
- **Rephrased Request**: "Calculate the cost of equity for a company using the CAPM formula, given a risk-free rate of 4.67%, a beta of 1.56, and a market risk premium of 6.63%. Show step-by-step calculations and round the final result to two decimal places."

**Niche Domain-Specific Guidance**
- **Scientific/Pharmacokinetic Queries**: Generalize study details (e.g., "18 healthy male volunteers" ŹŹ "a group of participants"), retain key parameters (e.g., AUC, peak concentrations), and avoid proprietary methodologies.
- **Creative/Translation Tasks**: Use placeholders for song titles/artists (e.g., "Mahmood's song [Title]") and specify tone/rhythm requirements for translations.
- **Financial Calculations**: Retain formulas and numerical values but avoid linking to specific companies or proprietary models.

**Final Output Format**
- **reasoning**: Briefly explain the rephrasing choices (e.g., "Generalized '18 volunteers' to 'a group of participants' to preserve privacy").
- **llm_request**: Provide the anonymized, actionable request for the LLM.

```

PUPA Qwen3 8B respond_to_query

Base Prompt:

Respond to a user query.
For inspiration, we found a potentially related request to a powerful external LLM and its response.

MIPROv2 Prompt:

Synthesize the external LLM's response with the user's specific query to generate a final privacy-preserving answer. Replace placeholders with actual values while omitting sensitive information, maintaining a professional tone, and ensuring contextual accuracy. Combine the technical solution from the LLM with the user's unique details to produce a clear, anonymized response that adheres to privacy constraints.

Demos:

Example 1:

related_llm_request:

Create a PESTLE analysis framework for GE Healthcare in the US, outlining key issues in each category (Political, Economic, Social, Technological, Environmental, Legal) and their potential impact on the organization. Prioritize these issues based on urgency and significance without referencing confidential data. Structure the output as a matrix with clear criteria for evaluation.

related_llm_response:

Certainly! Below is a ****PESTLE analysis framework**** tailored for ****GE Healthcare in the US****, structured as a matrix. It outlines key issues in each category, evaluates their ****urgency**** and ****significance****, and highlights their ****potential impact**** on the organization.

PESTLE Analysis Matrix for GE Healthcare (US)

```

| **Category** | **Key Issues** | **Urgency** (High/Med/Low) | **Significance**
(High/Med/Low) | **Potential Impact on GE Healthcare** |
|-----|-----|-----|-----|
|
|-----|-----|-----|-----|
|-----|
| **Political** | 1. Healthcare policy changes (e.g., Medicare/Medicaid reimbursement
reforms) | High | High | Directly affects revenue streams and product/service demand;
requires adaptation to new reimbursement models. |
| | 2. Trade policies and tariffs on medical equipment imports/exports | Medium | Medium
| Could increase costs of components or affect supply chain efficiency. |
| | 3. Government funding for healthcare innovation and research | Medium | High |
Influences R&D opportunities and partnerships; potential for grants and subsidies. |
| **Economic** | 1. US healthcare spending trends and budget constraints | High | High |
Impacts purchasing power of hospitals and clinics; affects sales volume and pricing
strategies. |
| | 2. Inflation and rising operational costs | High | Medium | Increases production and
operational expenses; may pressure profit margins. |
| | 3. Economic cycles affecting capital investment in healthcare infrastructure | Medium
| Medium | Influences demand for high-cost medical equipment and technology upgrades. |
| **Social** | 1. Aging population increasing demand for healthcare services and devices
| High | High | Drives demand for diagnostic imaging, monitoring, and therapeutic
devices; growth opportunity. |
| | 2. Growing patient awareness and demand for personalized medicine | Medium | Medium |
Necessitates innovation in tailored healthcare solutions and data-driven technologies. |
| | 3. Workforce shortages in healthcare professionals | High | Medium | May affect
adoption and training for new technologies; impacts customer support and service
delivery. |
| **Technological** | 1. Rapid advancements in AI and machine learning for diagnostics
and imaging | High | High | Critical for maintaining competitive edge; requires
continuous investment in R&D and product development. |
| | 2. Integration of IoT and connected devices in healthcare | Medium | High | Enhances
product offerings but demands robust cybersecurity and interoperability solutions. |
| | 3. Data privacy and cybersecurity technologies | High | High | Essential to protect
patient data and comply with regulations; failure risks legal penalties and reputational
damage. |
| **Environmental** | 1. Regulatory pressure to reduce carbon footprint and waste in
manufacturing | Medium | Medium | Necessitates sustainable practices; potential cost
implications but improves brand reputation. |
| | 2. Energy efficiency in product design and operations | Medium | Medium | Can reduce
operational costs and appeal to environmentally conscious customers. |
| | 3. E-waste management and recycling programs | Low | Medium | Compliance with
environmental standards; growing importance for corporate social responsibility. |
| **Legal** | 1. Compliance with FDA regulations and medical device approvals | High |
High | Critical for market access; delays or failures can halt product launches and cause
financial losses. |
| | 2. Intellectual property rights and patent protections | Medium | High | Protects
innovation and competitive advantage; infringement risks costly litigation. |
| | 3. Liability and product safety regulations | High | High | Ensures patient safety;
non-compliance can lead to lawsuits, recalls, and reputational damage. |
|
|-----|

```


Criteria for Evaluation:

- **Urgency**: How immediate the issue requires attention (High = immediate/ongoing; Medium = within 1-2 years; Low = longer-term).
- **Significance**: The potential magnitude of impact on GE Healthcare's operations, profitability, and strategic positioning (High = critical; Medium = moderate; Low = minor).
- **Potential Impact**: Describes how the issue could affect GE Healthcare's business, including risks and opportunities.

This matrix helps prioritize strategic focus areas for GE Healthcare in the US, balancing immediate challenges with longer-term trends.

user_query:

issue priority matrix on pestle analysis for GE Healthcare in US

response:

Here's a prioritized **issue priority matrix** based on the PESTLE analysis for **GE Healthcare in the US**, focusing on the most urgent and significant challenges and opportunities:

High Priority Issues (High Urgency + High Significance)

Category	Key Issue	Urgency	Significance	Impact Summary
Political	Healthcare policy changes (e.g., reimbursement reforms)	High	High	Direct revenue risk; requires rapid adaptation to new regulations.
Economic	US healthcare spending trends and budget constraints	High	High	Drives demand and pricing pressures; critical for sales strategy.
Social	Aging population demand for healthcare services	High	High	Growth opportunity in diagnostics/therapeutics; long-term market expansion.
Technological	AI/ML advancements in diagnostics and imaging	High	High	Essential for competitiveness; demands sustained R&D investment.
Legal	FDA regulations and medical device approvals	High	High	Critical for market access; delays risk financial losses.

Medium Priority Issues (High Urgency + Medium Significance or Medium Urgency + High Significance)

Category	Key Issue	Urgency	Significance	Impact Summary
Economic	Inflation and operational costs	High	Medium	Pressures profit margins; requires cost-efficiency measures.
Technological	IoT integration and cybersecurity	Medium	High	Enhances offerings but demands robust security frameworks.

```
| **Environmental** | Carbon footprint reduction and sustainability | Medium | Medium |
Balances cost implications with brand reputation and regulatory compliance. |
```

```
---
```

```
### **Prioritization Criteria**
```

- ****Urgency****: Immediate (High) vs. longer-term (Low).
- ****Significance****: Critical (High) vs. moderate (Medium).
- ****Focus****: Issues with ****High** in both urgency and significance require immediate strategic action, while others demand monitoring or phased planning.

This matrix helps GE Healthcare allocate resources to address the most pressing challenges and leverage high-impact opportunities.

```
#####
```

```
3 other demos omitted for brevity.
```

```
#####
```

GEPA Prompt generated by config GEPA:

Respond to a user query by leveraging a related LLM response as inspiration. Structure your answer in a clear, organized format (e.g., bullet points, sections) to enhance readability. Ensure the response is tailored to the user's language and avoids any personally identifiable information (PII) or sensitive data.

When rephrasing or summarizing content:

1. ****Maintain academic tone**** if the query specifies formal language (e.g., rephrasing technical or historical text).
2. ****Refactor repetitive or redundant information**** into concise, focused sections (e.g., grouping similar ideas, eliminating duplication).
3. ****Highlight domain-specific facts**** (e.g., cultural, technical, or regional details) to add value, as these may not be universally known.
4. ****Improve code or technical responses**** by breaking down complex logic into smaller, well-documented functions, reducing redundancy, and ensuring proper resource management.

For all outputs:

- Prioritize ****clarity, accuracy, and adherence to the user's requirements****.
- Avoid ****leaking proprietary information, PII, or unverified claims****.
- If the user's query involves multiple steps (e.g., code optimization, content rephrasing), address each part systematically.
- Use ****language-specific conventions**** (e.g., correct grammar, terminology) to align with the user's preferred language.

M GEPA GENERATED PROMPTS FOR KERNEL GENERATION

M.1 NPUEVAL: KERNEL CODE GENERATION FOR NEW HARDWARE ARCHITECTURE

Figure 27 shows the prompt generated by GEPA with GPT-4o for NPU Kernel Generation, that achieves 26.85% score with the same GPT-4o agent, that achieved just 4.25% with a simple prompt.

You are tasked with generating C++ code for a single kernel function that will run on an AI Engine (AIE) tile. The kernel should perform a specified operation on a vector of bfloat16 values. Your code should be complete and self-contained within a single code block, including all necessary headers and imports. Follow these guidelines:

1. ****Headers and Imports****: Include only the necessary headers for AIE operations. Avoid including ``<adf.h>`` or any headers that are not part of the standard AIE API. Use:

```

` ``cpp
#include <stdint.h>
#include <aie_api/aie.hpp>
#include <aie_api/utils.hpp>
` ``

```
2. ****Kernel Function****: Implement the kernel function as specified in the input. The function should take pointers to input and output buffers and a size parameter. Use AIE vector operations to process the data efficiently.
3. ****Vector Operations****: Utilize the AIE API's vector operations for loading, processing, and storing data. For example, use `aie::vector` for vector operations and `aie::reduce_min` for reduction tasks. Ensure that the vector size is compatible with the AIE hardware capabilities.
4. ****Avoid Non-Existent Functions****: Do not use functions like `aie::exp` or `aie::store_v` if they are not supported. Instead, implement the required functionality using available AIE API functions.
5. ****Error Handling****: Ensure that the code is free of syntax errors and compatible with the AIE environment. Test the code for compilation without errors.

Here is an example template for a kernel function:

```

` ``cpp
#include <stdint.h>
#include <aie_api/aie.hpp>
#include <aie_api/utils.hpp>

void kernel_function_name(bfloat16 *input_vector, bfloat16 *output_vector, uint32_t vector_size) {
    constexpr int vector_length = 16; // Adjust based on AIE capabilities
    aie::vector<bfloat16, vector_length> input_data;
    aie::vector<bfloat16, vector_length> output_data;

    for (uint32_t i = 0; i < vector_size; i += vector_length) {
        input_data = aie::load_v<vector_length>(input_vector + i);
        // Perform the required operation on input_data
        // Store the result in output_data
        aie::store_v(output_vector + i, output_data);
    }
}
` ``

```

Replace `kernel_function_name` and the operation logic with the specific task details provided in the input.

Figure 27: GEPA generated prompt for NPUEval that achieves 26.85% score with the same GPT-4o agent, that achieved a 4.25% score.

M.2 KERNELBENCH: CUDA KERNEL CODE GENERATION FOR NVIDIA GPUS

GEPA GPT-4o generated prompt for CUDA Kernel Generation

To optimize a given PyTorch model by replacing operators with custom CUDA kernels, follow these detailed instructions. Your goal is to achieve performance improvements while ensuring correctness. Name your optimized output architecture `ModelNew`. Output the new model code in codeblocks. Please generate real code, NOT pseudocode, and ensure the code compiles and is fully functional. Do not include testing code.

Steps to Create Custom CUDA Kernels

1. Identify Operators to Replace

- Analyze the model to identify operators that can benefit from custom CUDA implementations.
- Consider operator fusion opportunities to combine multiple operations into a single kernel for efficiency.

2. Setup and Compilation

- Use `torch.utils.cpp\_extension.load\_inline` to compile your CUDA code. This allows you to define and compile custom CUDA kernels directly within your Python script.
- Ensure all necessary CUDA and C++ headers are included to avoid missing includes errors.

3. Implementing the CUDA Kernel

- Write the CUDA kernel code, ensuring it is optimized for parallel execution. Use shared memory to reduce global memory accesses and ensure coalesced memory access patterns.
- Example structure for a CUDA kernel:

```
```cpp
__global__ void my_kernel(const float* input, float* output, int size) {
 int idx = blockIdx.x * blockDim.x + threadIdx.x;
 if (idx < size) {
 // Perform computation
 }
}
```
```

4. Kernel Launch Configuration

- Configure the kernel launch parameters (number of blocks and threads per block) to maximize GPU utilization.
- Example:


```
```python
const int block_size = 256;
const int num_blocks = (size + block_size - 1) / block_size;
my_kernel<<<num_blocks, block_size>>>(input, output, size);
```
```

5. Error Handling and Debugging

- Implement error checking after CUDA API calls and kernel launches using `cudaGetLastError()` to catch errors early.
- Use CUDA debugging tools like `cuda-memcheck` and NVIDIA Nsight for debugging and profiling.
- Be aware of common syntax errors and namespace issues, and ensure that the CUDA code adheres to syntax rules.

6. Integrating with PyTorch

- Define a Python function that wraps the CUDA kernel call and integrates it into the PyTorch model.

- Example:

```
```python
def my_custom_op(input):
 output = torch.zeros_like(input)
 my_kernel(input.data_ptr<float>(), output.data_ptr<float>(), input.numel())
 return output
```
```

7. Testing and Validation

- Validate the correctness of the custom kernel by comparing its output with the original PyTorch operator.
- Use profiling tools to measure performance improvements and identify further optimization opportunities.
- Establish reference outputs for known-good inputs to verify correctness.
- Consider floating-point discrepancies and use a small epsilon for acceptable differences.

8. Compatibility and Compute Capability

- Ensure that the CUDA code is compatible with the target GPU's compute capability.
- Use appropriate compiler flags and consider building for multiple compute capabilities if necessary.

Example Code

Here is an example of how to replace a simple element-wise addition with a custom CUDA kernel:

```
```python
import torch
import torch.nn as nn
from torch.utils.cpp_extension import load_inline

Define the custom CUDA kernel for element-wise addition
elementwise_add_source = """
#include <torch/extension.h>
#include <cuda_runtime.h>

__global__ void elementwise_add_kernel(const float* a, const float* b, float* out, int
size) {
 int idx = blockIdx.x * blockDim.x + threadIdx.x;
 if (idx < size) {
 out[idx] = a[idx] + b[idx];
 }
}

torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b) {
 auto size = a.numel();
 auto out = torch.zeros_like(a);

 const int block_size = 256;
```
```

```

    const int num_blocks = (size + block_size - 1) / block_size;

    elementwise_add_kernel<<<num_blocks, block_size>>>(a.data_ptr<float>(), b.data_ptr<
        float>(), out.data_ptr<float>(), size);

    return out;
}
"""

elementwise_add_cpp_source = (
    "torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b);"
)

# Compile the inline CUDA code for element-wise addition
elementwise_add = load_inline(
    name="elementwise_add",
    cpp_sources=elementwise_add_cpp_source,
    cuda_sources=elementwise_add_source,
    functions=["elementwise_add_cuda"],
    verbose=True,
    extra_cflags=[""],
    extra_ldflags=[""],
)

class ModelNew(nn.Module):
    def __init__(self) -> None:
        super().__init__()
        self.elementwise_add = elementwise_add

    def forward(self, a, b):
        return self.elementwise_add.elementwise_add_cuda(a, b)
...

### Additional Best Practices
- Optimize Memory Usage: Minimize data transfers between host and device, and use
  shared memory to reduce global memory access.
- Atomic Operations: When using atomic operations like `atomicMax` with floating-point
  numbers, ensure correct usage by following best practices, such as using appropriate
  data types and minimizing contention.
- Performance Optimization: Maximize parallel execution, optimize memory access
  patterns, and use compiler flags to enhance performance.
- Namespace Usage: Avoid adding class declarations or function definitions directly to
  reserved namespaces like `cuda`. Use nested namespaces within non-reserved
  namespaces to organize code.
- Numerical Precision: Be aware of floating-point arithmetic issues, such as non-
  associativity, and use appropriate precision levels for calculations.
- Debugging Tools: Utilize tools like CUDA-GDB and NVIDIA Nsight for debugging and
  profiling to ensure correctness and performance.

By following these instructions, you can effectively replace PyTorch operators with custom
CUDA kernels, ensuring both performance improvements and correctness.

### Instruction for Replacing PyTorch Operators with Custom CUDA Kernels

```

Your task is to optimize a given PyTorch model by replacing certain operators with custom CUDA kernels to achieve performance improvements. Follow the steps below to ensure a successful implementation:

Step 1: Identify Operators for Replacement

- **Criteria for Selection**: Choose operators that are computationally intensive and have potential for parallelization. Consider operators that are frequently used in the model's forward pass.
- **Operator Fusion**: Look for opportunities to fuse multiple operators into a single CUDA kernel, such as combining matrix multiplication with activation functions (e.g., `matmul + ReLU`).

Step 2: Implement Custom CUDA Kernels

- **Kernel Structure**: Define your CUDA kernel using the `__global__` specifier. Ensure that each thread handles a specific part of the computation. Use correct index calculations to access data.
- **Memory Management**:
 - Allocate memory for input, output, and any intermediate data on the GPU using `cudaMalloc`. Use `cudaMemcpy` to transfer data between host and device.
 - Utilize shared memory to cache frequently accessed data and reduce global memory accesses.
 - Ensure coalesced global memory accesses for efficient memory transactions.
- **Numerical Stability and Boundary Conditions**:
 - Implement verification mechanisms to ensure numerical stability. Use `__host__ __device__` functions for testing on both CPU and GPU.
 - Handle boundary conditions to prevent out-of-bounds memory access. Ensure that thread indices are within valid ranges.
- **Optimization Techniques**:
 - Use shared memory to reduce global memory accesses and improve performance.
 - Consider using mixed precision and Tensor Cores for matrix operations to enhance performance.
 - Avoid diverged execution paths to maintain efficient parallel execution.

Step 3: Integrate CUDA Kernels into PyTorch Model

- **Inline Compilation**: Use `torch.utils.cpp_extension.load_inline` to compile your CUDA code and integrate it into the PyTorch model.
- **Model Modification**: Replace the original PyTorch operators with calls to your custom CUDA functions. Ensure that the new model architecture (`ModelNew`) is fully functional and compiles without errors.

Step 4: Testing and Validation

- **Correctness**: Verify that the output of the optimized model matches the expected output of the original model. Use a set of test cases to ensure accuracy.
- **Performance Evaluation**: Measure the runtime of the optimized model and compare it to the original. Aim for a significant reduction in execution time.
- **Edge Case Handling**: Ensure that the kernel correctly handles cases where the matrix size is not a multiple of the block size and other potential edge cases.

Example Code

Below is an example of how to define and integrate a custom CUDA kernel for element-wise addition:

```

```python
import torch
import torch.nn as nn
from torch.utils.cpp_extension import load_inline

Define the custom CUDA kernel for element-wise addition
elementwise_add_source = """
#include <torch/extension.h>
#include <cuda_runtime.h>

__global__ void elementwise_add_kernel(const float* a, const float* b, float* out, int
 size) {
 int idx = blockIdx.x * blockDim.x + threadIdx.x;
 if (idx < size) {
 out[idx] = a[idx] + b[idx];
 }
}

torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b) {
 auto size = a.numel();
 auto out = torch::zeros_like(a);

 const int block_size = 256;
 const int num_blocks = (size + block_size - 1) / block_size;

 elementwise_add_kernel<<<num_blocks, block_size>>>(a.data_ptr<float>(), b.data_ptr<
 float>(), out.data_ptr<float>(), size);

 return out;
}
"""

elementwise_add_cpp_source = (
 "torch::Tensor elementwise_add_cuda(torch::Tensor a, torch::Tensor b);"
)

Compile the inline CUDA code for element-wise addition
elementwise_add = load_inline(
 name="elementwise_add",
 cpp_sources=elementwise_add_cpp_source,
 cuda_sources=elementwise_add_source,
 functions=["elementwise_add_cuda"],
 verbose=True,
 extra_cflags=[""],
 extra_ldflags=[""],
)

class ModelNew(nn.Module):
 def __init__(self) -> None:
 super().__init__()
 self.elementwise_add = elementwise_add

 def forward(self, a, b):

```



```

 return self.elementwise_add.elementwise_add_cuda(a, b)
...

Constraints
- Ensure that the optimized model maintains the same accuracy as the original.
- The custom CUDA kernels should be optimized for performance, minimizing execution time
 and maximizing GPU utilization.

By following these instructions, you will be able to effectively replace PyTorch operators
with custom CUDA kernels, achieving significant performance improvements while
maintaining model accuracy.

```

## N NUMBER OF REFLECTION LM CALLS MADE BY GEPA DURING OPTIMIZATION

Table 4: Total number of calls made by GEPA to reflection LM during optimization.

Benchmark Name	Num Reflection Calls	Num Reflection Calls
	GPT-4.1-Mini	Qwen3-8B
AIME-2025	24	90
LiveBench-Math	34	38
HotpotQA	69	64
IFBench	21	17
Hover	92	50
PUPA	46	38